

Functional Requirements Specification

Version 1.0

for

eMaap Platform

Submitted to:

Legal Metrology Division
Department of Consumer Affairs

Submitted by:



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1. Introduction

The Legal Metrology Division functions as a key regulatory arm under the Department of Consumer Affairs, operating within the administrative framework of the Ministry of Electronics and Information Technology (MeitY). Its primary mandate is to ensure accuracy and fairness in trade and commerce by regulating weights and measures and maintaining standards of metrology across the country. Legal metrology plays a critical role in safeguarding consumer interests, facilitating trade, and promoting transparency in transactions involving measurement-based products and services.

In a dynamic and rapidly evolving marketplace where consumer confidence is deeply intertwined with the integrity of measurements, the Legal Metrology Division serves as the watchdog and custodian of standardized measurement practices. Its responsibility includes formulating and enforcing legal requirements for measuring instruments, pre-packaged commodities, and commercial transactions involving weights and measures. It works in tandem with state legal metrology departments to ensure a harmonized, nation-wide approach to compliance and enforcement.

Traditionally, the regulatory and administrative processes under legal metrology have involved extensive paperwork, manual verification, and region-specific procedures. This has posed challenges in terms of transparency, efficiency, and accessibility. Recognizing the need for a digitally driven and citizen-centric approach, the division has embarked on a mission to modernize and streamline its operations through a centralized, end-to-end digital platform.

This proposed system, referred to as **eMaap (Electronic Metrology Administration and Automation Portal)**, aims to automate and integrate the complete lifecycle of legal metrology services: **user registration; application submission; verification, approval; and license issuance**. The platform is envisioned as a unified digital interface catering to all 36 states and Union Territories, offering seamless integration with their respective systems and departmental workflows. It will serve both business entities and regulatory authorities, acting as a bridge between consumers, traders, manufacturers, and government enforcement bodies.

The system will be designed to ensure full compliance with the Legal Metrology Act, 2009, and the associated rules and regulations. It will incorporate approval workflows that reflect the hierarchical structure of both central and state legal metrology authorities. By digitizing the documentation, verification, and approval processes, the platform will reduce turnaround time, minimize manual errors, and eliminate redundancy in data management.

A key component of the system is its **robust API integration with state departments, enabling real-time data exchange, status updates, and validation of documents submitted by users**. This will empower the Legal Metrology Division to maintain a centralized database of registered entities, approved licenses, and enforcement actions, enhancing traceability and accountability at all levels.

The system will support various services including, but not limited to: -

1. Registration of manufacturers, dealers, importers, and repairers of weights and measures
2. Issuance and renewal of licenses
3. Online verification and stamping of measuring instruments
4. Management of pre-packaged commodity declarations and approvals
5. Complaint management and inspection tracking
6. Centralized data analytics and reporting dashboards

From a user perspective, the platform will deliver a streamlined and user-friendly experience, supporting multilingual interfaces, mobile responsiveness, and automated notifications. From the regulator's standpoint, it will provide a scalable, secure, and policy-compliant digital ecosystem that can evolve with future legislative changes or administrative needs.

Moreover, the system aligns with the vision of the Digital India initiative, promoting digital governance, ease of doing business, and increased transparency in public service delivery. It is not just a technological upgrade but a transformational shift in how metrology-based regulatory functions is administered in India.

1.1 Purpose

The purpose of this Functional Requirement Specification (FRS) document is to define the detailed functional requirements for the development and implementation of the eMaap (Electronic Metrology Administration and Automation Portal). This system is being designed under the Legal Metrology Division, Department of Consumer Affairs, within the administrative scope of the Ministry of Electronics and Information Technology (MeitY). The FRS serves as a foundational blueprint for all stakeholders involved in the system's development, ensuring a clear and shared understanding of the system's functionalities, workflows, and performance expectations.

The eMaap portal is envisioned as a comprehensive digital platform that will unify and streamline all regulatory services related to legal metrology across India. These services include user registration, licensing, document verification, approval workflows, inspection tracking, and compliance management. The platform aims to eliminate redundant manual processes and improve transparency, consistency, and efficiency in legal metrology operations. Given the diversity of procedures followed by various state governments and the need for a centralized approach, the FRS document plays a critical role in aligning system functionalities with both central and state-specific regulatory requirements.

This document outlines the intended behavior of the eMaap system in various scenarios, specifying how users will interact with it and how the system will respond. It identifies and describes system users, their roles, input and output data, and the approval hierarchies required for different services. It also highlights the integration points with external APIs from state legal

metrology departments, which are essential for real-time validation and seamless service delivery across 36 States and Union Territories.

Moreover, the FRS ensures that the system is designed in compliance with the Legal Metrology Act, 2009, and associated rules, providing a reference point for regulatory consistency and technical accuracy. By documenting the functional requirements, this specification facilitates better planning, design, development, testing, and deployment phases of the project.

In summary, the FRS document serves as a vital communication tool between the project's technical and non-technical stakeholders. It ensures that all functional expectations are captured comprehensively, and that the resulting system delivers a robust, user-friendly, and policy-aligned solution to support the legal metrology ecosystem in India.

1.2 Scope

The eMaap platform is a centralized, state-integrated digital solution envisioned by the Legal Metrology Division under the Department of Consumer Affairs to enhance transparency, efficiency, and effectiveness in governance by automating and digitizing legal metrology services across India.

The objective of this initiative is to streamline the entire regulatory framework concerning weights and measures by creating a single, unified platform that provides end-to-end automation for services like licensing, registration, verification, enforcement, and analytics.

Key features of the software include

- modular architecture enabling flexibility and scalability,
- API-driven integration for real-time data sharing between Centre and States,
- role-based access control, blockchain-enabled certificate issuance,
- AI-powered dashboards for analytics and forecasting,
- mobile app support, and
- robust enforcement tracking with geo-tagging.

Benefits of the platform include

- enhanced ease of doing business,
- standardized and paperless workflows,
- improved compliance monitoring,
- faster service delivery,
- secure data handling, and
- effective decision-making through comprehensive MIS and predictive analytics.

By consolidating disparate systems into a single digital ecosystem, eMaap empowers stakeholders with transparent, accessible, and efficient services while ensuring national-level regulatory consistency.

1.3 Definitions, Acronyms, and Abbreviations

In this document, the following terms shall have respective meanings as indicated:

Table of Definitions	
Terms	Definitions
Legal Metrology	The branch of law and science that governs weights and measures and ensures accuracy and fairness in commercial transactions involving measurement.
Legal Metrology Division (LMD)	A division under the Department of Consumer Affairs responsible for enforcing the Legal Metrology Act, 2009. It regulates and standardizes measurements and measuring instruments used in trade.
eMaap Portal	The Electronic Metrology Administration and Automation Portal – a centralized digital platform designed to automate services under the Legal Metrology framework across India.
Licensee	An individual or business entity that holds a valid license issued by the Legal Metrology Department to manufacture, repair, import, or deal in weights and measures.
Stakeholder	Any individual or organization with an interest in or influence over the functioning of the Legal Metrology Division or eMaap system, including government bodies, businesses, and consumers.
Pre-packaged Commodity	Any product that is packed without the purchaser being present and is meant for retail sale with a declared weight or measure.
Verification	The process of checking and confirming the accuracy, authenticity, and compliance of submitted data, documents, or measuring instruments with legal standards.
Inspection	An official review conducted by legal metrology officers to assess compliance with applicable legal metrology standards and licensing requirements.
Approval Workflow	A pre-defined set of steps and decision points through which a request (e.g., license application) passes for review and approval at various administrative levels.
API Integration	Application Programming Interface integration refers to the connection between the eMaap portal and external systems (such as state-level databases) for real-time data exchange.
Compliance	Adherence to rules, regulations, and standards as prescribed by the Legal Metrology Act and related guidelines.
Central Repository	A centralized data store within the eMaap system that holds all records of users, licenses, inspections, and approvals.
Dashboard	A user interface component that displays key performance indicators, summaries, AI alerts to users based on their roles and permissions.

Table of Definitions	
Terms	Definitions
Role-Based Access Control (RBAC)	A security approach where users are assigned roles, and permissions are granted based on these roles to ensure data access is restricted to authorized personnel only.
Authentication	The process of confirming a user's identity, typically through credentials such as a username, password, OTP, or biometric verification
Authorization	The process of granting a user permission to access specific resources or perform certain actions within the system after successful authentication.
Encryption	The practice of encoding data to protect it from unauthorized access, ensuring secure transmission and storage.
Audit Trail	A chronological record of all user actions and system activities used for monitoring, security auditing, and compliance tracking.
Checklist Validation	A verification step that ensures all mandatory fields and required documents have been submitted before an application progresses in the workflow.
Supporting Document	Any required attachment that provides evidence or substantiates an application (e.g., PAN card, trade license, utility bill).
Data Validation	A systematic check to ensure that data submitted through forms meets required formats, ranges, and logic rules before processing.
e-Signature	A digital signature used to authenticate and authorize documents or applications electronically.
System Integration Testing (SIT)	A testing phase that checks whether multiple integrated components of the system work together as expected.
Regression Testing	A type of testing performed to ensure that new updates or changes do not break existing functionalities.
Test Case	A set of conditions or inputs used to test a particular feature or function of the software to confirm it behaves as expected.

Table of Acronyms		
Acronyms	Full Form	Description
FRS	Functional Requirement Specification	A document that outlines the functional requirements and features to be developed in the system.
LMD	Legal Metrology Division	The central authority under the Department of Consumer Affairs that governs weights and measures.

Table of Acronyms		
Acronyms	Full Form	Description
MeitY	Ministry of Electronics and Information Technology	The Government of India ministry responsible for promoting digital governance and IT infrastructure.
DoCA	Department of Consumer Affairs	A department under the Ministry of Consumer Affairs, Food and Public Distribution responsible for consumer protection and regulatory oversight.
eMaap	Electronic Metrology Administration and Automation Portal	The proposed web-based platform for managing legal metrology services.
API	Application Programming Interface	A set of rules and protocols for integrating different software systems and enabling data exchange.
OTP	One-Time Password	A time-sensitive code used for verifying user identity during registration or login.
UID	Unique Identification	A unique number assigned to users, licenses, or applications to ensure traceability within the system.
SLA	Service Level Agreement	A defined agreement between service providers and users outlining the expected performance and response times.
UI	User Interface	The visual component of the system with which users interact to access services and information.
UX	User Experience	The overall experience of a user when interacting with the system, including ease of use and satisfaction.
PDF	Portable Document Format	A file format used for document uploads and downloads, including application forms, licenses, and certificates.
SMS	Short Message Service	A text messaging service used for sending system-generated alerts or notifications to users.
PAN	Permanent Account Number	A unique 10-digit alphanumeric identifier issued by the Indian Income Tax Department for tax and identity purposes.
GST	Goods and Services Tax	A tax registration number required for businesses, which may be used during application and compliance verification.
QR Code	Quick Response Code	A scannable code that may be included on licenses or certificates issued by the system for verification purposes.

Table of Acronyms		
Acronyms	Full Form	Description
UAT	User Acceptance Testing	A phase in the software development lifecycle where end users test the system to ensure it meets their expectations and requirements.
SRS	Software Requirements Specification	A more detailed technical specification document derived from the FRS.
MIS	Management Information System	Tools and reports that provide insights into system operations and performance metrics.

Table of Abbreviation	
Terms	Definition
e-Governance	ICT (Information and Communication Technology) projects implemented in the government sector.
NICSI	National Informatics Centre Services Inc., New Delhi.
NIC	National Informatics Centre, New Delhi.
SI (System Integration)	The agency responsible for the end-to-end design, development, deployment, integration, operations, and maintenance of the eMaap platform
DoCA (Department of Consumer Affairs)	The nodal ministry under which the Legal Metrology Division functions.
DoLM (Department of legal metrology)	The division responsible for enforcing the Legal Metrology Act and associated rules.
MIS (Management Information system)	A reporting module within eMaap providing performance, inspection, and regulatory compliance data.
API (Application programming Interface)	Used to integrate eMaap with external platforms such as state systems or national services.
RRSL (Regional Reference Standard Laboratories)	Institutions responsible for testing and approving weighing and measuring instruments.
CLM (Central Legal Metrology)	The central division responsible for certifications and approvals integrated with eMaap.
SLA (Service Level Agreement)	A contractual document defining expected performance standards and penalties for non-compliance.

1.4 References

References	Descriptions
Legal Metrology Act, 2009	The central legislation that governs weights and measures and the legal standards for trade in India. This Act is the primary regulatory framework for the portal.
Digital India Program	A flagship initiative of the Government of India promoting e-Governance and digitization of public services. The eMaap portal is aligned with this mission.
Ministry of Electronics and Information Technology (MeitY) Guidelines	Includes policies and recommendations on digital infrastructure, software development, data security, and citizen service platforms.
e-Governance Standards and Guidelines by MeitY	Refers to common frameworks and standards such as interoperability, API guidelines, metadata schemas, and authentication mechanisms.
State-Specific Legal Metrology Guidelines	Individual rules or administrative workflows issued by state governments. These influence the configuration of state-level workflows within the system.
UIDAI Guidelines	Standards for Aadhaar authentication, used in user verification processes.
Information Technology Act, 2000 (with amendments)	Governs electronic records, digital signatures, and data protection—important for system compliance and legal validity of digital interactions.
GST Registration Rules	Referenced during business verification for dealers and manufacturers using GSTIN.
PAN Verification	Guidelines for validating PAN data used in user registration and compliance.
RTI Act, 2005	Provides users the right to request information from the system, which impacts audit trails and record-keeping functionality.

1.5 Overview

This document provides a high-level summary of the functional requirements, data collection processes, and associated systems relevant to the design and operation of the eMaap portal. The system is proposed as a unified platform to digitally manage all regulatory and administrative functions under the Legal Metrology framework, offering seamless services to stakeholders and ensuring alignment with national and state-level legal requirements.

Functional Requirements Identified and Proposed to be Implemented:

The eMaap system is designed to support the end-to-end digital processing of services provided by the Legal Metrology Division (LMD), Department of Consumer Affairs. Based on stakeholder consultations and statutory mandates, the following functional requirements have been identified and proposed:

- User Registration and Authentication: Online registration of businesses (manufacturers, dealers, importers, and repairers of weights and measures). Includes OTP-based verification, Aadhaar/PAN/GSTIN-based validation, and secure login.
- License Management: Issuance, amendment, renewal, duplication, suspension, and cancellation of licenses for various categories of stakeholders based on rules under the Legal Metrology Act.
- Document Verification Workflow: Automated verification of supporting documents using APIs and internal validation logic.
- Pre-Packaged Commodity Approval: Online application, assessment, and approval process for packaged commodities, as per the LM Packaged Commodities Rules.
- Approval Hierarchies: Configurable workflows mapped to central and state-specific administrative hierarchies, ensuring correct routing and authorization at each level.
- Inspection Scheduling and Reporting: Assignment of inspectors, scheduling of physical inspections, and submission of inspection reports with geo-tagging and photographs.
- Complaint and Grievance Management: A dedicated module for lodging and tracking complaints by end users, along with escalation matrix and resolution workflows.
- State API Integrations: API-based integration with state legal metrology systems for real-time data exchange, application syncing, and inter-jurisdictional updates.
- Notifications and Alerts: Automated notifications to users and officials at each stage of the process via SMS, email, and portal alerts.
- Dashboard and Analytics: Role-specific dashboards for central, state, and field-level users with analytics on applications, verifications, inspections, and compliance metrics.

Proposed Data Collection:

The eMaap system is expected to collect, validate, and manage a variety of data types as part of its service delivery and compliance framework. Data collection will occur during registration, licensing, inspections, verification, and complaint management processes. The proposed data types include:

- User Identification and Profile Data:
 - Name, address, email, mobile number
 - Aadhaar, PAN, GSTIN (as applicable)
 - Type of applicant (manufacturer, dealer, etc.)
- Business Details:
 - Nature of business (weights, measures, commodities)
 - Premises address and geo-location
 - Trade license and firm registration documents

- License Data:
 - Type and category of license
 - Validity period, fee details, issuance authority
 - Status (issued, suspended, cancelled, expired)
- Application Documents:
 - Uploads in PDF/JPEG format (proof of identity, ownership, compliance certificates)
 - System-generated forms, declarations, and undertakings
- Verification & Inspection Records:
 - Assigned inspector details
 - Date, time, and outcome of inspection
 - Non-compliance observations, if any
- Complaint Data:
 - Nature of grievance, location, and stakeholder involved
 - Timeline of response and resolution actions taken
- System Metadata and Logs:
 - Time stamps for every transaction
 - Role-based user actions (e.g., who approved what, and when)
 - Communication logs (SMS, email)

Integration with eMaap and Other Systems:

The eMaap portal is designed to function both as a standalone system and as an integrated component within a broader digital governance ecosystem. It may interoperate with several external systems to ensure seamless service delivery, verification, and compliance.

Identified and Potentially Integrated Systems:

- State Legal Metrology Portals: eMaap will integrate via APIs with existing state-level legal metrology portals for license verification, approval tracking, and enforcement data exchange.
- UIDAI (Aadhaar): For user identity verification during registration or application processes.
- GST Network (GSTN): To validate GSTIN and ensure the applicant is a registered business entity.
- PAN Verification System (CBDT): For validating Permanent Account Numbers submitted by users.
- e-Sign and DigiLocker Services: To enable digital signing of applications and storage or retrieval of documents from verified government repositories.
- CPGRAMS or Departmental Grievance Portals: For aligning citizen grievance redressal systems if required.
- Geo-Tagging and Location APIs: For real-time capture of inspection locations and mapping of business premises.

2. Overall Description Methodology

This section outlines the structured and collaborative approach used in the development of the Functional Requirement Specification (FRS) for the eMaap system. It details the planning, information gathering, requirement modeling, validation, and documentation processes that were followed to ensure the system's design is both comprehensive and aligned with the needs of all stakeholders.

The goal of this methodology is to ensure clarity, transparency, and completeness in capturing what the system must do, how it should behave, and how it aligns with statutory, administrative, and user needs. The methodology was designed to be understandable by both technical and non-technical readers, as the system touches multiple levels of users—from central ministries to local inspectors, and from manufacturers to individual traders.

2.1 Objective of the Methodology

- Understand and document the current challenges in Legal Metrology processes across India.
- Identify the expectations of users at various levels—central, state, district, and citizen.
- Create a digital blueprint for seamless delivery of licensing, registration, verification, and complaint resolution services.
- Provide a clear and unambiguous guide for developers, testers, project managers, and stakeholders.

2.2 Key Stakeholder Engagement

- Central Legal Metrology Division (Department of Consumer Affairs)
- State Legal Metrology Departments (36 States/UTs)
- Legal Metrology Inspectors and field officers
- Manufacturers, dealers, repairers, and importers of weights and measures
- System integrators and developers
- IT consultants from MeitY and NIC

2.3 The process included

- **Workshops and Consultations:** State-wise sessions were conducted to gather localized workflows, challenges, and expectations.
- **Surveys and Feedback:** Structured questionnaires were shared with users to understand the pain points in the current manual or semi-automated processes.
- **Document Review:** Existing state systems, forms, licensing documents, and legal frameworks were studied in depth.
- **Site Observations:** Visits to legal metrology offices and field inspection sites to witness real-world processes.

3. Functional Requirements

The proposed system is designed as a fully online platform, eliminating the need for any physical touchpoints for processes such as registration, license issuance, renewal, amendments, document submission, payments of penalties across all modules. It aims to integrate and streamline operations by collecting codification from states and districts, mapping them with LGD codes, and creating various interconnected modules. The system will facilitate seamless registration and licensing processes for manufactures, dealers, repairers, packers, importers, and manufacturers of packaged commodities. The following section details the modules to be developed, which will incorporate enhanced features: -

3.1 Registration Process on eMaap Portal

The applicant registration process on the eMaap portal begins with the user providing a valid mobile number, email address, or other identity credentials to initiate registration. Once the initial details are submitted, the system generates a unique user ID for the registrant. To verify the user's identity, an OTP (One Time Password) is sent to the provided mobile number and email address. After the user successfully enters the OTP, the system verifies the credentials and assigns the user a secure login. This secure login grants the user access to the platform's services. While the image focuses on applicant registration, it implies that stakeholders such as dealers, manufacturers, or other user roles will undergo a similar identity-based registration and verification process, ensuring secure and role-appropriate access to the portal's functionalities.

3.1.1 New Registration #FR001

Description	This functionality allows users to register through a dedicated link. Upon clicking the "Register New User" link, the interface will present a choice of user types via either distinct UI tabs or a dropdown menu. User can select their appropriate role from options such as inspector, stakeholders, or admin to proceed with their registration.
User	1. Inspector: Legal Metrology officers responsible for inspections, verification and enforcement. 2. Stakeholder: External users such as manufacturer, dealer, packers or importers who require registration and access to licenses or certificates. 3. Admin: For personnel managing roles, approvals and user configurations.

Pre-Condition	1. The system should be online and reachable through the designated URL. 2. The user must access the portal via a supported web browser. 3. The “Register New User” link must be visible and accessible on the login or home page. 4. User must possess a valid email ID and mobile number.
Sequence of Operations	1. User clicks the “Register New User” link. 2. A registration form loads with UI tab/dropdown to select the User Type (Inspector / Stakeholder / Admin). 3. Based on the selection, specific fields are dynamically shown (e.g., Employee ID for Inspector, Business License Number for Stakeholder). 4. User fills the required details (name, contact, address, role-specific fields). 5. Captcha and agreement to terms & conditions are presented. 6. User submits the form. 7. System performs: a. Validation checks b. Duplicate check (email, mobile, document IDs) 8. If all validations pass: a. Registration is saved b. Confirmation screen is shown c. Verification email/SMS is sent
Post Condition	1. On successful registration, a confirmation message is shown, and an email/SMS with login credentials or activation link is sent. 2. User data is stored securely in the system and awaits further approval (if applicable, based on role). 3. A verification token is logged for audit and security tracking.
Validity Checks	1. Email format must follow RFC 5322 standard. (It means the rules email programs must follow so that message is understood and delivered correctly across different mail servers and clients (Like Gmail, outlook etc.)). 2. Mobile number must be 10 digits and start with valid prefixes (Length Always 10 Digits excluding country code, starting digits – Most common and currently valid for Indian mobile numbers are 6,7,8 and 9.) 3. Mandatory fields must not be null or empty.

	- Captcha must be correctly matched. - Unique constraints must be enforced on email, mobile number, and business ID fields.
Input	User type (from dropdown) - Full name - Email address - Mobile number - Address Role-specific data: - Inspector: Employee ID, Department Name - Stakeholder: Business/License Number, Company Name - Captcha code
Output	- Success message upon form submission - Email/SMS confirmation - Error message for invalid or duplicate entries - System logs and audit trail entries
Exception Handling	Email/Mobile already registered: Display “User with this email/mobile already exists.” Missing mandatory field: Highlight missing field(s) with “This field is required.” Invalid email/phone format: Display “Please enter a valid email/phone number.” Backend/server error: Show “We’re experiencing technical difficulties. Please try again later.” Verification link expired: Prompt “Verification link has expired. Please re-register.”

3.1.2 Role Based Registration #FR002

Description	The Role-Based Form Rendering feature in the eMaap platform is designed to dynamically present customized registration forms based on the user type selected during onboarding. This functionality ensures that users from different categories—Inspector / Field Officer, Stakeholder, and Admin / Legal Officer—are provided with relevant data entry fields, validation logic, and document upload options tailored to their specific roles. Upon clicking the “Register New User” link, users are prompted to select their role via a dropdown or tab interface. Based on this selection, the system loads a dedicated form:
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	1. Inspectors / Field Officers are required to provide government credentials such as Employee ID, Department, Official Email, and upload ID proof. 2. Stakeholders, such as manufacturers or dealers, must submit business details like Trade License, GSTIN/PAN, and business category along with the upload of supporting documents. 3. Admin / Legal Officers have a separate form designed for internal access request, tied to administrative responsibilities within the Legal Metrology system. Upon successful form submission and validation, the data is stored in the system, and an approval workflow is initiated for administrative review. This role-based rendering improves user experience, enhances system security, and ensures only authorized and verified users are onboarded into the eMaap platform.
User	1. Inspector / Field Officer Registration Form 2. Stakeholder Registration Form 3. Admin/Legal Officer Request Form
Pre-Condition	1. User must access the "Register New User" link from the portal's home page: a. The system must be online and fully functional. b. The role selection (Inspector / Stakeholder / Admin) must be active and responsive. c. User must have a valid email ID and mobile number for OTP-based verification. d. Required documents should be ready for upload.
Sequence of Operations	User accesses registration form: 1. Selects role from dropdown: Inspector / Stakeholder / Admin. 2. System dynamically renders role-specific form 3. User fills the required fields and uploads documents. 4. OTP is sent to provided mobile number; user enters OTP. 5. System validates input and OTP. 6. On success, form is submitted, and user is notified. 7. Workflow triggers review and approval process by admin/legal officer.

Post- Condition	Upon successful submission, the system: 1. Stores form data in a secure database. 2. Sends verification/confirmation via SMS/email. 3. Triggers workflow for approval by appropriate authority (Admin or DoCA). 4. Tags user under the selected role with appropriate access rights.
Validity Checks	1. Email ID: Must follow standard email format 2. Mobile Number: Must be 10 digits; OTP required. 3. Government Employee ID / Trade License No.: Alphanumeric validation; mandatory uniqueness. 4. GSTIN / PAN: Format-validated as per Indian regulatory standards. (For GSTIN – It includes 15 Characters, first 2 Digit should be state code, Next 10 Character will be PAN number, 13th character Entity number, 14th character usually blank by default or “Z”, 15th character will be checksum digit, For PAN – It includes 10 Characters, first 5 characters should be alphabetic, next 4 character will be digits, last character will be alphabetic). 5. File Uploads: Accept only PDF, JPG, PNG formats; maximum file size (e.g., 2MB). 6. Dropdown selections (Business Category): Must be from a pre-configured list; cannot be blank.

Input	Inspector / Field Officer Form Inputs: - 1. Full Name (First Name, Middle Name, Last Name) 2. Government Employee ID (We can validate through official letterheads, department seals, barcodes, Match employee ID against department issued documentation, and by suing RTI last report) 3. Department 4. Designation 5. Office Location 6. Official Email ID 7. Mobile Number (OTP verified) 8. ID Proof (Upload PDF/Image) a. Aadhar Card b. PAN Card c. Official Govt. ID card d. Duty Order (Optional) e. Inspection Notice (If applicable) f. Appointment letter Stakeholder Form Inputs: - 1. Business Name 2. Trade License Number 3. GSTIN / PAN 4. Contact Person Name 5. Business Address 6. Email ID 7. Mobile Number (OTP verified) 8. Business Category (Dropdown) 9. Upload Trade License & ID Proof Admin/Legal Officer form: - 1. Name 2. Employee ID/Authorization code 3. Department 4. Role Request: (Admin, Regional admin, Legal officer) 5. Upload nomination / Authorization letter 6. Email and mobile (OTP Verified)
Output	1. Registration confirmation message 2. OTP confirmation for mobile verification 3. Email notification for successful submission 4. Admin dashboard update for pending approval

Exception Handling	1. Invalid OTP: “Incorrect OTP entered. Please try again.” 2. Duplicate registration (based on email, mobile, or ID): “User already exists with provided credentials.” 3. Missing or invalid mandatory fields: Highlight and show “This field is required.” 4. Upload failure or unsupported format: “File must be a valid PDF or image and under size limit.” 5. Backend error (e.g., database failure): “System error. Please try again later or contact support.”
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3.1.3 OTP Verification (Email & SMS) #FR003

Description	This feature ensures secure user onboarding through two-factor authentication by triggering One-Time Passwords (OTPs) to both email and mobile number upon registration form submission, (OTP Length should contain for Mobile – 6 numeric value and for email – 6 to 8 alphanumeric value). Users must verify both their email and mobile within a stipulated time to complete registration. The system tracks verification status through flags (mobile verified, email verified) stored in the database and restricts progression until both are successfully confirmed.
User	1. Inspectors / Field Officers 2. Stakeholders (e.g., dealers, manufacturers, importers) 3. Admins / Legal Officers 4. All user types registering on the platform must pass dual OTP verification.
Pre-Condition	1. User has completed the registration form and submitted it. 2. Valid and unique email and mobile number are provided. 3. System is connected to email and SMS gateway services.
Sequence of Operations	1. User fills and submits the registration form. 2. System generates OTPs: a. One for mobile (via SMS) (6 digit numeric) b. One for email (6-8 alphanumeric) 3. OTPs are sent via integrated SMS and email gateways. 4. User enters received OTPs in respective fields. 5. System validates each OTP: a. If correct and within 5 minutes, marks as verified. b. If incorrect, allows up to 3 attempts per channel.

	6. Once both verifications succeed: a. Flags are saved in DB: mobile verified = true, email verified = true. b. Registration is considered complete and allowed to proceed.
Post Condition	1. On successful verification: a. Flags (email verified, mobile verified) are updated in the database. b. User is allowed to proceed to the next step (e.g., dashboard access or document upload). 2. On failure or timeout: a. System prompts user to re-initiate the registration or verification process.
Validity Checks	1. OTP must match the generated token associated with user session. 2. OTP must be entered within 5 minutes of being sent. 3. User is allowed a maximum of 3 retry attempts for both email and mobile separately. 4. Both OTP verifications must be successful before proceeding.
Input	1. Mobile number 2. Email address 3. OTPs entered by user
Output	1. OTPs sent via SMS and email 2. Success/failure message 3. Verification status update in DB 4. Proceed to next step if successful
Exception Handling	1. OTP not received: Option to resend OTP after a cooldown (e.g., 30 seconds). 2. Incorrect OTP: "Invalid OTP. You have X attempts remaining." 3. OTP expired (after 5 mins): "OTP expired. Please request a new one." 4. Max retries exceeded: "Maximum attempts reached. Please restart the verification process." 5. Gateway failure: "Unable to send OTP due to network error. Try again later."

3.1.4 Document Upload Handling #FR004

Description	The Document Upload Handling feature enables users to upload necessary supporting documents (e.g., ID proof, trade license) in secure and approved formats during registration or service application. The system supports file types of PDF, JPEG, and PNG with a maximum size of 5MB. Uploaded files are stored in a secure cloud directory, organized by user ID and document type, ensuring traceability, data security, and compliance with government standards.
User	1. Inspectors / Field Officers – Upload government-issued ID proofs. (Govt. Photo ID, Aadhar Card, PAN, Appointment Letter) 2. Stakeholders – Upload business licenses, ID proofs, GSTIN/PAN documents, etc. 3. Admins / Legal Officers – May upload authorization or verification documents during internal registration.
Pre - Condition	1. User is authenticated (or undergoing registration). 2. Required fields in the form have been filled. 3. Document is available in one of the allowed formats and within the size limit. 4. Cloud storage service is active and integrated.
Sequence of Operations	1. User reaches the document upload section. 2. System prompts for document selection/upload. 3. User selects a file (PDF, JPEG, or PNG ≤ 5MB). 4. System validates format and size. 5. If valid, the document is uploaded to a cloud folder: Folder structure: /cloud/documents/{User_ID}/ {Document Type} 6. File path is stored in the database linked to the user record. 7. User sees a confirmation message, and the form proceeds to the next step.
Post - Condition	1. Document is successfully uploaded to a cloud location with path structured by User_ID/Document Type/. 2. System confirms the upload and stores the file path in the database. 3. Document is accessible to authorized roles for processing and review.

Validity Checks	1. File type must be strictly .pdf, .jpeg, or .png. 2. File size must be $\leq$ 5MB. 3. System must reject any file that: a. Has an invalid extension b. Has an unsupported MIME type c. Exceeds size limit 4. Folder structure must match: /User_ID/document Type/ for consistent document management. 5. Only authenticated or actively registering users can upload. Submission and system notification
Input	1. File from user's device (PDF, JPEG, PNG) 2. Metadata (User ID, document type/category)
Output	1. Upload success/failure message 2. Secure cloud file path saved in the database 3. Preview or download link (if applicable for review) 4. If preview is not satisfied, there should be provision for re-upload the documents.
Exception Handling	1. Invalid file format: "Only PDF, JPEG, and PNG formats are allowed." 2. File size exceeded: "File exceeds the 5MB limit. Please upload a smaller file." 3. No file selected: "Please select a file to upload." 4. Upload error/cloud failure: "Upload failed due to network error. Try again later." 5. Access denied (on retrieval): "You do not have permission to view this document."

3.1.5 Submission & System Notification #FR005

Description	This feature handles the final step of form submission in the eMaap platform. Once a user completes and submits a registration or service request form, the system performs key actions to ensure successful communication and acknowledgment. It displays a confirmation screen to the user, sends a notification via email and SMS to the applicant confirming receipt of the application, and simultaneously alerts the admin or designated authority that a new registration is awaiting review. This enhances transparency, real-time tracking, and workflow automation.
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User	1. Applicants (Inspectors, Stakeholders, Admin/Legal Officer) – Users submitting the registration or service application. 2. System Admin / Review Officer – Users responsible for reviewing and approving submitted applications.
Pre - Condition	1. The user has completed all required fields and successfully uploaded necessary documents. 2. Both email and mobile number have been OTP verified. 3. The system is integrated with active email and SMS gateways.
Sequence of Operations	1. User submits the completed form. 2. System validates and stores the form data in the database. 3. A confirmation screen is rendered to the applicant with an acknowledgment message and reference ID. 4. System triggers: a. Email/SMS to applicant: “Your application has been received.” b. Email/system alert to Admin: “New registration pending review.” 5. The application is marked with status: Pending Review.
Post - Condition	1. User sees a confirmation message or screen with reference number. 2. Applicant receives SMS and email stating “Application received.” 3. Admin is notified via system-generated email or dashboard notification about “New registration pending review 4. Application data is logged into the backend and queued for review.
Validity Checks	1. Ensure all required fields are completed before enabling submission. 2. Validate email and mobile number before allowing notifications. 3. Confirm form data is successfully stored before sending notifications. 4. Notification delivery logs must be maintained with timestamp and status (sent/failed).
Input	1. Submitted form data (user info, documents, contact details) 2. Verified email and mobile number 3. Role-specific metadata (e.g., user type, department)
Output	1. Confirmation screen with acknowledgment ID 2. Email and SMS notification to applicant 3. System/email notification to admin 4. Application status update in system

Exception Handling	1. Form submission error (e.g., network timeout): “Submission failed. Please try again.” 2. Notification gateway failure: Retry notification; log the failure and alert system admin. 3. Email/SMS not sent: “Application submitted”. However, we were unable to send confirmation. Please contact support. 4. Database error: “System error”. Please retry after some time.”
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3.1.6 Admin Review Panel #FR006

Description	The Admin Review Panel enables authorized administrators to manage and act upon pending user registration requests. Upon logging into the system, the admin can access the “Pending Registrations” section, filter applications by role, submission date, or status, and perform detailed evaluations. Each registration entry includes form data, document uploads, and OTP verification logs. The admin can choose to approve or reject the application. Approved users are granted system access with login credentials and active status, while rejected applicants receive a notification with the mandatory rejection reason.
User	System Admin / Legal Officer – Authorized personnel responsible for reviewing, approving, or rejecting user registrations.
Pre - Condition	1. Admin is logged into the system with valid credentials. 2. There are pending registrations stored in the system. 3. Form submissions have been verified with OTP and document uploads. 4. Notifications services (email/SMS) are active.

Sequence of Operations	- Admin logs into the system. - Navigates to “Pending Registrations” from the dashboard. - Uses filters (Role, Date, Status) to locate applications. - Clicks on a specific application to open Detailed View. Reviews: - Submitted form data - Uploaded documents - OTP verification logs Admin selects Approve or Reject: If Approved: - Record moves to user table - Login credentials are generated - Account status is set to “Active” - Notification is sent to the applicant If Rejected: - Admin enters mandatory rejection reason - Application status set to “Rejected” - Rejection reason is sent to the applicant
Post - Condition	- Approved applications are transferred to the live user database and marked Active. - Login credentials are generated and sent to the approved user. - Rejected applications are logged with reason, and the applicant is notified. - Status updates are reflected in the system and accessible in dashboards.
Validity Checks	- Ensure all form fields and documents are fully loaded before allowing action. - Ensure OTP logs exist for the record. - Rejection cannot proceed without a reason (mandatory validation). - Only authorized roles (Admin/Legal Officer) can access and act on this panel. - Check for duplicate approvals to prevent multiple activations.
Input	Admin filters: Role, Date, Status Admin action: Approve / Reject - Rejection reason (mandatory if rejecting) System-fetched applicant details: Form, Docs, OTP log
Output	- Updated application status (Approved/Rejected) - System-generated login credentials (on approval) - Email/SMS notification to applicant - Audit trail and action logs

Exception Handling	1. Missing rejection reason: “Please enter a reason to reject the application.” 2. Document not accessible: “Unable to retrieve uploaded document. Try again.” 3. Credential generation error: “Error assigning login credentials. Please retry.” 4. System timeout: “Session expired. Please log in again.” 5. Notification service failure: Retry logic and fallback alert to system admin.
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3.1.7 User Login, Activation, and Role-Based Dashboard #FR007

Description	This functionality governs the user login and activation process post-approval. Once login credentials or activation links are sent to the user, the system ensures secure first-time access via password reset or activation. After successful authentication, users are redirected to dashboards based on their assigned roles—Inspector, Stakeholder, Admin, or Legal Officer—each tailored to display relevant modules and functionalities.
User	1. Inspector / Field Officer – Accesses inspection tasks and compliance records. 2. Stakeholder – Views notices, case status, and application tracking. 3. Admin – Manages users, configurations, and master data. 4. Legal Officer – Handles legal filings and court-related actions.
Pre - Condition	1. User has been approved by admin and credentials or activation link have been sent via email/SMS. 2. The system's authentication and dashboard modules are operational. 3. User has access to a secure browser/device.

Sequence of Operations	If login credentials sent: - User visits login page and enters username/password. - System prompts user to reset password (first login). - New password is set and stored securely. - Role-based dashboard is displayed. If activation link sent: - User clicks on activation link from email/SMS. - System prompts to set a new password. - Once set, user is logged in and redirected to the appropriate dashboard. Role-based Dashboard Redirection: - Inspector → Inspection Dashboard - Stakeholder → Notice & Case Status - Admin → Master Data and User Management - Legal Officer → Court Filing Interface
Post - Condition	- User sets a new password after first login or activation. - Role-based dashboard is rendered upon successful login. - Login session is established and logged. - Unauthorized roles or access attempts are blocked.
Validity Checks	- Password reset fields must match and meet policy (e.g., 8+ chars, alphanumeric, special character). - Activation link must be time-bound (e.g., valid for 24 hours). - System must verify role from database before dashboard routing. - User session token must be securely stored and time limited. - Dashboard components must load only if user has valid, active session and matching role permissions.
Input	- Username & password (or activation token) - New password (entered twice for confirmation) - User role (determined at registration approval)
Output	- Confirmation of password reset or activation - Session initiation - Role-based dashboard interface - Log entry for login and redirection
Exception Handling	Wrong credentials: "Invalid username or password." Expired activation link: "Activation link expired. Please request a new one." Mismatch in password reset: "Passwords do not match." Inactive account: "Your account is not yet activated. Contact admin." Unauthorized access: "Access denied. Insufficient permissions."

3.1.8 Use Cases (Registration Process)

Use Case 1: Stakeholder (Offender/Trader) Registration

Field Name	Description
Use Case ID	UC_REG_001
Use Case Name	Stakeholder (Offender/Trader) Registration
Created By (Date)	eMaap Functional Team (16-June-2025)
Last Update (Date)	16-June-2025
Primary Actor	Business Entity Representative / Trader
Secondary Actor	Admin (Reviewer), OTP Service, Email Gateway
Trigger Activating Use Case	User clicks on “Register New User” → selects “Stakeholder” from the role dropdown
Inter Linkages	OTP Verification, Document Upload, Admin Review Panel, Role-Based Dashboard
Normal Event Flow	1. User opens Stakeholder registration form 2. Fills in Business Name, Trade License, GSTIN, etc. 3. Verifies mobile via OTP 4. Uploads Trade License/ID Proof 5. Submits form 6. Admin reviews and approves 7. Credentials are shared
Alternate Event Flow	• OTP fails → Retry up to 3 times • Missing document → Rejection or resubmission • Invalid business details
Frequency of Use	Once per stakeholder during account setup
Assumptions	Business has required documents, valid mobile/email
Additional Note	Business Category is selected via dropdown; dashboard includes notice and case tracking functionality

Use Case 2: Admin (Super / Regional Admin) Registration

Field Name	Description
Use Case ID	UC_REG_002
Use Case Name	Admin (Super/Regional) Registration
Created By (Date)	eMaap Functional Team (16-June-2025)
Last Update (Date)	16-June-2025
Primary Actor	Admin User (Applicant)
Secondary Actor	Approving Authority, OTP Service, Email Gateway

Trigger Activating Use Case	Admin applicant selects "Admin" as role during registration
Inter Linkages	OTP Verification, Document Upload, Nomination Letter Review, Role Assignment
Normal Event Flow	1. Applicant fills admin registration form with employee ID, authorization letter, etc. 2. Mobile/email OTP 3. Submit nomination 4. Admin approval 5. Credentials assigned
Alternate Event Flow	• Nomination not valid → rejection • OTP fails • Missing employee code or incorrect role selected
Frequency of Use	As needed for new admin onboarding
Assumptions	Role is approved internally before registration; proper authorization letter present
Additional Note (if any)	Access includes Master Data and User Management modules

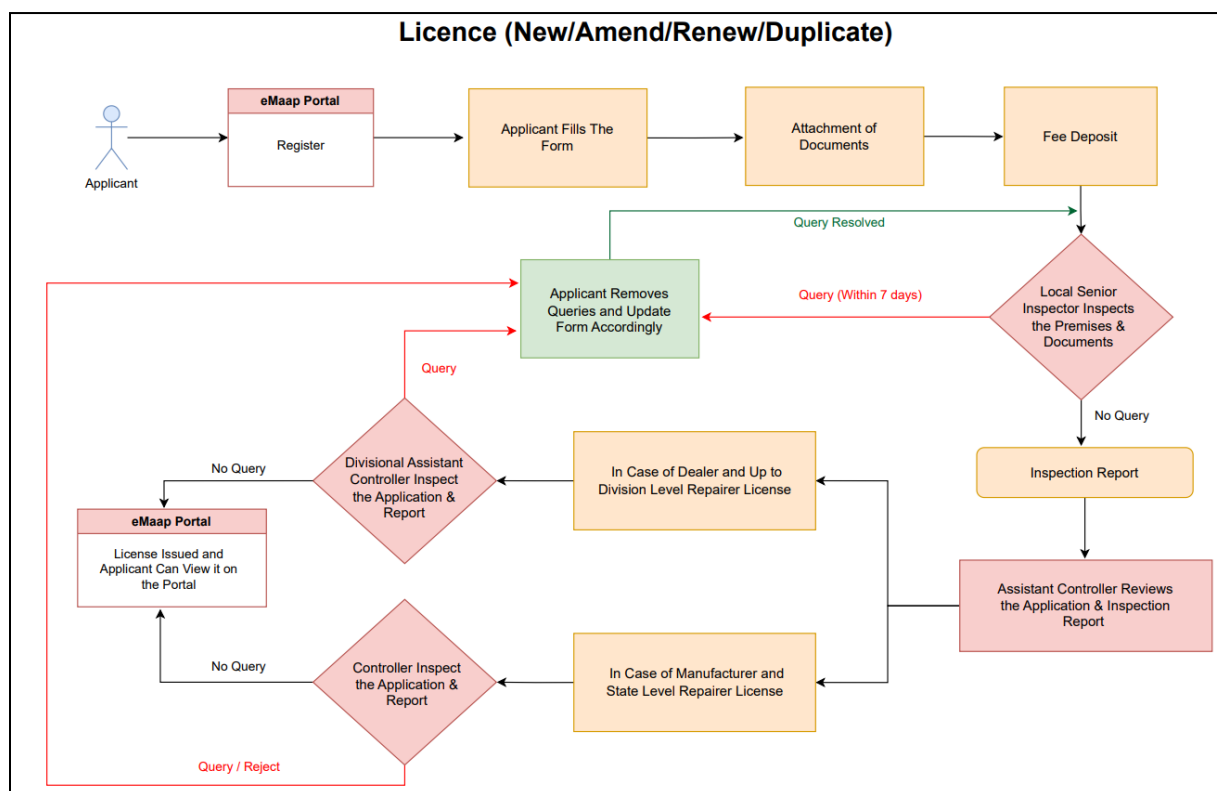
Use Case 3: Legal Officer Registration

Field Name	Description
Use Case ID	UC_REG_003
Use Case Name	Legal Officer Registration
Created By (Date)	eMaap Functional Team (16-June-2025)
Last Update (Date)	16-June-2025
Primary Actor	Legal Officer / Judicial Liaison
Secondary Actor	Admin, OTP Service, Approving Authority
Trigger Activating Use Case	User selects "Legal Officer" role during registration
Inter Linkages	OTP Verification, Document Upload, Admin Review Panel
Normal Event Flow	1. Officer fills registration form 2. Enters employee ID and uploads authorization letter 3. OTP verification 4. Admin approves request 5. Account activated
Alternate Event Flow	• Invalid authorization • OTP expired • Rejected due to incomplete form

Frequency of Use	Low (only during onboarding of new Legal Officers)
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3.2 Licenses for Manufacturer/ Dealer/ Repairer for center and states

This module will be developed to streamline the licensing process for Manufacturers, Dealers, Repairers, and other relevant stakeholders. It will encompass functionalities such as the issuance for new license, renewal of licenses, issue of duplicate license and amendments to existing licenses (e.g., changes to the firm's name, address, articles, or key personnel). The system will adhere to the provisions of the applicable Acts and Rules, ensuring compliance at every stage. Stakeholders will be able to select the desired service category, complete the application form prescribed by the Department on the eMaap portal, upload scanned copies of the required documents, and make online fee payment, as applicable. Upon thorough processing by the department, an electronic copy of the license will be issued.



The process begins with the applicant registration on the eMaap portal. Upon successful registration, the applicant proceeds to fill out the license application form relevant to their service requirements. After completing the form, they are required to attach necessary supporting documents. These documents typically include identification, proof of premises, previous licenses (If any), and other statutory certifications. Once documents are uploaded, the applicants must proceed to deposit the application fee through the portal to complete the initial submission.

Following this, the application enters the inspections phase, where a Local Senior Inspector is assigned to inspect the premises and verify the submitted documents. The inspector evaluates the authenticity and compliance of the applicant's establishment and the correctness of the submitted information. If any discrepancies or missing information are observed, a query is raised within 7 days, and the application is reverted to the applicant for correction. The applicant is then required to address the queries and update the application accordingly to proceed further.

If no query arises during the inspection, or once all queries are satisfactorily resolved, the inspection report is submitted by the Local Inspector. This report then moves forward to the Assistant Controller, who reviews the application and inspection report in detail. At this stage, the workflow diverges based on the type of license requested by the applicant.

In the case of dealer licenses or division-level repairer licenses, the application is routed to the Divisional Assistant Controller. The Divisional Assistant Controller inspects the application and the inspection report to ensure all standards and statutory requirements are met. If no issues are found, the process moves ahead; otherwise, a query is raised for further clarification or correction.

For applications involving manufacturer licenses or state-level repairer licenses, which are more critical in nature, the documents and report are escalated to the Controller. The Controller performs a final inspection of the application and the attached inspection report. If the application is found to be satisfactory and compliant with the legal norms, the process moves to the final stage. However, if there are any non-compliances, errors, or unresolved queries, the Controller has the authority to either raise a query for correction or reject the application outright.

If all inspections are completed successfully and no queries are raised by either the Divisional Assistant Controller or the Controller, the license is issued via the eMaap portal. The applicant is notified of the issuance and can view and download their approved license directly from the portal dashboard.

This end-to-end workflow ensures that every license application undergoes due diligence, proper inspection, and tiered review based on the type and complexity of the license. The process also provides multiple checkpoints for correcting or updating information, making it a user-centric and error-resilient system. The integration of query mechanisms at each stage adds a layer of transparency and allows applicants to actively participate in correcting or clarifying their submissions.

3.2.1 Login and Dashboard Access #FR001

Description	This feature enables applicants to access a personalized dashboard immediately after a successful login. From this dashboard, users can initiate actions related to the license management process, including applying for a New License, Amendment, Renewal, or Duplicate Copy. The interface provides a user-friendly experience tailored to each user's role and license category.
User	1. Stakeholders: Manufacturers, Dealers, Repairers, Packers, and Importers applying for licenses. 2. Inspectors / Field Officers: For follow-up and inspections. 3. Admins / Controllers: For workflow and dashboard configuration.
Pre - Condition	1. User is successfully registered and activated on the eMaap portal. 2. The user has valid login credentials. 3. Role-based access control is enabled.
Sequence of Operations	1. User logs into the eMaap portal using their credentials. 2. System validates credentials and loads role-based UI. 3. Applicant is redirected to a personalized dashboard. 4. Dashboard displays four key options: a. Apply for New License b. License Amendment c. Renew License d. Request Duplicate License 5. User selects a service type and is redirected to the corresponding application module.
Post - Condition	1. User sees a personalized dashboard with active options relevant to their profile. 2. License application data is stored and tracked per category. 3. Actions taken by the user are logged in the system.
Validity Checks	1. Login must be authenticated using verified credentials. 2. Role must be validated against user profile stored in the system. 3. Dashboard options must be displayed only if user profile is Active. 4. License options should correspond with the user's registered license type/category. 5. Prevent duplicate submissions by checking for ongoing applications in the systems.

Input	1. User credentials (username/password or token) 2. User role (stakeholder, inspector, admin) 3. Selection of license process type
Output	1. Personalized dashboard view 2. Dynamic links/buttons to license service forms 3. Access to license history or draft applications
Exception Handling	1. Invalid login credentials: "Incorrect username or password. Try again." 2. Unauthorized access: "You are not permitted to access this section." 3. Session timeout: "Your session has expired. Please log in again." 4. Dashboard not loading: Show fallback error: "Dashboard could not be loaded. Please try again later or contact support."

3.2.2 Form Filling and selection #FR002

Description	This functionality enables the eMaap portal to dynamically render appropriate application forms based on the license type and user role (e.g., dealer, manufacturer, or repairer). The system provides inline guidance via tooltips and applies field-level validations to ensure accuracy and completeness of the data entered. Additionally, users can save their application as a draft, allowing flexibility to resume and complete the form at a later time without losing progress.
User	1. Stakeholders (Dealers, Manufacturers, Repairers) applying for licenses. 2. Inspectors / Admins (view submitted forms during review but do not fill them).
Pre - Condition	1. User is logged in with valid credentials. 2. User's role is identified and mapped to appropriate license types. 3. The user selects a license action (New, Amendment, Renewal, or Duplicate).
Sequence of Operations	1. User logs into the eMaap portal and accesses the license module. 2. User selects the desired service: New License, Renewal, Amendment, or Duplicate Copy. 3. System identifies the user's role and license category. 4. A dynamic application form is loaded corresponding to the license type.

	5. User fills in the required fields; tooltips assist in understanding inputs. 6. Form fields are validated in real-time (e.g., mandatory checks, format restrictions). 7. User can: a. Save as draft to complete later. b. Proceed to submit if all validations passes.
Post - Condition	1. A dynamic form is rendered based on selected license type and role. 2. Data entered is validated and either saved as draft or submitted for processing. 3. A saved draft is accessible from the user's dashboard.
Validity Checks	1. License type and user role must be matched from backend metadata. 2. Mandatory fields must be filled before submission (not required for drafts). 3. Email, PAN, mobile number, and GSTIN fields must follow correct formats. 4. Drop-down fields must restrict user input to predefined options. 5. Maximum character limits and allowed formats enforced per field definition.
Input	1. License type (selected by user) 2. User role (from profile) 3. Form data (e.g., name, business info, contact details, license details) 4. Uploaded documents (optional at draft stage)
Output	1. Dynamic application form view 2. Real-time validation feedback 3. Tooltip text for field clarification 4. Saved draft (if selected) 5. Confirmation message (on save or submission)
Exception Handling	1. Form rendering error: "Unable to load the form. Please try again or contact support." 2. Missing mandatory fields on submission: "Please complete all required fields before submitting." 3. Invalid input format: "Enter a valid [e.g., PAN number, email address]." 4. Save failure due to session timeout: "Session expired. Please log in and try again." 5. Draft save error: "Draft could not be saved due to system error. Please retry."

3.2.3 Attachment and Document Upload #FR003

Description	The document upload feature allows applicants to upload supporting documents required for license applications in PDF or JPEG formats. The system dynamically displays the mandatory documents list based on the selected license category (e.g., manufacturer, dealer, repairer) and validates each file's type and size before allowing submission. All uploaded documents are linked to the specific application for secure storage, future reference, and audit purposes.
User	1. Stakeholders (Dealers, Manufacturers, Repairers) submitting license applications. 2. Admins / Inspectors / Legal Officers (for viewing and reviewing submitted documents as part of the approval process).
Pre - Condition	1. The user is logged in and has selected a license type to apply for. 2. The license application form is in progress or ready for submission. 3. Required document metadata (types, descriptions, mandatory/optional) is configured in the system.
Sequence of Operations	1. User accesses the license application form. 2. Based on license type, the system lists mandatory and optional document types. 3. User selects and uploads documents in supported formats. 4. System performs validation: a. File type (PDF/JPEG) b. File size (as per limit, e.g., ≤ 5MB) 5. If validations pass, the document is uploaded and linked to the application. 6. User can preview or delete the uploaded document before final submission. 7. On form submission, all valid documents are stored and locked in review.
Post - Condition	1. Uploaded documents are validated and stored securely in the backend. 2. Each document is linked to the corresponding application record. 3. Audit logs are generated for each upload action.
Validity Checks	1. Validate that file format is .pdf or .jpeg (or .jpg). 2. Enforce maximum file size (e.g., 5MB). 3. Check that all mandatory documents listed for the selected license type are uploaded. 4. Prevent submission of application if mandatory document uploads are incomplete.

	5. Uploaded files should be scanned for malware and stored with access control in secure cloud storage, linked using application ID and user role.
Input	1. License type and role (to determine document list) 2. File uploads (PDF or JPEG) 3. Metadata (e.g., document name/type, applicant ID)
Output	1. Document preview (thumbnail or filename) 2. Upload success/failure messages 3. Secure storage path or document ID 4. Linkage of documents with application for reference
Exception Handling	1. Invalid file type: "Only PDF and JPEG formats are allowed." 2. File size exceeded: "File size must not exceed 5MB." 3. Missing mandatory document: "Please upload all required documents to proceed." 4. Upload error: "Document upload failed. Please try again." 5. Duplicate file upload: "A document with the same name already exists. Rename and retry."

3.2.4 Fee calculation and Payment #FR004

Description	The Fee Calculation and Payment functionality automates the computation of application fees based on the applicant's license category (e.g., manufacturer, dealer, repairer) and geographical location. Once the fee is calculated, the system provides a secure interface for payment through multiple modes such as UPI, debit/credit cards, and net banking. Upon successful payment, a digital receipt is generated and made available for download. The system also stores the payment status and links it with the applicant's reference number for audit and tracking.
User	1. Stakeholders (Manufacturers, Dealers, Repairers, Packers, Importers) applying for new licenses or services. 2. Admins / Finance Officers (for payment reconciliation and transaction audit).
Pre - Condition	1. The applicant has completed the application form and uploaded the required documents. 2. License category and location details are correctly entered. 3. The payment gateway is integrated and functional.

Sequence of Operations	1. Applicant fills out the license application form. 2. Based on input (category + location), the system auto-calculates the applicable fee. 3. The calculated amount is displayed to the user with a "Proceed to Pay" option. 4. Applicant selects a preferred payment method: UPI, debit/credit card, or net banking. 5. System redirects to the payment gateway. 6. Upon successful transaction: a. A digital receipt is generated and shown/downloaded. b. Payment status is updated in the system. c. Receipt and status are linked to the application record. 7. Applicant is redirected to a confirmation screen.
Post - Condition	1. Fee is successfully paid, and a digital receipt is generated. 2. Payment status is updated to "Success" and linked to the application reference. 3. The application moves to the next stage (e.g., review by admin).
Validity Checks	1. Fee must be accurately calculated using a configuration matrix based on category and location. 2. Only valid payment methods should be accepted through integrated gateway services. 3. Payment status must be stored securely and updated only upon confirmation from the gateway. 4. Duplicate transaction checks should be enforced using application reference and timestamp. 5. Receipt must include transaction ID, applicant ID, fee amount, date/time, and payment status.
Input	1. License type and applicant location 2. Selected payment method 3. Payment gateway response (success/failure) 4. Application reference ID
Output	1. Auto-calculated fee display 2. Secure payment interface 3. Digital receipt (PDF) 4. Payment status (Success/Failed) linked to the application

Exception Handling	1. Payment failed: “Transaction unsuccessful. Please try again or use a different payment method.” 2. Gateway timeout or error: “Unable to connect to payment service. Please try after some time.” 3. Invalid payment details: “Entered card/UPI details are incorrect. Please verify.” 4. Duplicate payment attempt: Prevent multiple charges for the same application. 5. Receipt generation failure: “Payment successful, but receipt could not be generated. Please contact support.”
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3.2.5 Application submission and acknowledgement #FR005

Description	The Application Submission and Acknowledgment feature enables users to formally submit a license application after completing all required fields and uploading supporting documents. Upon submission, the system generates a unique Application Reference Number (ARN), sends an acknowledgment via email and SMS, and provides a downloadable receipt for the applicant’s records. This process ensures traceability, formal confirmation, and audit-readiness of every submission.
User	1. Stakeholders (e.g., Dealers, Manufacturers, Repairers, Packers, Importers) submitting license or service applications. 2. Admins / Inspectors / Legal Officers (view application status and ARN for further processing).
Pre - Condition	1. User is logged in and authenticated. 2. All mandatory fields in the application form are completed. 3. Required supporting documents have been uploaded and validated. 4. Payment has been successfully completed (if applicable).
Sequence of Operations	1. User completes all fields and uploads all required documents. 2. System validates form data, file attachments, and payment status. 3. On validation success, user clicks “Submit Application.” 4. System generates an Application Reference Number (ARN). 5. The application is saved with status = “Submitted.” 6. System sends acknowledgment to applicant via: a. Email: with ARN and receipt b. SMS: brief confirmation with ARN 7. A downloadable PDF receipt is made available on the dashboard.

Post - Condition	1. Application status is updated to “Submitted” in the system. 2. A unique ARN is generated and linked to the application. 3. Acknowledgment email/SMS is sent to the applicant. 4. Downloadable receipt with ARN and submission details is made available.
Validity Checks	1. Ensure all mandatory fields and documents are validated before submission is allowed. 2. Each ARN must be system-generated, unique, and non-editable. 3. Email and mobile number must be verified prior to submission. 4. Payment status (if required) must be marked as “Success” before submission. 5. Receipt must contain verified data: ARN, applicant details, timestamp, and application summary.
Input	1. Finalized application form data 2. Uploaded documents 3. User ID and Role 4. Payment confirmation (if required)
Output	1. Application Reference Number (ARN) 2. Application status update in database 3. Acknowledgment message (email/SMS) 4. Downloadable receipt (PDF with ARN, timestamp, license category)
Exception Handling	1. Incomplete application: “Please complete all required fields and uploads before submission.” 2. Validation error: “Some fields contain invalid data. Please correct and try again.” 3. System error during submission: “Application submission failed. Please try again later.” 4. Email/SMS gateway error: “Submission successful, but acknowledgment could not be sent. Please check your dashboard.” 5. Receipt generation failure: “Application submitted. Receipt will be available shortly on your dashboard.”

3.2.6 Inspection Assignment #FR006

Description	The Inspection Assignment feature automates the routing of submitted applications to the appropriate Local Senior Inspector. The assignment logic is based on the applicant’s geographical jurisdiction and license category (e.g., manufacturer, dealer, repairer). Once assigned, the inspector receives a portal/dashboard notification indicating a new inspection task. This automation streamlines workload distribution and ensures timely processing in accordance with regulatory guidelines.
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User	1. Applicants (Stakeholders such as Dealers, Manufacturers, Repairers) – Submit applications requiring inspection. 2. Local Senior Inspectors – Receive assigned applications for physical inspection or document verification. 3. Admins / Controllers – Manage inspection parameters, override or reassign tasks if required.
Pre - Condition	1. The applicant has successfully submitted a complete and valid license application. 2. The jurisdictional mapping of inspectors is already configured in the system. 3. Inspector accounts are active and linked with their operational zones and categories. 4. Notification services (in-app or email) are operational.
Sequence of Operations	1. Applicant submits a license application. 2. System checks the applicant's location and license category 3. System searches inspector master data to identify the corresponding Local Senior Inspector. 4. Application is auto assigned to the inspector. 5. Inspector receives a notification via the portal/dashboard. 6. Assignment details are stored in the database with application ID and inspector ID.
Post - Condition	1. The application is routed to the correct inspector's dashboard. 2. Inspector is notified of a new inspection assignment. 3. The system logs the assignment with a timestamp and linked application reference.
Validity Checks	1. Ensure inspector is mapped to applicant's location and license category. 2. Inspector's account must be active, and role must match inspection criteria. 3. One-to-one application-to-inspector assignment must be logged with a unique reference. 4. Notification status must be recorded (delivered, failed). 5. Inspectors should not receive assignments outside their jurisdiction unless manually overridden by admin.
Input	1. Application details (location, license type) 2. Inspector master data (jurisdiction, role, active status) 3. Notification configurations
Output	1. Assigned inspection task in inspector's dashboard 2. Notification to inspector (e.g., "New inspection assigned: Application [ARN]") 3. Audit record of assignment (applicant ID, inspector ID, date/time)

Exception Handling	1. No inspector mapped for jurisdiction: “No inspector available for the selected region. Please assign manually.” 2. Inspector profile inactive: “Assigned inspector is inactive. System will escalate to admin.” 3. Notification failure: “Assignment complete, but notification could not be sent. Please check inspector dashboard.” 4. Multiple inspectors found: Apply priority rules (e.g., round-robin or load balancing); log decision.
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3.2.7 Premises and Documents Inspection #FR007

Description	The Premises and Documents Inspection functionality enables the assigned Inspector to conduct a site visit to the applicant’s premises to verify adherence to Legal Metrology Act standards. Through the eMaap portal, the Inspector can upload photographs, notes, and a checklist of verified documents as evidence of inspection. The system also allows the inspector to mark the application’s inspection status as "Verified", "Pending Verification", or "non-compliant", which determines the next steps in the application workflow.
User	1. Local Senior Inspectors / Field Officers – Conduct inspections and update compliance status. 2. Stakeholders (applicants) – Receive updates based on inspection outcome. 3. Admins / Legal Metrology Controllers – Monitor inspection reports and escalate non-compliant cases.
Pre - Condition	1. The application has been submitted and auto assigned to an inspector. 2. The inspector has access to a valid login and an active assignment in their dashboard. 3. The inspection checklist and upload functionality are enabled on the portal.

Sequence of Operations	1. Inspector logs into the eMaap portal and views assigned inspection task. 2. Visits the applicant's premises. 3. Verifies the physical setup, instruments, and documents against legal metrology norms. 4. Uploads: a. Photos of premises/equipment b. Inspection notes/remarks c. Document verification checklist 5. Selects inspection status: a. Verified b. Pending Verification c. non-compliant 6. Saves and submits the inspection report through the portal.
Post - Condition	1. The inspection data (photos, notes, checklist) is stored in the system. 2. Inspection status is updated to one of the defined categories. 3. Application workflow progresses based on inspection status: a. Verified → Forwarded for final approval. b. Pending Verification → Returned for re-inspection. c. non-compliant → Notification sent to applicant with findings.
Validity Checks	1. Only the assigned inspector should be allowed to submit the inspection report. 2. Uploaded files must be in accepted formats (JPG, PNG) and within size limits (e.g., ≤ 5MB). 3. All mandatory checklist fields must be completed before submission. 4. One inspection status must be selected per submission. 5. Inspection data must be timestamped and securely stored against the Application Reference Number.
Input	1. Inspector's notes and comments 2. Uploaded images (JPG/PNG formats) 3. Verified checklist (tick-boxes or form fields) 4. Selected status (Verified, Pending, Non-compliant)
Output	1. Digital inspection report 2. Uploaded files linked to the application ID 3. Application status update 4. Notification to applicant (if required)

Exception Handling	1. Photo or file upload failure: “Unable to upload file. Check internet connection and retry.” 2. Incomplete checklist submission: “Please complete the checklist before submitting the report.” 3. Status not selected: “Inspection status must be selected to proceed.” 4. Permission denied: “You are not authorized to perform this action on this application.” 5. Save error/system timeout: “Inspection report could not be saved. Please try again later.”
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3.2.8 Query generation and submission #FR008

Description	The Query Generation and Submission functionality allows Inspectors to raise queries in the eMaap portal when discrepancies or missing documents are found during or after inspection. The system enables the inspector to log detailed remarks and automatically notifies the applicant of the query. A strict 7-day window from the date of inspection is enforced by the system to maintain process integrity and timely resolution. This ensures transparent communication and proper documentation of issues prior to further application processing.
User	1. Inspectors / Field Officers – Raise queries for incomplete or non-compliant submissions. 2. Applicants (Stakeholders) – Receive and respond to queries within the portal. 3. Admins / Controllers – Monitor query logs and timelines; intervene if deadlines are missed.
Pre - Condition	1. The inspection must have been conducted and recorded in the system. 2. The application must be in an “Under Inspection” or “Pending Clarification” state. 3. The inspector must be authorized and assigned to the specific application. 4. The query is being raised within 7 calendar days of inspection.
Sequence of Operations	1. Inspector logs into the portal and opens the inspected application. 2. Selects “Raise Query” option and enters remarks describing the discrepancy or missing document. 3. System checks the inspection date to verify it is within 7 days. 4. On confirmation, the query is submitted and stored with a timestamp.

	- The applicant receives notification (email/SMS and dashboard alert). - Application status is updated to “Query Raised.”
Post - Condition	- The query is recorded in the system and linked to the application. - Applicant is notified via email/SMS with the query details and response instructions. - Application status is updated to “Query Raised”. - A deadline for applicant response may be tracked
Validity Checks	System must validate: - Query raised within 7 calendar days from inspection date. - Remarks field is mandatory. - Inspector is assigned to the application. - Each query must be uniquely linked to an Application Reference Number (ARN). - Timestamp and inspector ID must be stored with the query for audit trail. - Duplicate or repeated queries for the same issue must be flagged or prevented unless re-inspection occurs.
Input	- Inspector ID and Role - Application ID - Inspection date - Query remarks (free text) - Optional document references or checklist tags
Output	- Logged query details - Notification to applicant (with remarks) - Updated application status - Timestamped query record for audit tracking
Exception Handling	Query raised after 7-day window: “Query period has expired. Contact admin for manual intervention.” Empty remarks: “Please provide remarks before submitting the query.” Notification failure: “Query submitted, but applicant could not be notified. System will retry.” Unauthorized access: “Only the assigned inspector can raise a query on this application.” System error during submission: “Unable to raise query at the moment. Please try again later.”

3.2.9 Applicant Query Resolution #FR009

Description	The Applicant Query Resolution functionality enables stakeholders (applicants) to view and respond to queries raised by Inspectors during the inspection or document verification process. Through their portal dashboard, applicants receive a clear and itemized list of all queries. The system allows applicants to take corrective actions by uploading missing documents, updating incorrect form fields, or providing explanatory remarks. This ensures transparent communication and supports a structured workflow for resolving compliance issues efficiently.
User	1. Applicants (Stakeholders including Manufacturers, Dealers, Repairers) – View and respond to queries. 2. Inspectors / Admins – Review applicant responses to determine compliance. 3. Controllers / Legal Metrology Officers – Monitor resolution status and enforce timelines.
Pre - Condition	1. At least one query must have been raised and logged by the Inspector against the application. 2. Applicant is authenticated and has access to the eMaap dashboard. 3. The application status is “Query Raised”.
Sequence of Operations	1. Applicant logs into the eMaap portal. 2. On the dashboard, system displays an itemized list of raised queries linked to the application. 3. For each query, applicant can: a. Upload the required/missing documents b. Edit incorrect data in editable fields c. Provide a written clarification or explanation 4. Applicant submits the response. 5. System logs the response and notifies the Inspector/Admin for follow-up review.
Post - Condition	1. Applicant submits response and uploads required documents or updates data. 2. Query resolution is logged with timestamps. 3. Application moves to “Response Submitted” or similar state for inspector review.

Validity Checks	1. Ensure applicant is responding to their own application queries only. 2. File uploads must pass file type and size validations (PDF/JPEG/PNG ≤ 5MB). 3. At least one valid action (upload/update/remark) must be performed before allowing submission. 4. Each response must be logged with: a. User ID b. Query ID c. Timestamp d. Status (e.g., Resolved, Pending Review)
Input	1. Query ID and application reference 2. Uploaded documents (PDF, JPEG, PNG) 3. Updated form fields 4. Clarification remarks (text field)
Output	1. Confirmation of successful submission 2. Updated query status (e.g., "Response Submitted") 3. Notification to inspector/admin 4. Response log entry for audit and tracking
Exception Handling	1. No action taken on query: "Please update the required fields or upload documents to proceed." 2. Invalid file format or size: "Only PDF/JPEG/PNG files under 5MB are allowed." 3. Field update not permitted: "This field cannot be modified after submission. Contact support." 4. Network or save error: "Unable to submit your response. Please try again later." 5. Multiple submission attempts: "Response already submitted. Await review."

3.2.10 Resubmission and Inspector Acknowledgement #FR010

Description	The Resubmission and Inspector Acknowledgment feature manages the workflow following a query raised by the Inspector. Once the applicant responds by updating data or uploading required documents, they may resubmit the application through the portal. The assigned Local Senior Inspector is then notified and allowed to review the updated application and query responses. The inspector must determine whether the issues are resolved. If yes, the query is marked as "Resolved", and the application progresses. If not, the inspector may raise a repeat query, but the system limits this to a maximum of two cycles per application to prevent indefinite reprocessing.
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User	1. Applicants (Stakeholders) – Resubmit corrected applications after addressing queries. 2. Local Senior Inspectors / Field Officers – Review resubmitted applications and confirm resolution. 3. Admins / Controllers – Oversee query cycles, enforce limits, and escalate unresolved cases.
Pre - Condition	1. An initial query must have been raised by the inspector and assigned to the applicant. 2. Applicant has submitted a query response (updated data, documents, or clarifications). 3. Inspector has an active account and access to the application.
Sequence of Operations	1. Applicant resubmits the updated form with query responses via the portal. 2. System logs the resubmission and notifies the assigned Inspector. 3. Inspector opens the application and reviews: a. Updated form data b. Uploaded documents c. Applicant's clarification remarks 4. Inspector selects: a. "Mark as Resolved" → query closes, application progresses b. "Raise Repeat Query" → system checks query cycle count c. If under limit, query is raised again d. If limit reached (2 cycles), system blocks further query and flags admin 5. Application status is updated accordingly.
Post - Condition	1. If query is resolved: Application proceeds to the next stage (e.g., final verification or approval). 2. If not resolved: Inspector raises a repeat query (max two cycles). 3. If maximum query cycles are exceeded: Application is flagged for administrative review or rejection.

Validity Checks	1. Ensure application resubmission is permitted only after initial query response is submitted. 2. Query cycle counter must not exceed 2 repeat queries. 3. Only the assigned Local Senior Inspector can review and act on the application. 4. Inspector's action must be logged with: a. Timestamp b. Remarks c. Application and Query ID 5. Application cannot proceed unless marked as "Resolved" by the inspector.
Input	1. Resubmitted application with changes 2. Inspector decision (Resolved / Raise Repeat Query) 3. System-tracked query cycle count
Output	1. Updated application status 2. Notification to applicant (resolved or new query raised) 3. Log entry of inspector action and remarks 4. Escalation alert (if cycle limit exceeded)
Exception Handling	1. Query cycle exceeds limit: "Maximum query attempts reached. Application will be escalated for review." 2. Missing inspector input: "Please review and select an action (Resolve or Raise Query)." 3. Data mismatch or conflict: "Submitted data does not match prior inspection findings. Please verify." 4. System error during resubmission: "Unable to submit. Please try again later." 5. Inactive inspector account: "Assignment inactive. Contact admin for reassignment."

3.2.11 Assistant controller review #FR011

Description	The Assistant Controller Review functionality is a critical step in the license processing workflow, initiated once the inspection is completed and no query is pending. The Assistant Controller receives the full application dossier, including uploaded documents, inspection remarks, and query status. Their task is to verify the applicant's eligibility under the Legal Metrology Act and classify the license based on the nature and scale of the applicant's operations into either: a. Dealer / Division-level Repairer, or b. Manufacturer / State-level Repairer. This classification determines the type and jurisdiction of the license to be issued.
User	1. Assistant Controller – Reviews completed applications, verifies compliance, and classifies license type. 2. Applicants (Stakeholders) – Receive notification of license classification. 3. Admins / Controllers – Monitor review progress and override decisions if needed.
Pre - Condition	1. The application has passed the inspection phase and has no pending queries. 2. Inspector has marked the inspection status as Verified. 3. All required documents and inspection remarks are available and linked to the application. 4. Assistant Controller has an active login and access to the assigned applications.
Sequence of Operations	1. System identifies applications with inspection marked as "Verified" and no open queries. 2. Application and inspection report are routed to the Assistant Controller's dashboard. 3. Assistant Controller opens the application and reviews: a. Uploaded documents b. Inspector's remarks and findings c. Applicant's submitted data (license type, business activity) 4. Assistant Controller evaluates eligibility under Legal Metrology Act. 5. Controller selects classification: a. Dealer / Division-level Repairer, or b. Manufacturer / State-level Repairer 6. Classification is saved, and application proceeds to the next workflow stage.

Post - Condition	1. Assistant Controller verifies the application and selects the license classification. 2. Application is updated with classification status. 3. The workflow proceeds to final approval or certificate generation. 4. Applicant is notified of the classification outcome.
Validity Checks	1. System must confirm that: a. Inspection status = Verified b. Query status = None/Open Queries = 0 2. Classification must be selected before proceeding. 3. Review must be timestamped and linked to reviewer ID. 4. Controller cannot modify documents, only view and verify 5. Only applications meeting eligibility under Legal Metrology Act are allowed to proceed.
Input	1. Application ID and reference 2. Uploaded documents 3. Inspector's report and verification status 4. Applicant data (license category, business type)
Output	1. Verified classification (Dealer/Repairer or Manufacturer) 2. Updated application status 3. Notification to applicant 4. Audit log of Assistant Controller's review decision
Exception Handling	1. Inspection incomplete or pending query: "Application cannot be reviewed. Please ensure all queries are resolved." 2. Missing or unreadable documents: "Document [filename] could not be opened. Request resubmission." 3. Invalid license category: "License type does not align with submitted business activity. Escalate to Admin." 4. Controller inaction timeout: Alert generated if no action is taken within predefined SLA period. 5. System error during classification save: "Unable to save review. Please retry or contact support."

3.2.12 Routing Based on license Type #FR012

Description	The Routing Based on License Type feature ensures that applications are automatically routed to the appropriate authority based on the classification determined during Assistant Controller review. The system distinguishes between: a. Dealer and Division-Level Repairer licenses – routed to the Divisional Assistant Controller, and b. Manufacturer and State-Level Repairer licenses – routed to the State Controller for final inspection and approval. This automated routing mechanism streamlines workflow, ensures jurisdictional compliance, and eliminates manual errors in application forwarding
User	1. Assistant Controllers – Initiate license classification after inspection. 2. Divisional Assistant Controllers – Review and approve Division-level licenses. 3. Controllers (State-Level) – Review and approve Manufacturer and State-level Repairer licenses. 4. Applicants – Monitor status changes in their application workflow.
Pre - Condition	1. Application must have been reviewed by the Assistant Controller. 2. A valid license type/classification must have been selected (Dealer, Division-Level Repairer, Manufacturer, State-Level Repairer). 3. No pending queries or inspection issues remain.
Sequence of Operations	1. Assistant Controller completes license classification. 2. System checks the selected license type. 3. If type = Dealer / Division-Level Repairer: a. Route application to Divisional Assistant Controller. 4. If type = Manufacturer / State-Level Repairer: a. Route application to Controller (State-Level). 5. System updates application status and triggers notification to the assigned authority.
Post - Condition	1. The application is automatically routed to the correct authority. 2. Status is updated to reflect the new review stage. 3. Assigned authority receives a notification and gains dashboard access to the application.

Validity Checks	1. License classification must be completed and match system-configured routing rules. 2. Only verified applications (no open queries, inspection complete) are eligible for routing. 3. System must verify reviewer availability and jurisdiction before routing. 4. Only one route per license type must be active to avoid duplication. 5. All routing actions must be logged with: a. Timestamp b. Origin and destination user roles c. Application reference ID
Input	1. License type/classification selected by Assistant Controller 2. Application reference number 3. Jurisdictional mapping data
Output	1. Routed application to appropriate authority 2. Updated application status (e.g., “Under Final Review – Division” or “Under Final Review – State”) 3. Notification to the relevant reviewer
Exception Handling	1. Missing or invalid license classification: “Unable to route application. License type not specified.” 2. Jurisdiction not mapped for license type: “No destination authority configured for selected license category.” 3. Routing logic failure: “System error during routing. Please contact system admin.” 4. Unassigned reviewer account: “Reviewer for this jurisdiction is inactive or unassigned. Escalate to admin.”

3.2.13 Divisional Assistant Controller Review #FR013

Description	The Divisional Assistant Controller Review functionality facilitates the final review and decision-making process for applications routed under Dealer and Division-Level Repairer license categories. Upon receiving the application and inspection report, the Divisional Assistant Controller verifies compliance with local jurisdictional norms. Based on the review, the application may be Approved, Rejected, or sent back with a Query Raised. If a query is raised, the system automatically routes it back to the applicant using the standard query handling module.
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User	1. Divisional Assistant Controller – Final reviewer for Division-level applications. 2. Applicants (Stakeholders) – Receive feedback if a query is raised. 3. Inspectors / Admins – View and track review status in the workflow.
Pre - Condition	1. Application has been classified as Dealer or Division-Level Repairer. 2. Application has been routed to the Divisional Assistant Controller. 3. All inspections are complete, and inspection status is Verified. 4. No open or pending queries exist.
Sequence of Operations	1. Divisional Assistant Controller logs into the eMaap portal. 2. Opens the assigned application from the dashboard. 3. Reviews: a. Submitted application form b. Uploaded documents c. Inspection report and notes 4. Verifies compliance with local legal metrology norms. 5. Selects one of the following actions: a. Approved → Application moves to license issuance. b. Rejected → Application is closed; reason recorded. c. Query Raised → Application routed back to applicant via the query module. 6. System updates status and logs the decision.
Post - Condition	1. Application status is updated based on the review action (Approved, Rejected, or Query Raised). 2. If query is raised, the application is routed back to the applicant. 3. If approved, application moves forward for license generation. 4. If rejected, applicant is notified with remarks.
Validity Checks	1. Application must be classified under a division-level category. 2. Only applications with Verified inspection status and no active queries are eligible for final review. 3. Remarks are mandatory for both Rejection and Query Raised actions. 4. Only the assigned Divisional Assistant Controller can act on the application. 5. All decisions must be timestamped, linked to reviewer ID, and stored in the audit log.

Input	1. Application reference ID 2. Inspection report and remarks 3. Jurisdictional compliance criteria 4. Review decision (Approved, Rejected, Query Raised) 5. Optional remarks (mandatory for query/rejection)
Output	1. Updated application status 2. Notification to applicant (approval, rejection, or query) 3. Query (if applicable) logged in the system 4. Audit trail of decision
Exception Handling	1. Missing inspection report: "Inspection report not found. Cannot proceed with review." 2. No action selected: "Please select an action (Approve, Reject, or Raise Query) to proceed." 3. Query routing failure: "Unable to route query to applicant. Please contact system admin." 4. Invalid jurisdiction assignment: "You are not authorized to review this application."

3.2.14 Controller Review #FR014

Description	The Controller Review functionality governs the final stage of evaluation for higher-category licenses, such as those for Manufacturers and State-Level Repairers. The Controller is responsible for validating the application integrity, optionally conducting a physical verification, and deciding whether to approve or reject the application. If discrepancies or missing information are identified, the Controller can raise queries with remarks, which are then routed back to the applicant for resolution through the standard query handling module.
User	1. Controller (State-Level Authority) – Final reviewer and decision-maker for higher-category license applications. 2. Applicants (Manufacturers, State-Level Repairers) – Receive query feedback or final decision. 3. Inspectors / Admins – View application status and support with physical verification logistics if needed.
Pre - Condition	1. Application has been classified as Manufacturer or State-Level Repairer. 2. Application has been routed to the Controller's dashboard. 3. Inspection is marked as Verified and no open queries exist. 4. All supporting documents are uploaded and available.

Sequence of Operations	1. Controller logs into the eMaap portal. 2. Opens the assigned application from their dashboard. 3. Reviews: a. Application form and submitted data b. Uploaded documents c. Inspection report and remarks 4. Optionally initiates a physical verification if necessary. 5. Chooses one of the following actions: a. Approve – Application is marked as Approved and forwarded for license generation. b. Reject – Application is closed with mandatory reason recorded. c. Raise Query – System prompts for remarks and routes the query back to the applicant. 6. Application status is updated accordingly in the system.
Post - Condition	1. Application status is updated (Approved, Rejected, or Query Raised). 2. If approved, the application moves to license issuance. 3. If rejected, the applicant is notified with reasons. 4. If a query is raised, the system routes it back to the applicant with Controller's remarks.
Validity Checks	1. Application must be classified as Manufacturer or State-Level Repairer. 2. Inspection must be verified, and there must be no unresolved queries. 3. Only the assigned Controller can take final action on the application. 4. Remarks are mandatory for Rejection and Query actions. 5. All actions must be logged with: a. Controller ID b. Timestamp c. Decision status d. Application reference number
Input	1. Application ID and license type 2. Uploaded documents 3. Inspection findings 4. Controller's decision (Approve, Reject, Raise Query) 5. Remarks (mandatory for query/rejection) 6. Optional verification request (flag or note)

Output	1. Updated application status 2. Notification to applicant (Approval, Rejection, or Query) 3. Query log entry (if raised) 4. Audit trail of Controller's action
Exception Handling	1. Missing inspection report or documents: "Required files are not available. Please verify before proceeding." 2. No action selected: "Please approve, reject, or raise a query to complete the review." 3. Query routing error: "System failed to send query to applicant. Please try again." 4. Invalid license classification: "You are not authorized to process this license type."

3.2.15 License Issuance #FR015

Description	The License Issuance functionality finalizes the licensing workflow by auto-generating a license document in PDF format once an application is approved by the respective authority (Divisional Assistant Controller or Controller). The license includes a unique License Number and Validity Date and is automatically stored in the applicant's "My Licenses" section on the portal. The system also sends an email/SMS notification to the applicant and records the entire issuance event in the audit log for traceability and compliance.
User	1. Applicants (Stakeholders) – Receive and access the issued license from their portal dashboard. 2. Divisional Assistant Controllers / Controllers – Approve the application triggering license issuance. 3. Admins – Monitor license generation and audit records for governance.
Pre - Condition	1. The application has been approved by the appropriate authority. 2. No pending queries or compliance issues remain. 3. All application data and inspection records are complete and verified. 4. License configuration templates and numbering logic are pre-defined in the system.

Sequence of Operations	1. Application is approved by Divisional Assistant Controller or Controller. 2. System triggers the license generation module. 3. A unique License Number is generated using pre-configured logic (e.g., category + region + year + serial). 4. License PDF is created, including: a. Applicant details b. License number c. Validity period d. Issue date e. Digital seal or signature (if applicable) 5. License is saved to the applicant's dashboard under "My Licenses." 6. System sends email/SMS to applicant confirming license issuance. 7. Audit log entry is created with timestamp, license number, and user ID.
Post - Condition	1. A digitally signed license PDF is generated. 2. The license is stored in the applicant's "My Licenses" portal section. 3. Notifications are sent to the applicant. 4. Issuance is recorded in the system audit trail.
Validity Checks	1. License number must be system-generated and unique across all license types. 2. Validity period must conform to license category configuration (e.g., 1 year, 3 years). 3. All required data fields (applicant name, category, address, approval authority) must be present before license generation. 4. Only applications with Approved status can trigger license issuance. 5. All issuance actions must be timestamped and recorded in the system audit trail with the user and application ID.
Input	1. Approved application data 2. License category and jurisdiction 3. Validity configuration (license duration) 4. Applicants contact information
Output	1. PDF license document with unique number and validity date 2. Entry in "My Licenses" dashboard 3. Email/SMS notification to applicant 4. Audit log entry (event type: "License Issued")

Exception Handling

1. **License number generation failure:** “Unable to generate license number. Contact admin.”
2. **PDF generation error:** “License document creation failed. Please retry.”
3. **Notification failure:** “License issued, but notification not delivered. Please check dashboard.”
4. **Storage issue:** “License generated but not saved to user profile. Retry or contact support.”
5. **Template misconfiguration:** “License template is missing or invalid.”

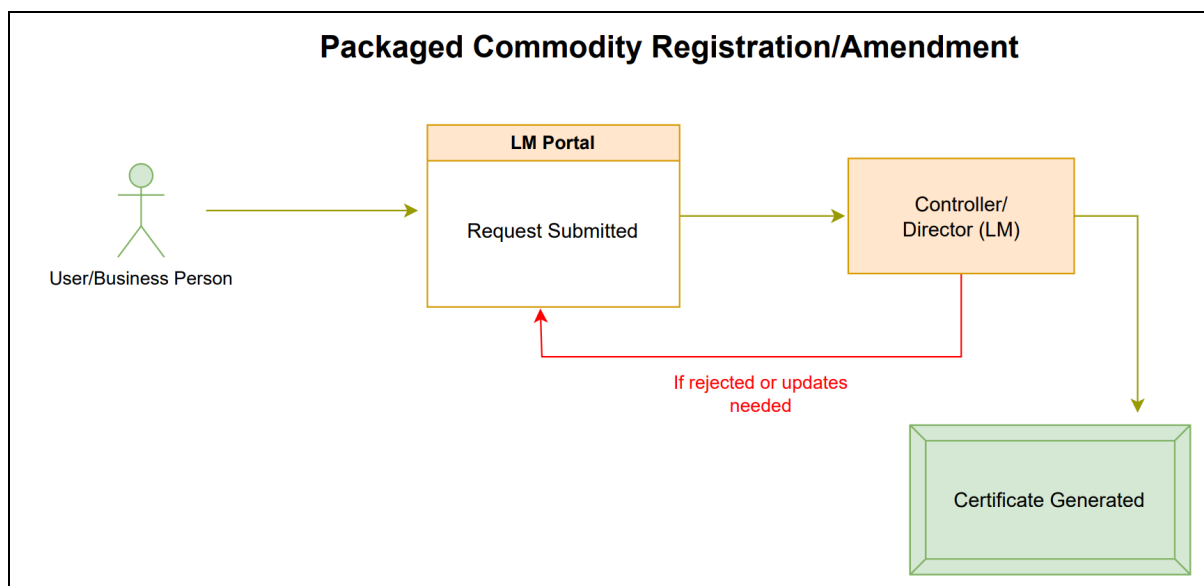
3.2.16 Use Cases (License Module)

Field Name	Description
Use Case ID	UC_eMaap_LIC_004
Use Case Name	License Application, Inspection & Issuance
Reference	Based on FRS Numbers: FR001 to FR015
Created By (Date)	eMaap Functional team (16-June-2025)
Last Updated (Date)	16-June-2025
Primary Actor	Applicant (Manufacturer / Dealer / Repairer)
Secondary Actor	Inspector, Assistant Controller, Divisional Assistant Controller, Controller
Trigger Activating Use Case	User login to eMaap portal and chooses to apply for a new license or related service
Inter Linkages	Dashboard > Form Filling > Document Upload > Payment > Inspection > Review > Issuance
Normal Event Flow	1. Applicant logs into portal 2. Selects license type (New / Amendment / Renewal / Duplicate) 3. System displays dynamic form based on role (Manufacturer/Dealer/Repairer) 4. Applicant fills and saves form (drafts supported) 5. Applicant uploads required documents 6. System validates document types/sizes 7. Fee auto-calculated and user pays (via UPI, Net banking, etc.) 8. Application Reference Number (ARN) generated 9. Inspector assigned based on jurisdiction 10. Inspector inspects premises, uploads report 11. Status marked as Verified / Pending / Non-compliant 12. If query arises, system notifies applicant 13. Applicant addresses and resubmits

	14. Inspector reviews response and resolves 15. Application forwarded to appropriate Controller (based on type) 16. Controller reviews and approves or raises query 17. Final license document auto generated and stored in dashboard
Alternate Event Flow	1. If incomplete or incorrect document: system raises error before submission 2. If query is raised post-inspection: applicant receives notification and has 7 calendar days to respond 3. If query not resolved in two cycles, application may be rejected or reinitiated 4. In case of manual override required by Controller, physical verification can be initiated
Frequency of Use	Daily, depending on new applications and renewals across 36 states + Centre
Assumptions	• Applicant has access to required documents • Internet access and OTP/email verification functional • Jurisdiction mapping of inspectors and controllers is maintained in the backend
Additional Note	• Auto-trail maintained throughout for audit • Email/SMS updates sent to applicant after every major step • License is stored in applicant's dashboard under "My Licenses" section

3.3 Package Commodities registration for Manufacturers / Packers / Importers for center and states

This module will be developed to enable registration for those businesses where packaged commodities are manufactured, packed or imported. With this module, the Legal Metrology Division will ensure compliance with established standards, guaranteeing that packaged commodities—such as soap, biscuits, bread, pulses, rice, spices, as well as items sold by volume like milk, kerosene, petrol, and other packaged goods—are supplied to the consumers with accurate measurements of weight, volume, or count as per regulatory guidelines.



The process for Packaged Commodity Registration or Amendment through the LM Portal begins with a user or business person submitting a request online. Once the request is submitted, it is received and processed by the Controller or Director of Legal Metrology, who reviews the application for completeness and compliance. If the application is found to be in order, the process proceeds to the generation of a registration certificate. However, if the application is rejected or requires updates, the user is notified through the portal and must take the necessary corrective action before resubmitting. Upon successful approval, the certificate is automatically generated, completing the registration or amendment process. This digital workflow ensures efficiency, transparency, and proper documentation at each stage.

3.3.1 Request submission by Business user #FR001

Description	This functionality allows Manufacturers, Packers, and Importers to submit registration or amendment requests for packaged commodities under the Legal Metrology (Packaged Commodities) Rules, 2011. The process is initiated through the eMaap portal, where authenticated business users fill in product, packaging, and regulatory compliance information, upload required documents and submit their applications. The system records this submission as “Request Submitted”, triggering further review by the appropriate Centre or State authority.
User	1. Business Users (Manufacturers / Packers / Importers) – Submit registration or amendment applications. 2. State and Central Legal Metrology Officers – Review and process submitted applications. 3. System Admins – Oversee workflow and resolve submission issues.

Pre - Condition	1. The user is authenticated and logged in with valid credentials. 2. The user has access to the “Packaged Commodities Registration/Amendment” module. 3. All required data fields and documents are prepared in advance (e.g., manufacturer license, previous approvals).
Sequence of Operations	1. Business user logs into the eMaap portal. 2. Navigates to “Packaged Commodity Registration/Amendment”. 3. Selects the type of request (New Registration or Amendment). 4. Fills out the digital form with: a. Product details (name, type, category) b. Packaging information (quantity, units, material) c. Legal declarations and compliance details 5. Uploads required supporting documents (PDF/JPEG). 6. Clicks “Submit” to complete the request. 7. System assigns a unique application ID and logs the submission with timestamp. 8. Status is marked as “Request Submitted.”
Post - Condition	1. Application status is updated to “Request Submitted.” 2. The submitted request is visible in the user’s dashboard and pending review by the respective authority. 3. Notifications or acknowledgments may be triggered (depending on configuration).
Validity Checks	1. All mandatory fields (product name, packaging type, quantity, manufacturer details) must be completed. 2. File uploads must meet file type (PDF/JPEG) and size constraints (e.g., ≤ 5MB). 3. Only authenticated users with appropriate business roles can access the module. 4. System must verify no duplicate submission is made for the same product in the same period (for new registrations). 5. Submission must be logged with: a. Application ID b. User ID c. Timestamp d. Request Type (New/Amendment)
Input	1. User login credentials 2. Product and packaging details 3. Legal declarations (checkboxes or signed fields) 4. Supporting documents (license, prior certificates, etc.) 5. Type of request (Registration / Amendment)

Output	1. Application ID and status = "Request Submitted" 2. Entry in the business user's application history 3. System log entry for audit purposes 4. Optional email/SMS notification (configurable)
Exception Handling	1. Missing mandatory fields: "Please complete all required fields before submission." 2. File upload error: "Document upload failed. Please check the format and try again." 3. Invalid data format: "Entered value is not in the correct format (e.g., invalid quantity or date)." 4. Submission timeout or server error: "Unable to process your request. Please try again later." 5. Authentication failure: "Session expired, or user not authorized."

3.3.2 System Notification and Logging #FR002

Description	This functionality ensures that after a Manufacturer, Packer, or Importer submits a packaged commodities registration request, the system automatically logs the request in the database and notifies the appropriate authority (Controller or Director of Legal Metrology) based on the applicant's state or central jurisdiction. This ensures prompt attention and transparent tracking of submissions within the regulatory workflow.
User	1. Business Users (Manufacturers / Packers / Importers) – Submit the registration request. 2. Controllers / Directors of Legal Metrology (State/Centre) – Receive notifications of new submissions for review and action. 3. System Admins – Monitor notification dispatch and log audit trails.
Pre - Condition	1. A valid registration request has been successfully submitted by the business user. 2. The user's jurisdiction and license category are mapped correctly in the system. 3. Notification services (email/SMS/in-app alerts) are enabled and configured. 4. The designated Controller/Director has an active user account and assigned state/center access.

Sequence of Operations	1. Business user completes and submits the packaged commodity registration form. 2. System validates submission and stores all application data in the backend database. 3. System identifies the appropriate regulatory authority based on the jurisdiction (state or center). 4. A notification is triggered and sent to the designated Controller/Director of Legal Metrology via: a. Email and/or SMS b. Portal dashboard alert 5. Submission status is marked as “Pending Review” in the authority’s interface.
Post - Condition	1. Submission is logged in the system’s central database with an application ID. 2. A notification (email/SMS or in-app alert) is sent to the designated Controller/Director. 3. The new request appears in the reviewing authority's dashboard for further processing.
Validity Checks	1. Authority mapping must be based on: 2. User’s selected state or central jurisdiction 3. License type (Manufacturer, Packer, Importer) 4. Notifications must only be sent to active, assigned users with the appropriate role and access rights. 5. The application must be fully submitted before notification is triggered (status = “Request Submitted”). 6. All notification events must be timestamped and logged in the audit trail with: a. User ID b. Application ID c. Notification delivery status
Input	1. Submitted application data (product type, jurisdiction, documents) 2. User profile and jurisdiction mapping 3. Role-based notification settings
Output	1. Application entry in the central database 2. Unique Application ID generation 3. Notification message sent to assigned authority 4. Entry logged in system audit trail

Exception Handling	1. Notification failure (email/SMS): “Notification could not be delivered. System will retry and alert admin.” 2. No authority mapped to jurisdiction: “Unable to identify reviewing officer for the selected state. Contact admin.” 3. Controller/Director account inactive: “Designated authority is not active in the system. Escalate to admin.” 4. Database submission failure: “Application not saved. Please retry submission.”
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3.3.3 Application Review #FR003

Description	The Application Review functionality enables the Controller or Director of Legal Metrology (State or Centre) to review packaged commodities registration requests submitted by Manufacturers, Packers, or Importers. The officer assesses the application against defined regulatory parameters under the Legal Metrology (Packaged Commodities) Rules, 2011, including packaging declarations and compliance standards. Based on the review, the officer may Approve, Reject, or Send for Updates with remarks for minor corrections.
User	1. Controller/Director of Legal Metrology (State or Centre) – Reviews and takes action on applications. 2. Business Users (Manufacturers, Packers, Importers) – Submit applications and respond to review outcomes. 3. System Admins – Monitor review activity and workflow integrity
Pre - Condition	1. A registration or amendment request has been submitted and logged. 2. Notification has been sent to the appropriate regulatory officer. 3. Application is accessible on the reviewer’s dashboard with complete data and documents.
Sequence of Operations	1. Controller/Director logs into the eMaap portal. 2. Accesses the list of applications pending review from their dashboard. 3. Opens an individual application and reviews: a. Product and packaging details b. Declarations and compliance fields c. Supporting documents 4. Based on review, selects one of the following: a. Approve → If application meets all compliance requirements. b. Reject → If there are major errors or non-compliance issues. c. Send for Updates → If minor changes are needed; adds remarks.

	5. System updates the application status and notifies the applicant accordingly.
Post - Condition	1. Application status is updated based on action: Approved, Rejected, or Sent for Updates. 2. Notification is triggered to the applicant regarding the review decision. 3. Remarks (where applicable) are saved for reference and audit.
Validity Checks	1. Only applications with status = "Request Submitted" are eligible for review. 2. Reviewer must be mapped to the correct jurisdiction (State or Centre) based on the applicant's location. 3. Remarks are mandatory for Rejection and Send for Updates actions. 4. Application must contain: a. All required fields filled b. All mandatory documents uploaded 5. All review actions must be logged with: a. Reviewer ID b. Timestamp c. Decision status and remarks (if any)
Input	1. Application ID and reference data 2. Regulatory compliance parameters 3. Officer's action (Approve / Reject / Send for Updates) 4. Remarks (mandatory for Reject and Send for Updates)
Output	1. Updated application status 2. Notification to applicant (with remarks if applicable) 3. Decision log in system audit trail
Exception Handling	1. Incomplete application data: "Unable to review application. Required fields or documents missing." 2. No action selected: "Please select an action to complete the review." 3. Invalid reviewer access: "You do not have permission to review this application." 4. System error during status update: "Error saving review decision. Please try again." 5. Applicant notification failure: "Review completed, but applicant could not be notified. Please inform manually."

3.3.4 Rejection and Update loop #FR004

Description	The Rejection or Update Loop module manages the feedback and correction process following a review decision by the Controller/Director of Legal Metrology. If an application is Rejected or sent back for Updates, the system changes the application status accordingly and notifies the applicant via the portal and email. The applicant can then view the officer's comments, make necessary corrections, and resubmit the application. This loop ensures that applicants have a structured opportunity to address compliance issues without starting a new application.
User	1. Business Users (Manufacturers / Packers / Importers) – Receive feedback, correct data, and resubmit applications. 2. Controller/Director (State/Centre) – Reviews resubmissions and finalizes decisions. 3. System Admins – Oversee the process and ensure workflow integrity.
Pre - Condition	1. The application has undergone a review and was either rejected or marked as Update Required. 2. The reviewing authority has provided comments. 3. The user's account is active and authorized to access the application dashboard.
Sequence of Operations	1. Reviewing officer selects Reject or Send for Updates during application review and adds remarks. 2. System: a. Updates the application status to "Rejected" or "Update Required" b. Sends a notification (portal + email) to the applicant 3. Applicant logs in and views: a. Application status b. Officer's remarks/comments 4. If status = "Update Required", user is allowed to: a. Edit form data b. Replace/upload additional documents 5. Applicant resubmits the updated application. 6. System logs the new submission and updates status to "Resubmitted".
Post - Condition	1. The applicant receives a notification with actionable feedback. 2. The application status is changed and editable if updates are required.

	3. Resubmitted applications are re-queued for regulatory review.
Validity Checks	1. Remarks must be provided when rejecting or sending an update request. 2. Only applications with status = Update Required can be resubmitted. 3. Resubmission must include at least one updated field or document. 4. Rejected applications should be view-only and not editable. 5. All status transitions and actions must be logged with: a. Application ID b. User ID c. Reviewer ID d. Timestamp e. Remarks
Input	1. Reviewer's decision (Reject / Send for Updates) 2. Remarks added by reviewer 3. Updated data and documents from applicant 4. Resubmission action
Output	1. Application status change (Rejected / Update Required / Resubmitted) 2. Notification to applicant 3. Log entry in audit trail with timestamp and remarks 4. Updated application view for the reviewer
Exception Handling	1. No remarks provided by reviewer: "Remarks are mandatory for rejection or update requests." 2. User tries to edit a rejected application: "This application is rejected and cannot be edited." 3. Resubmission without corrections: "No changes detected. Please update the required fields/documents before resubmitting." 4. Document upload failure during update: "File could not be uploaded. Check format and size." 5. Notification failure: "Application updated, but user could not be notified. Alert system admin."

3.3.5 Certificate Generation #FR005

Description	The Certificate Generation functionality finalizes the packaged commodities registration process by automatically generating a digital certificate once the application is approved by the Controller/Director of Legal Metrology. The certificate includes essential details such as the Application ID, Applicant's Name, Description of the Packaged Commodity, Approval Date, and a QR code for digital authenticity verification. The certificate is made available in the applicant's dashboard for secure download.
User	1. Business Users (Manufacturers / Packers / Importers) – Download the registration certificate from their dashboard. 2. Controllers/Directors (Centre and State) – Trigger approval, which initiates certificate generation. 3. System Admins – Ensure system availability and resolve generation issues if any.
Pre - Condition	1. Application has been approved by the relevant regulatory authority. 2. All required information (product details, applicant info, approval date) is available in the system. 3. QR code generation and certificate template are configured and functional.
Sequence of Operations	1. Application is approved by Controller/Director. 2. System automatically triggers certificate generation process. 3. Certificate content is populated with: a. Application ID b. Applicant name c. Packaged commodity details d. Approval date 4. QR code is generated and embedded in the certificate for verification. 5. Certificate is saved as a PDF and linked to the applicant's profile. 6. System sends notification (portal/email) to the applicant. 7. Certificate is made available under "My Certificates" or "My Licenses" section.
Post - Condition	1. Certificate is generated in PDF format and stored in the applicant's dashboard. 2. QR code is embedded for authenticity verification. 3. The issuance is logged in the system audit trail. 4. Certificate becomes available for download at any time.

Validity Checks	1. Certificate must only be generated after official approval. 2. Each certificate must include: a. Unique Application ID b. Verified applicant details c. Exact packaged commodity information d. Date of approval e. Functional and scannable QR code 3. File format must be non-editable PDF. 4. Certificate generation event must be logged with: a. Certificate ID b. Linked Application ID c. Timestamp d. Issuing authority ID
Input	1. Approved application data (ID, applicant name, product info) 2. Date of approval 3. System-generated QR code
Output	1. Digitally generated certificate (PDF format) 2. QR code embedded for verification 3. Certificate accessible in applicant dashboard 4. Notification sent to applicant 5. Audit log of certificate issuance
Exception Handling	1. Template error: "Certificate template not found. Contact system admin." 2. Missing data field: "Certificate generation failed due to incomplete application data." 3. QR code generation failure: "Certificate generation incomplete. QR code could not be embedded." 4. File not saved: "Certificate generated but not stored. Please regenerate or contact support." 5. Notification failure: "Certificate issued but user notification not delivered. Retry or alert admin."

3.3.6 Use Cases (Package Commodities registration)

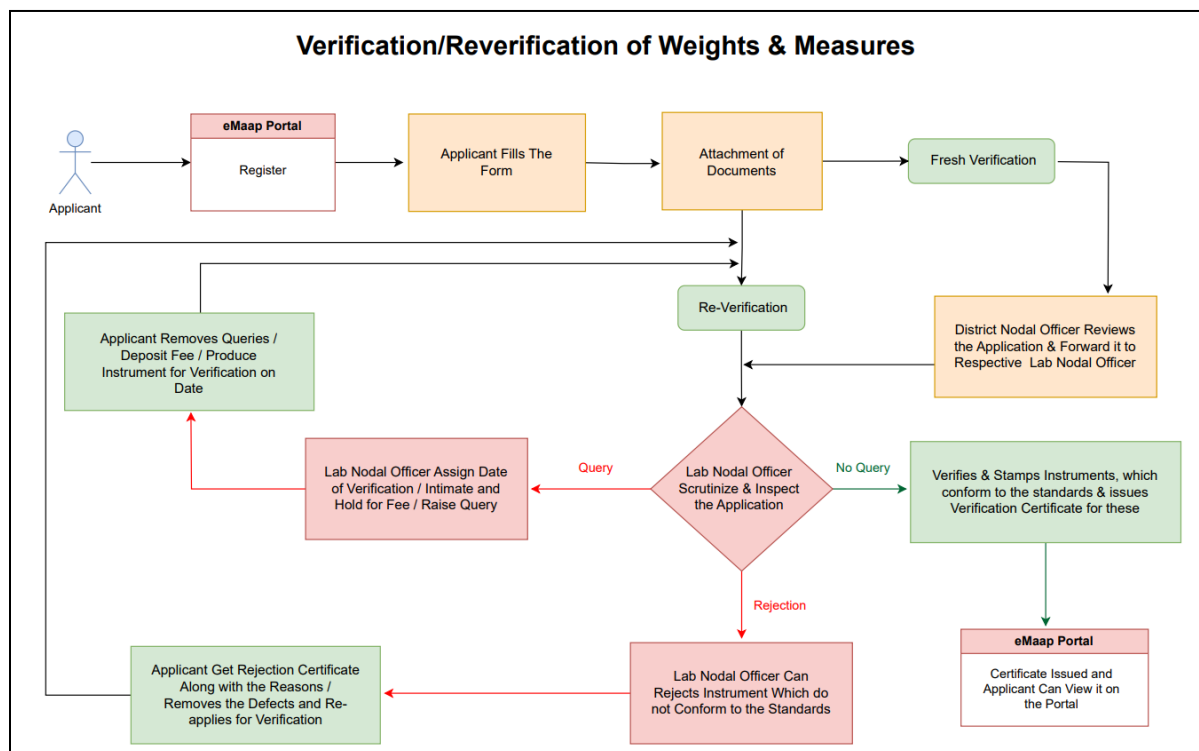
Field Name	Description
Use Case ID	eMaap_PC_005
Use Case Name	Registration / Amendment of Packaged Commodities
Reference	FRS Numbers: FRS_001 to FRS_005
Created By (Date)	eMaap Functional team (16-June-2025)
Last Update (Date)	16 June 2025

Primary Actor	Business User (Manufacturer, Packer, Importer)
Secondary Actor	Controller / Director, Legal Metrology (Centre or State)
Trigger Activating Use Case	Business user initiates application for registration or amendment of packaged commodities.
Inter-linkages	1. User Authentication System 2. Dashboard and Notification Engine 3. Certificate Generation Module 4. Document Management System 5. Audit Trail and Logging
Normal Event Flow	1. User logs in via eMaap portal 2. Navigates to “Packaged Commodity Registration/Amendment” module 3. Fills application form with details 4. Uploads necessary documents 5. Submits application (eMaap_RS_005) 6. System logs and notifies Controller/Director 7. Controller reviews application 8. If approved, certificate is auto generated 9. Certificate is available for download in user dashboard
Alternate Event Flow	1. Application is Rejected or requires Update 2. Status is updated to “Rejected” or “Update Required” 3. User receives comments, resubmits updated application 4. Application is re-reviewed for final decision
Frequency of Use	As required per business need – typically during: 1. New product registration 2. Packaging changes 3. Regulatory updates 4. Amendments or renewals
Assumptions	1. User is a valid registered business entity with login credentials 2. Controller/Director has appropriate permissions for review 3. Portal has uptime and document upload functionality 4. Regulatory rules are configured in the system
Additional Note	1. The QR code on the certificate provides a mechanism for field officers and consumers to verify the authenticity of the registration. 2. System maintains full audit logs for all approvals, rejections, and modifications.

3.4 Verification and Re-verification of weights and measures for center and states

The Department plays a crucial role in protecting consumers from unethical trading practices. To uphold the provisions of the relevant Acts and Rules, the officers of Legal Metrology ensure the accuracy of weights and measures through regular verification, re-verification, and stamping of devices used by traders, such as weighbridges, dispensing pumps at petrol outlets, and equipment at LPG depots. This module will facilitate verification and re-verification requests submitted by repairers, manufacturers, or dealers, enabling real-time processing and database updates with verification details.

This module will also support the generation of online verification certificates with unique codes for each certificate. In cases where violation is detected, the Legal Metrology Officer (LMO) will levy a compounded fee specifying the reasons. The system will also empower the LMO to perform user verification throughout the application lifecycle, fostering greater transparency and efficiency.



The process of **Verification/Reverification of Weights & Measures** through the eMaap Portal begins with the applicant registering on the portal and filling out the required application form. The applicant then attaches all necessary documents before choosing between either a fresh verification or a reverification, depending on the nature of the instrument. Once submitted, the application is reviewed by the District Nodal Officer, who then forwards it to the appropriate Lab Nodal Officer for further action. The Lab Nodal Officer scrutinizes the application and inspects the instrument. If no queries arise, the officer proceeds to verify and stamp the instrument, ensuring it conforms to prescribed standards, and then issues a verification certificate. If there

are any queries, the Lab Nodal Officer raises them, and the applicant must respond by removing the queries, depositing the required fee, or producing the instrument for verification on the assigned date. In cases where the instrument does not conform to standards, the officer rejects the application, and a rejection certificate is issued with stated reasons. The applicant may then rectify the defects and reapply. Finally, once the instrument is successfully verified, the certificate is issued and made available to the applicant through the eMaap Portal for download.

3.4.1 Applicant Login #FR001

Description	This functionality allows applicants (business users) to securely log into the eMaap portal and access the Verification and Re-verification module for weights and measures. Upon login, the applicant is presented with a personalized dashboard that displays active applications, saved drafts, and an option to initiate a new verification or re-verification request. This access point serves as the gateway for users to manage regulatory compliance under the Legal Metrology Act for both Centre and State jurisdictions.
User	1. Applicants (Manufacturers, Dealers, Repairers, Importers, Packers) – Log in to initiate, view, and manage verification/re-verification requests. 2. System Admins – Maintain account access and monitor login activity.
Pre - Condition	1. The applicant is a registered and approved user on the eMaap portal. 2. The user holds valid login credentials (username/password or OTP-based access). 3. The system's login service and dashboard components are operational.
Sequence of Operations	1. Applicant accesses the eMaap portal and enters credentials. 2. System authenticates the login. 3. Upon successful login: a. The applicant is redirected to the main dashboard. 4. The dashboard displays: a. Active Applications (with current status) b. Draft Applications c. New Request option (button/link) 5. Applicant can select any of these to view, edit, or initiate requests.

Post - Condition	1. User is redirected to the Verification/Re-verification dashboard 2. The user can view: a. Active requests (submitted and under process) b. Draft applications (saved but not submitted) c. New request button to begin a verification/re-verification application
Validity Checks	1. Only approved and registered applicants can access this module. 2. Login must be validated via secure authentication (password policy or OTP). 3. Session must be encrypted and logged. 4. Dashboard must dynamically load relevant data based on the logged-in user's ID and jurisdiction. 5. All login attempts must be recorded with: a. User ID b. Timestamp c. IP address d. Login status (success/failure)
Input	1. Username and password (or OTP) 2. Applicant role and jurisdiction profile
Output	1. Login success or failure message 2. Redirect to dashboard (if login successful) 3. Visible sections: Active Requests, Drafts, New Application
Exception Handling	1. Incorrect credentials: "Invalid username or password. Please try again." 2. Inactive account: "Your account is not active. Please contact support." 3. Session timeout: "You have been logged out due to inactivity." 4. Dashboard not loading: "Unable to load dashboard. Please refresh or try again later."

3.4.2 Application form submission #FR002

Description	This module enables registered applicants to submit requests for fresh verification or re-verification of weights and measuring instruments as per the Legal Metrology Act. Through the eMaap portal, users can select the verification type, provide instrument and usage details, and choose the jurisdictional laboratory. The system validates all mandatory fields before allowing the applicant to either save the form as a draft or submit it for further processing by the department.
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User	1. Applicants (Manufacturers, Dealers, Repairers, Importers, Packers) – Fill and submit verification/re-verification forms. 2. Legal Metrology Officers / Inspectors – Review and process submitted applications. 3. System Admins – Monitor submission workflows and address technical issues.
Pre - Condition	1. The applicant is logged in with valid credentials. 2. The applicant has access to the Verification/Re-verification module. 3. Jurisdictional mapping of labs and instrument types is pre-configured in the system.
Sequence of Operations	1. Applicant selects “Verification/Re-verification” from the dashboard. 2. System loads the application form. 3. Applicant selects: a. Type of request: Fresh / Re-verification b. Instrument type (dropdown or category list). 4. Applicant fills in: a. Instrument details: Make, Model, Serial Number, Capacity b. Usage information: Location, Business Type, Purpose 5. System validates mandatory fields. 6. Jurisdictional lab is auto populated based on location (or manually selected). 7. Applicant chooses to: a. Save as Draft, or b. Submit the application 8. If submitted, system: a. Assigns a unique application reference number b. Logs submission date and status 9. Triggers acknowledgment notification
Post - Condition	1. Application is saved either as a draft or submitted for review. 2. A unique reference number is generated upon submission. 3. The application status is set as “Request Submitted” if finalized.

Validity Checks	Mandatory fields include: - Instrument type, make, model, serial number, capacity - Usage location and business purpose - Jurisdictional lab - Only registered users with active roles can submit the form. - System must enforce field validation (e.g., non-empty, numeric ranges for capacity). - Duplicate serial numbers within the same business and category must be flagged. Application submission must generate a: - Unique reference ID - Timestamp - Draft or submitted status - Record in audit logs
Input	- Verification type (Fresh / Re-verification) - Instrument details (Make, Model, Serial No., Capacity) - Usage details (Location, Business Type, Purpose) - Lab selection (Auto or Manual)
Output	- Draft saves confirmation or submission success message - Unique Application Reference Number - Updated application status - Dashboard entry under “Active Applications”
Exception Handling	Missing mandatory fields: “Please complete all required fields before proceeding.” Invalid input format: “Check entered values for serial number or capacity.” No lab mapped: “Jurisdictional lab not found. Please select manually or contact support.” Save failure: “Draft could not be saved. Try again later.” Submission failure: “Application could not be submitted. Contact system admin.”

3.4.3 Document Upload #FR003

Description	The Document Upload feature facilitates the submission of required supporting documents—such as Invoice, Calibration Certificate, or other compliance materials—as part of the verification or re-verification application process. The system prompts applicants to upload specific documents in PDF or JPG formats, validates the file type and size, links the documents to the corresponding Application ID, and allows users to preview files before final submission.
User	1. Applicants (Manufacturers, Dealers, Repairers, Importers, Packers) – Upload supporting documents for verification or re-verification requests. 2. Inspectors / Legal Metrology Officers – Review uploaded documents during the inspection and approval stages. 3. System Admins – Monitor document uploads and manage format or size validations.
Pre - Condition	1. Applicant is logged into the eMaap portal and has filled out the application form. 2. Application is at the document upload stage (pre-submission). 3. Mandatory document types are configured in the system based on instrument category.
Sequence of Operations	1. System displays a list of mandatory documents required (e.g., Invoice, Calibration Certificate). 2. Applicant selects and uploads each file (PDF/JPG format). 3. System performs validation: a. File format check (PDF/JPG only) b. File size check (e.g., ≤ 5MB) 4. If valid, the file is uploaded and linked to the Application ID. 5. System shows a preview (or filename link) of uploaded files. 6. Applicant can: a. Replace or delete files b. Proceed to submit the application
Post - Condition	1. Uploaded documents are linked to the application and stored securely. 2. The applicant can view, update, or delete documents before final submission. 3. System allows the applicant to proceed to the next step (submission or fee payment).

Validity Checks	System must enforce: - Allowed formats: Only PDF, JPG/JPEG - File size limits: Max 5MB per document - Mandatory document types: Configurable by application type and jurisdiction Each uploaded document must be: - Linked to the correct Application ID - Logged with metadata: document type, user ID, timestamp - Preview or file verification feature must be enabled before submission. - Application cannot proceed without fulfilling all mandatory uploads.
Input	- Document files (PDF/JPG) - Application reference number - Document category/type
Output	- Upload confirmation - Document preview (filename, thumbnail, or download link) - Documents linked to the application record - Option to proceed to final submission
Exception Handling	- Unsupported file format: "Invalid file format. Only PDF and JPG files are allowed." - File size exceeded: "File exceeds the 5MB size limit. Please upload a smaller file." - Missing mandatory document: "Please upload all required documents to proceed." - Upload error: "Document upload failed. Please check your connection and retry." - Duplicate file name or mismatch: "A document with this name already exists. Please rename or replace."

3.4.4 Submission & Acknowledgement #FR004

Description	The Submission and Acknowledgment feature finalizes the verification or re-verification application process. Once the applicant completes the form and uploads all required documents, the system performs final validation before accepting the application. Upon successful submission, the system generates a unique Application ID, stores the request in the database, and sends an acknowledgment notification via email/SMS to the applicant, confirming receipt of the request.
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User	1. Applicants (Manufacturers, Dealers, Repairers, Packers, Importers) – Submit verification/re-verification applications. 2. Legal Metrology Officials / Inspectors – Access submitted applications for inspection and review. 3. System Admins – Monitor submission events and ensure acknowledgment dispatch.
Pre - Condition	1. Application form is fully filled with valid data. 2. All mandatory documents are uploaded and validated. 3. Applicant is authenticated and has access to the eMaap portal.
Sequence of Operations	1. Applicant reviews the filled form and uploaded documents. 2. Clicks “Submit”. 3. System performs: a. Field-level validation (completeness and correctness) b. Document check (presence and compliance with format/size) 4. If validation passes: a. Application is saved in the backend with a unique Application ID b. Status is set to “Submitted” c. Acknowledgment message is generated 5. Email/SMS notification is sent to the applicant. 6. Application appears in the applicant’s dashboard for tracking.
Post - Condition	1. Application is stored with a unique Application ID. 2. Status is updated to “Submitted”. 3. Applicant receives an acknowledgment via registered email/SMS. 4. Application appears in the applicant’s dashboard under “Active Requests.”

Validity Checks	Form must be 100% complete, with: - Instrument details - Usage information - Lab selection All mandatory documents must be: - Present - In allowed formats (PDF/JPG) - Under size limit Application submission must: - Trigger unique Application ID generation - Be logged with timestamp and applicant ID - Update application status to "Submitted" Notification must include: - Application ID - Submission timestamp - Link to track status (if applicable)
Input	- Completed form data - Uploaded supporting documents - Applicant credentials and contact details
Output	- Unique Application ID - Submission status confirmation - Acknowledgment message (on screen and via email/SMS) - Application log entry in database and audit trail
Exception Handling	Validation failure: "Please complete all required fields and upload all mandatory documents." Document missing or corrupted: "One or more required documents are missing or unreadable." Database error: "Submission failed due to server error. Please try again later." Notification failure: "Application submitted, but acknowledgment could not be delivered. Please check your dashboard." Duplicate submission attempt: "This application has already been submitted."

3.4.5 District Nodal officer Review #FR005

Description	The District Nodal Officer Review feature is a key step in the verification/re-verification workflow. Upon application submission, the District Nodal Officer accesses the backend portal, reviews submitted applications and attached documents and assigns the application to the respective Lab Nodal Officer for technical processing. The system automatically updates the application status and notifies the applicant of the assignment.
User	1. District Nodal Officer – Reviews applications and routes them to Lab Nodal Officers based on jurisdiction. 2. Lab Nodal Officer – Receives assigned applications for inspection or verification execution. 3. Applicants – Are notified of status changes and progress. 4. System Admins – Monitor workflow routing and escalation logic.
Pre - Condition	1. Applicant has submitted a complete application with documents. 2. Application is available in the District Nodal Officer's pending queue. 3. District Nodal Officer has valid login credentials and active role permissions.
Sequence of Operations	1. District Nodal Officer logs into the backend portal. 2. The dashboard displays a list of pending applications. 3. Officer opens a specific application. 4. Reviews: a. Application form b. Uploaded documents (Invoice, Calibration Certificate, etc.) 5. Selects the appropriate Lab Nodal Officer from a mapped list. 6. Clicks "Assign to Lab." 7. System: a. Updates the application status (e.g., "Assigned to Lab Officer") b. Logs the assignment with timestamp c. Sends notification to the applicant
Post - Condition	1. Application is assigned to the correct Lab Nodal Officer. 2. Status is updated in the system (e.g., "Assigned to Lab"). 3. Notification is sent to the applicant confirming assignment.

Validity Checks	1. Only verified and complete applications can be assigned. 2. Only active and jurisdiction-mapped Lab Nodal Officers are eligible for assignment. 3. District Nodal Officer must be logged in and assigned to the applicant's district. 4. Each assignment must be: a. Logged with Application ID, District Officer ID, Lab Officer ID b. Timestamped c. Status updated accordingly 5. Notification to applicant must include: a. Application ID b. Status: "Assigned to Lab" c. Next steps (if any)
Input	1. District Nodal Officer login credentials 2. Application ID and data 3. Lab Nodal Officer selection 4. Review comments (optional)
Output	1. Updated application status 2. Assignment log with officer and lab details 3. Notification to applicant 4. Visibility of the application in Lab Nodal Officer's dashboard
Exception Handling	1. No lab officer mapped: "No Lab Nodal Officer available for the selected jurisdiction." 2. Incomplete application data: "Missing data or document. Please flag to applicant or escalate." 3. Assignment failure: "System error during assignment. Please retry or contact support." 4. Unauthorized action: "You are not authorized to reassign or modify this application."

3.4.6 Lab Nodal Officer Review #FR006

Description	The Lab Nodal Officer Review feature manages the technical assessment, scheduling, physical inspection, and outcome processing of applications for fresh verification or re-verification of weights and measures. Upon receiving an assigned application, the Lab Officer can raise a query, schedule a verification date, or issue a rejection. The system handles applicant communication, fee payment, scheduling, certification, and traceability of verification attempts, including rejection and resubmission cycles.
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User	1. Lab Nodal Officer – Handles technical review, scheduling, physical verification, and outcome decisions. 2. Applicants – Respond to queries, confirm schedule, pay fees, and download certificates. 3. System Admins – Monitor workflow integrity, handle exceptions, and manage traceability.
Pre - Condition	1. Application has been assigned to the Lab Nodal Officer by the District Nodal Officer. 2. Applicant has submitted all required documents and form fields. 3. Lab Officer has access to login and jurisdictional dashboard.
Sequence of Operations	1. Review and Decision: a. Lab Officer logs into portal and accesses assigned applications. b. Reviews instrument and applicant details. c. Chooses one of the following: 1. Raise Query 2. Schedule Verification 3. Reject Application 2. Query Raised Flow: a. Officer raises query with comments. b. System notifies applicant. c. Applicant logs in, views query, makes changes (form or documents). d. Applicant resubmits response. e. System notifies Lab Officer. f. Officer re-reviews and proceeds accordingly. 3. Verification Scheduling Flow: a. Officer selects date and time slot. b. System sends schedule to applicant via portal/email. c. Applicant confirms availability. d. Applicant pays verification fee via integrated payment gateway. e. System generates and stores digital receipt. 4. Physical Verification Flow: a. On the scheduled date, applicant presents instrument. b. Officer conducts inspection. c. Officer records results in the portal.

	5. Verification Outcomes: a. Pass: 1. Officer certifies instrument. 2. System generates digital certificate. 3. Certificate is uploaded to portal; applicant receives notification. b. Fail: 1. Officer rejects application with remarks. 2. Rejection certificate is generated. 3. Applicant is notified and may correct issues. 4. Applicant reapplies using the same flow. 5. System links new application to rejected one for traceability.
Post - Condition	1. Application status is updated based on action: Query Raised, Verification Scheduled, Rejected, or Certified. 2. System notifies the applicant accordingly. 3. Verification fee is paid, and receipt is stored. 4. Certificate (or rejection report) is generated and delivered.
Validity Checks	1. Officer can only access applications within assigned jurisdiction. 2. All key decisions (query, schedule, reject, certify) must be logged with: a. Officer ID b. Timestamp c. Action and remarks 3. Verification cannot proceed unless: a. Fee is paid b. Applicant confirms schedule c. Physical verification is recorded 4. Documents and instrument data must be validated at each stage. 5. Reapplications must link to previously rejected IDs for audit trail.
Input	1. Instrument details, documents, and verification type 2. Lab Officer decisions (Query/Reject/Schedule) 3. Applicant's updated responses or confirmation 4. Fee payment input via gateway 5. Physical inspection results

Output	1. Application status updates 2. Notifications to applicant 3. Verification schedule and payment receipt 4. Digital verification certificate or rejection certificate 5. Linked application chain (for re-verification attempts)
Exception Handling	1. Query without remarks: "Please provide remarks for the raised query." 2. Payment failure: "Payment not completed. Retry or contact support." 3. Schedule clash: "Selected time slot unavailable. Choose a different date." 4. No response from applicant: "Query deadline exceeded. Application will be flagged for closure." 5. Physical inspection failure: "Unable to update verification results. Please try again or contact admin." 6. QR code or certificate generation error: "Certificate not generated. Contact admin."

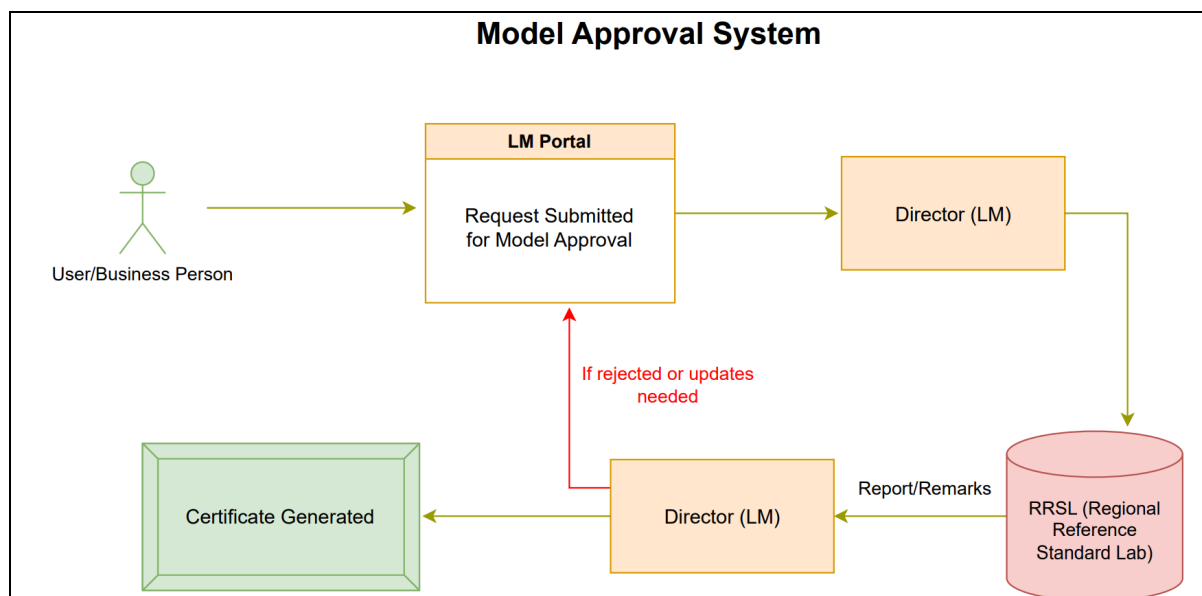
3.4.7 Use cases (Verification and Re-verification Module)

Field Name	Description
Use Case ID	eMaap_VR_006
Use Case Name	Verification & Re-verification of Weights & Measures
Reference	FRS Numbers: FR001 to FR006S
Created By (Date)	eMaap functional team (16 June 2025)
Last Update (Date)	16 June 2025
Primary Actor	Applicant (Repairer / Manufacturer / Dealer)
Secondary Actor	District Nodal Officer, Lab Nodal Officer, Legal Metrology Officer (LMO)
Trigger Activating Use Case	Applicant initiates a request for verification or re-verification of a weight or measure instrument.
Inter-linkages	1. User Authentication Module 2. Dashboard and Notification Engine 3. Document Management System 4. Payment Gateway 5. Certificate Generation Module 6. Audit Log and Rejection Tracking
Normal Event Flow	1. Applicant logs in and selects 'Verification/Reverification' 2. Fills instrument details, uploads documents, and selects jurisdiction 3. Documents are uploaded and validated

	4. System validates form and submits application 5. District Nodal Officer reviews and assigns Lab Nodal Officer 6. Lab Officer processes application, schedules verification, performs inspection 7. Upon conformity, certificate is generated and shared with applicant
Alternate Event Flow	1. If query is raised, applicant is notified and must update/resubmit 2. If instrument is non-compliant, rejection certificate issued 3. Applicant may correct and reapply; system maintains traceability to previous rejected application
Frequency of Use	As per statutory requirements – typically: 1. Fresh verification (prior to use) 2. Re-verification (annually or biennially based on instrument type) 3. Post repair or modification
Assumptions	1. Applicants are registered and verified on the eMaap Portal 2. Legal Metrology infrastructure and Labs are geo-tagged and functional 3. Officers are trained in digital verification flow 4. Payment integration and receipt generation systems are live
Additional Note	1. System logs every action by officer and applicant for audit 2. QR-enabled certificates facilitate field verification 3. Rejected applications are version-controlled and traceable

3.5 Modal approval system for Center

This module will be developed to ensure that weighing and measuring instruments are accurate and reliable, safeguarding consumer interests and promoting fair trade practices. Through this module, the businesses can obtain a certificate confirming that their weights and measuring instruments (like scales, meters etc.) comply with Indian metrological standards. The applications received will be sent to RRSL and approval certificate will be issued once clearance is received from RRSL.



The Model Approval System for the Legal Metrology (LM) department. The process begins with a User or Businessperson submitting a request for model approval through the LM Portal. This request is then forwarded to the Director (LM), who reviews the submission and directs it to the RRSL (Regional Reference Standard Laboratory) for technical examination. The RRSL evaluates the model and sends back a detailed report or remarks to the Director. Based on the RRSL's feedback, the Director either approves the request and initiates certificate generation, or if the submission requires modifications or is not acceptable, the request is rejected or sent back for updates, prompting resubmission through the portal. This workflow ensures that all models meet regulatory standards before certification.

3.5.1 User access and application submission #FR001

Description	The Model Approval System allows manufacturers and applicants to submit requests for new or revised model approvals for weights and measuring instruments under the Legal Metrology (General) Rules, 2011. The user accesses the eMaap (Legal Metrology) Portal, fills in detailed model and manufacturer information, uploads technical documents, selects the RRSL Lab for testing, pays the applicable fee, and submits the application. A unique Application ID is generated, and acknowledgment is sent via email and SMS.
User	1. Manufacturers / Authorized Applicants – Submit model approval applications for new or revised instruments. 2. RRSL Lab Officials – Receive and process applications for model testing. 3. Controllers (Centre) – Oversee application workflows and review/approve submissions.

	4. System Admins – Ensure platform functionality and resolve technical issues.
Pre - Condition	1. The applicant is a registered user with valid login credentials. 2. Manufacturer information is either newly entered or available in the system. 3. The application form and fee structure are configured in the backend.
Sequence of Operations	1. User logs into the eMaap (LM) Portal. 2. System verifies credentials and displays the user dashboard. 3. User selects “Apply for Model Approval.” 4. System displays a new Model Approval Application Form. 5. User enters: a. Model Name, Type, Category, and Specifications b. Manufacturer details (Name, License, Address, Contact Info) c. Chooses New or Revised model 6. If the manufacturer is previously registered, the system auto-fills details. 7. User selects the relevant RRS� Lab. 8. System prompts for required document uploads: a. Technical drawings b. User manuals c. Calibration data, etc. 9. System validates: a. File types (PDF, JPG, etc.) b. File sizes (e.g., ≤ 5MB) c. Mandatory uploads 10. Fee is calculated based on model category. 11. User proceeds to the integrated payment gateway. 12. Payment is processed, confirmation is logged, and a receipt is generated 13. System submits the application and generates a unique Application ID. 14. Acknowledgment is sent via email and SMS to the applicant.
Post - Condition	1. Application is submitted with a unique Application ID. 2. Acknowledgment is sent via SMS/email. 3. Application appears in the RRS� Lab and Controller dashboard for processing.

Validity Checks	1. Only authenticated users may access the Model Approval module. 2. Manufacturer details must be validated or auto filled from system records. 3. Mandatory fields: a. Model name, type, category, manufacturer details b. Selected RRSL lab c. Required document uploads 4. Fee must be calculated based on predefined category logic. 5. Application submission is only allowed after successful payment. 6. Every submitted application must be: a. Assigned a unique Application ID b. Logged with timestamp, user ID, and payment reference c. Linked to acknowledgment delivery status (email/SMS)
Input	1. Login credentials 2. Model and manufacturer information 3. Document uploads 4. RRSL lab selection 5. Fee payment
Output	1. Form validation messages 2. Payment receipt 3. Application ID 4. Submission status: "Submitted" 5. Email and SMS acknowledgment
Exception Handling	1. Invalid login credentials: "Incorrect username or password." 2. Missing mandatory fields: "Please complete all required sections before proceeding." 3. File format or size error: "Invalid file. Only PDF/JPG files under 5MB are accepted." 4. Fee mismatch: "Unable to determine fee category. Contact support." 5. Payment failure: "Payment unsuccessful. Please try again or use a different method." 6. Submission failure: "Application could not be submitted due to system error. Please try later."

3.5.2 Internal routing to LM Director #FR002

Description	This functionality manages the internal routing and review workflow for model approval applications by the Director of Legal Metrology (LM Director). Once an application is submitted by a manufacturer/applicant, it is routed to the LM Director's dashboard for initial screening. The Director reviews the application form and attaches documents and determines whether it is complete and eligible for technical evaluation. If issues are found, the application is either marked for resubmission or rejected with appropriate remarks. Resubmissions are tracked under the same Application ID.
User	1. LM Director (Centre) – Reviews submitted model approval applications for eligibility and completeness. 2. Applicants (Manufacturers / Authorized Representatives) – Receive feedback and resubmit corrected applications. 3. System Admins – Ensure routing, notification, and audit tracking are functioning correctly.
Pre - Condition	1. The application has been successfully submitted by the applicant. 2. All mandatory documents have been uploaded. 3. The LM Director has valid credentials and access to the backend portal. 4. Application appears under the "New Submissions" section of the Director's dashboard.
Sequence of Operations	1. LM Director logs into the eMaap backend portal. 2. Dashboard displays a list of newly submitted applications. 3. Director selects and opens an application. 4. Reviews: a. Application form fields (model name, type, manufacturer info, etc.) b. Uploaded documents (technical drawings, calibration data, manuals) 5. Based on review: a. If complete and valid, forwards to RRSL for technical testing. b. If incomplete or invalid, adds remarks and selects: c. Resubmit Required, or d. Rejected 6. System updates application status and logs the action. 7. Applicant receives a notification (email/SMS + portal alert). 8. Applicant logs in, views remarks, and edits application. 9. Applicant re-submits the corrected version. 10. System retains the same Application ID and logs the resubmission cycle.

Post - Condition	1. Application is marked “Approved for Testing”, “Resubmit Required”, or “Rejected”. 2. If marked for resubmission or rejection, a notification with remarks is sent to the applicant. 3. Applicant logs in, makes corrections, and resubmits. 4. Resubmission is tracked under the same Application ID for traceability.
Validity Checks	1. Only users with LM Director role may perform review and routing. 2. System must validate: a. Application completeness b. Required documents uploaded c. Mandatory remarks for rejection/resubmission 3. Application status and history must be: a. Logged with timestamp and user ID b. Traceable under the same Application ID for resubmission cycles 4. Application must not proceed to RRS� lab without LM Director’s approval.
Input	1. Application ID and form data 2. Attached documents 3. Director’s action (Approve / Resubmit / Reject) 4. Remarks (mandatory for Resubmit and Reject)
Output	1. Application status update 2. Notification to applicant 3. System log entry (timestamp, user ID, decision, remarks) 4. Linkage of original and resubmitted applications (under same Application ID)
Exception Handling	1. Document not accessible: “Unable to open attached file. Please contact system admin.” 2. No remarks entered for rejection/resubmission: “Remarks are required for this action.” 3. Duplicate action attempt: “This application has already been marked. Refresh the dashboard.” 4. Applicant edits after rejection: “Application is rejected and cannot be edited. Submit a new request.” 5. Notification failure: “Decision saved, but applicant notification not delivered. Please inform manually.”

3.5.3 Forwarding to RRSL #FR003

Description	This functionality manages the forwarding of model approval applications from the Legal Metrology Director (Centre) to the selected RRSL (Regional Reference Standard Laboratory) for technical testing. Once the LM Director reviews and approves an application for completeness, the system securely forwards the application along with all supporting documents to the selected RRSL. The RRSL technical team then schedules testing, records test results, and prepares a digitally signed report that is routed back to the LM Director through the eMaap portal.
User	1. LM Director (Centre) – Forwards completed applications to the selected RRSL lab. 2. RRSL Officers / Lab Staff – Review submissions, schedule testing, perform analysis, and submit reports. 3. Applicants (Manufacturers / Representatives) – Receive testing notifications and send physical models for evaluation. 4. System Admins – Manage workflow integrity, secure file routing, and audit trails.
Pre - Condition	1. Application has been submitted and reviewed by the LM Director. 2. All required documents have been uploaded. 3. LM Director has selected an RRSL lab. 4. Application is marked as “Approved for Testing”.
Sequence of Operations	1. Forwarding by LM Director: a. Director selects an RRSL lab from the list. b. System securely routes application and documents to the selected RRSL. 2. RRSL Review and Scheduling: a. RRSL Officer logs in and views assigned application. b. Technical team reviews design/specifications. c. Lab schedules the model test and enters the test date in the portal. - d.) System notifies the applicant of the scheduled date. 3. Model Receipt and Testing: a. Applicant dispatches the physical model to RRSL. b. RRSL logs receipt of model in the portal. c. Technical team conducts the test and enters results (compliant / non-compliant). d. Remarks are entered explaining test outcome. e. Final test report is digitally signed by authorized officer. 4. Submission to LM Director: a. Signed report is uploaded and forwarded to LM Director

	through the portal. b. Report is linked to the application record.
Post - Condition	1. Application is visible in RRSL dashboard. 2. Testing schedule is generated and shared with the applicant. 3. Testing is performed, and a digitally signed report is submitted to the LM Director. 4. Applicant is notified at each critical step.
Validity Checks	1. Only LM Director can forward applications to RRSL. 2. Application must be marked "Complete" and all required files attached. 3. RRSL officer must log: a. Test date b. Receipt of model c. Results and compliance remarks d. Digital signature 4. All submissions must be: a. Timestamped b. Linked to the correct Application ID c. Accessible in LM Director's dashboard after report submission 5. Testing report must include: a. Clear status (Pass/Fail) b. Reasons for failure (if any) c. Authenticated digital signature
Input	1. Application and uploaded documents 2. Selected RRSL lab 3. Testing schedule 4. Physical model/instrument 5. Test results and remarks 6. Digital signature of authorized RRSL officer
Output	1. Status updates: "Forwarded to RRSL", "Test Scheduled", "Model Received", "Report Submitted" 2. Notification to applicant (schedule, outcome) 3. Digitally signed test report 4. Log of all activities for audit
Exception Handling	1. RRSL lab not selected: "Please select a lab before forwarding." 2. Document upload failure: "Could not attach required file. Retry or contact admin." 3. Test schedule conflict: "Selected date unavailable. Choose another slot." 4. Model not received: "Applicant has not dispatched the model. Follow up required." 5. Incomplete test report: "Results or remarks missing. Cannot finalize."

	6. Signature failure: “Digital signature could not be verified. Please re-sign and upload.”
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3.5.4 Director’s Final Review #FR004

Description	The Director Final Review functionality enables the Legal Metrology Director (Centre) to make the final decision on model approval applications after receiving the RRSL testing report. The Director reviews the test results, lab remarks, and all associated documents to determine whether the model meets compliance standards. Based on the outcome, the Director either approves the model, or marks the application as “Rejected” or “Resubmit Required”. Applicants are notified accordingly and can reapply if necessary.
User	1. LM Director (Centre) – Performs final review and approval/rejection of the model. 2. Applicants (Manufacturers / Representatives) – Receive decision notifications and may reapply if required. 3. System Admins – Monitor decision logs, workflow completion, and resubmission tracking.
Pre - Condition	1. The application has been forwarded to RRSL and testing has been completed. 2. RRSL has submitted a digitally signed report along with compliance remarks. 3. The LM Director has valid login credentials and dashboard access.

Sequence of Operations	1. LM Director receives a notification that the RRSL report is available. 2. Logs into the portal and opens the application. 3. Reviews: a. RRSL test results and remarks b. Uploaded documents and certificates c. Historical application data (if reapplication) 4. Based on review, selects one of the following actions: a. Approve: If model meets all standards b. Reject: If model fails testing or lacks conformity c. Resubmit Required: If minor improvements are needed 5. Adds remarks for rejection or resubmission. 6. System updates application status. 7. Applicant is notified of the decision and remarks. 8. Applicant may reapply with modifications if applicable.
Post - Condition	1. Application is marked as Approved, Rejected, or Resubmit Required. 2. The applicant is notified of the outcome via portal/email/SMS. 3. In case of rejection/resubmission, the applicant may reapply referencing the original Application ID.
Validity Checks	1. Application must have a completed and signed RRSL report before Director review. 2. Only users with LM Director role can take final action. 3. Remarks are mandatory for “Reject” or “Resubmit Required” actions. 4. System must ensure: a. Each decision is timestamped b. Linked to the specific Application ID c. Logged with user ID of the reviewer 5. If rejected or resubmission is required: a. The applicant should be allowed to reapply with reference to the original application b. New submissions must maintain traceability for compliance and audit
Input	1. Application ID 2. RRSL test report and compliance status 3. Supporting documents 4. Director’s action and remarks
Output	1. Updated application status: Approved / Rejected / Resubmit Required 2. Remarks displayed in applicant’s dashboard 3. Notification via email/SMS 4. Logged action in system audit trail

Exception Handling	1. Missing RRSL report: “Cannot proceed—test report not found.” 2. No remarks entered for rejection/resubmission: “Remarks are mandatory for this action.” 3. System error during decision save: “Decision could not be saved. Please try again later.” 4. Applicant not notified: “Notification delivery failed. Retry or escalate to admin.”
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3.5.5 Certificate Generation #FR005

Description	This functionality supports the generation of a digital Model Approval Certificate once the Director (Legal Metrology) approves a model application. The certificate acts as official proof that the weighing/measuring instrument model complies with Indian metrological standards. It includes key approval details and is digitally signed by the Director. It is stored in the applicant’s dashboard, accessible for download, and logged in the system registry for compliance and traceability.
User	1. Applicant: Views and downloads the certificate from the dashboard 2. Director (LM): Reviews RRSL report and triggers certificate approval 3. System Administrator: Manages certificate templates, system logs, and security 4. RRSL Officer: Sends compliance test results to Director
Pre - Condition	1. Model application must be approved by the Director (LM) based on RRSL report 2. Mandatory fields in application must be filled and uploaded documents verified 3. Payment for the application must be successfully completed

Sequence of Operations	1. Director logs into eMaap and views RRSL report. 2. If report is compliant, Director approves the application. 3. System auto-generates the Model Approval Certificate with: a. Model Name b. Approval Number c. Validity Period d. Manufacturer Details e. Date of Issue f. Director's Digital Signature 4. Certificate is saved in applicant's dashboard. 5. System sends SMS/email notification to applicant. 6. Applicant downloads certificate in PDF format. 7. System logs the certificate issuance in the internal registry with timestamp and approval number.
Post - Condition	1. A digitally signed certificate is generated and stored in user dashboard 2. System sends email/SMS notifications to the applicant 3. Certificate is logged in system with approval number and issue date
Validity Checks	1. Certificate fields presence: All required fields (Model Name, Approval Number, etc.) must be non-empty 2. Unique Approval Number: Must not duplicate an existing entry in the registry 3. Validity Period: Should not exceed predefined policy duration (e.g., 5 years) 4. Digital Signature: Must be applied using authorized credentials 5. Document Format: PDF generation only; downloadable without corruption
Input	1. Approved model application details 2. Director's approval trigger 3. RRSL compliance report 4. Manufacturer details, model specifications
Output	1. Model Approval Certificate (PDF) 2. SMS/Email notification to applicant 3. Log entry in issuance registry 4. Dashboard certificate listing for applicant
Exception Handling	1. Director tries to approve without RRSL report: System prevents action and shows error message 2. Required certificate fields (e.g., Model Name) are missing System throws validation error, certificate generation is blocked 3. Technical failure during generation: Transaction rolled back; admin notified via alert

	4. Digital signature fails: System logs error and retries or notifies admin
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3.5.6 Post Certification #FR006

Description	The post-certification process of the Model Approval System ensures regulatory traceability, administrative transparency, and operational efficiency after a model is approved and the certificate is issued. This module allows inspection authorities and departmental users to access certified models, prevents duplicate submissions, maintains detailed audit logs, and provides reporting and MIS features for monitoring model approvals across regions and categories.
User	1. Director (LM): Oversees and approves final model certifications 2. Admin (Central): Views list of approved models 3. Inspection Authorities: Access approved models for enforcement and verification purposes 4. Department Users: Generate MIS and summary reports Applicant: Can no longer resubmit identical model without justification
Pre - Condition	1. Model Approval Certificate must have been generated and signed 2. System must log Director's approval and RRS� test results 3. Unique approval number assigned to the certificate
Sequence of Operations	1. System logs the issuance of the Model Approval Certificate. 2. Certificate is stored in applicant's dashboard and made accessible to inspection authorities via secure interface. 3. Admin panel updates the list of approved models for department users. 4. Audit trail records all actions performed by: a. Applicant b. Director (LM) c. RRS� officials 5. System validates new applications to prevent resubmission of identical models unless justification is provided. 6. Authorized users generate: a. Model-wise approval reports b. Periodic MIS reports based on filters like region, category, and manufacturer.

Post - Condition	1. Inspection authorities can access model certificate from central registry 2. Admin panel displays all approved model entries 3. Reports and audit logs are available and exportable for authorized users
Validity Checks	1. Duplicate model submission: System checks model name + specifications + manufacturer ID in registry 2. User role access to certificates: Role-based access control – only authorized users can view/download 3. Justification field for re-submission: Mandatory if identical model is detected during new application submission 4. Audit trail coverage: Each action must be time-stamped and linked to specific user ID and role. 5. Report data integrity: Reports must pull from finalized/approved entries only
Input	1. Approved model data and certificate 2. User login credentials (Inspector/Admin/Director) 3. Attempt to re-submit same model 4. Reporting filters (date range, manufacturer, category)
Output	1. Visible entry in Admin dashboard 2. Access to approved model list 3. Warning message: rejection unless justification is provided 4. Downloadable MIS and approval reports (PDF/Excel)
Exception Handling	1. Unauthorized user attempts to access certificate data: Access denied; user redirected to error page with message 2. Attempt to re-submit an already approved model without reason: System blocks submission and prompts for justification 3. Failure in loading audit or reports data: System shows error message and logs issue for admin review 4. Report generation fails due to large dataset: Notify user; suggest filtering or narrowing report parameters

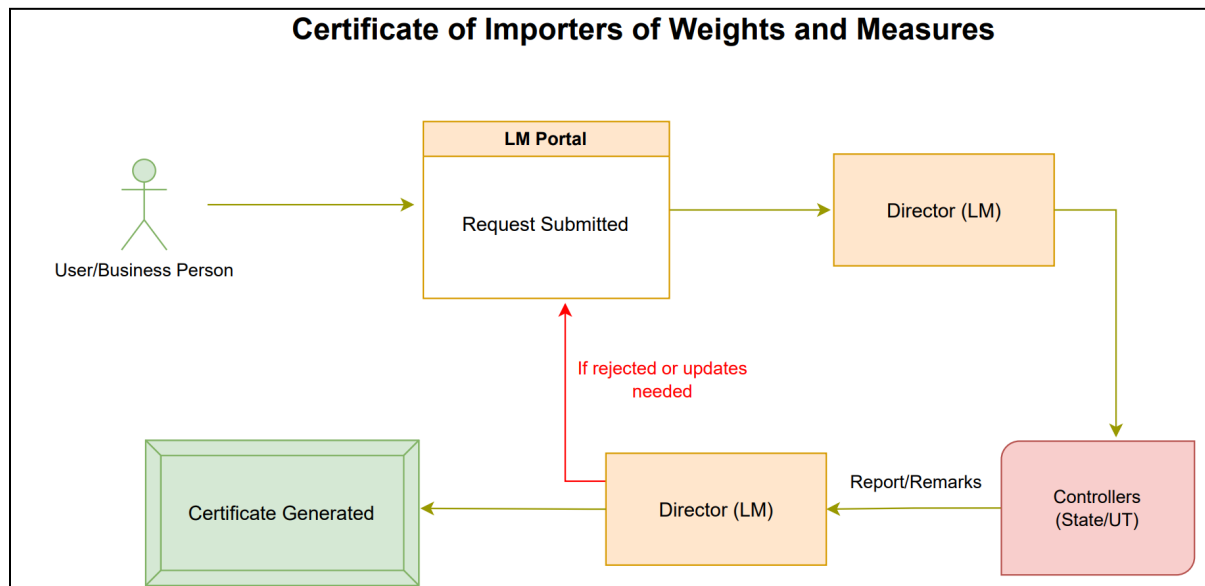
3.5.7 Use Cases (Modal approval system)

Field Name	Description
Use Case ID	eMaap_MA_007
Use Case Name	Model Approval System – Centre
Reference	FRS Number: FR001 to FR006
Created By (Date)	eMaap functional team (16 June 2025)
Last Update (Date)	16 June 2025
Primary Actor	Applicant (Manufacturer/Importer submitting model approval request)

Secondary Actor	Director, Legal Metrology (Centre); Regional Reference Standard Laboratory (RRSL)
Trigger Activating Use Case	Applicant logs in to the eMaap portal and submits a new model approval request
Inter-linkages	1. Dashboard Module 2. RRSL Interface Module 3. Certificate Management Module 4. Audit Trail 5. Notification Engine
Normal Event Flow	1. Applicant logs in and navigates to “Model Approval” 2. Fills out model details (e.g., instrument type, category, purpose) 3. Uploads technical documents 4. Submits the application 5. Director LM receives and reviews 6. Forwards to RRSL for evaluation 7. RRSL performs tests and submits report 8. Director LM takes final action (Approve/Reject/Send for update) 9. If approved, certificate is auto generated and available to applicant
Alternate Event Flow	1. If documents are incomplete or invalid, application is returned to applicant with comments 2. Applicant updates and resubmits 3. If RRSL report is negative, application is rejected
Frequency of Use	As and when a new model is to be introduced in the market (per manufacturer or importer)
Assumptions	1. RRSL is digitally integrated with the eMaap system 2. Users have access to full technical specs 3. Certificates are digitally signed and stored
Additional Note	1. Audit logs are maintained for every user action 2. Model once approved can be used for further manufacturer or import licenses 3. The certificate contains QR code for verification

3.6 Certificate of Importers of weight and measures for centers

This module will be designed to facilitate the registration process for businesses aiming to import goods into the country. It will collect comprehensive details about the products intended for import, along with other required supporting documents. Following successful verification, a certificate will be issued to the business, completing the registration process.



The **Certificate of Importers of Weights and Measures** workflow begins with a user or business entity initiating a request through the Legal Metrology (LM) Portal. Upon logging in, the user accesses the application module and submits the required documentation and information for obtaining the Importer Certificate. Once the application is successfully submitted, it is routed to the Director of Legal Metrology for the initial level of scrutiny. The Director examines the submission for completeness, accuracy, and compliance with applicable rules and standards. If the application is found lacking or requires updates, it is reverted back to the applicant with detailed remarks for corrections or additional inputs. The applicant is then required to log in, address the observations made by the Director, and re-submit the corrected application through the LM Portal. Once the Director is satisfied with the updated submission, they forward the application and associated remarks to the respective State or Union Territory Controller for final evaluation.

The Controller conducts a verification based on the Director's recommendations and may also perform additional assessments if needed. Upon satisfactory review, the Controller communicates the decision and remarks back to the Director of LM. If all conditions are met, the Director then authorizes the generation of the Certificate of Importer of Weights and Measures. The final certificate, once generated, becomes available on the applicant's LM Portal dashboard and is also digitally signed. Throughout this process, applicants are notified via automated emails or SMS regarding the status of their application, ensuring transparency and timely action at each stage of the workflow.

3.6.1 User/Businessperson Action #FR001

Description	This module allows importers of weights and measures to apply for a Certificate of Importer through the Legal Metrology (LM) Portal. Authorized business users log in, access the importer certificate module, fill in relevant importer and firm details, upload necessary documents (e.g., PAN, GST), and submit the application for approval. The system provides submission acknowledgment, real-time status tracking, and enables clarification or resubmission in case of query or rejection by the LM authority.
User	1. Business Users / Importers – Submit and manage applications for Importer Certificates. 2. LM Officers (Centre) – Review, approve, or return applications with remarks. 3. System Admins – Maintain workflow integrity and resolve exceptions.
Pre - Condition	1. The user is a registered and authenticated business entity on the LM Portal. 2. The Importer Certificate module is enabled and accessible to the user. 3. The system has defined required documents and fields.
Sequence of Operations	1. User logs in to the LM Portal with valid credentials. 2. Navigates to “Importer Certificate” module. 3. Fills in the application form, including: a. Importer name b. Firm details c. Contact information d. State/UT of operation 4. Uploads mandatory documents (e.g., PAN card, GST certificate). 5. Selects the appropriate State/UT for importer operations. 6. Clicks “Submit” to send the application. 7. System validates entries and generates a submission acknowledgment with a unique reference number. 8. Application appears in the user dashboard with status. 9. User receives notifications via email/SMS for: a. Submission confirmation b. Status changes c. Director queries or clarification requests 10. If returned for clarification, user: a. Reviews Director's remarks b. Edits the application and/or uploads corrected documents c. Resubmits the application

Post - Condition	1. Application is submitted and stored in the database. 2. User receives a unique application reference number. 3. Status is updated and tracked through the dashboard. 4. LM authority receives the application for review.
Validity Checks	1. Only authenticated users with business registration can apply. 2. Mandatory fields (firm name, PAN, GST, State/UT) must be completed. 3. Document uploads must match format and size requirements. 4. System must: a. Generate a unique application reference ID b. Log each action with timestamp and user ID c. Allow query response and resubmission only if the application was returned by LM Director 5. Application must be editable only during the clarification/resubmission stage.
Input	1. Login credentials 2. Importer and firm details 3. Uploaded documents (PAN, GST, etc.) 4. Selected State/UT
Output	1. Acknowledgment receipt with reference number 2. Application status updates in dashboard 3. Email/SMS notifications 4. Editable application in case of return
Exception Handling	1. Missing mandatory fields: "All required fields must be completed." 2. Invalid document format/size: "Only PDF/JPG under 5MB are accepted." 3. Login/session failure: "Your session has expired. Please log in again." 4. Submission error: "Unable to submit application. Please try again later." 5. Query response not submitted: "Clarifications must be submitted before resubmission." 6. Multiple submissions: "An active application for this entity already exists."

3.6.2 System (LM Portal) Functions #FR002

Description	This module outlines the automated system-level functions performed by the Legal Metrology (LM) Portal during the lifecycle of an Importer Certificate application. From validating submissions to tracking progress and maintaining audit trails, the system ensures secure, accurate, and transparent handling of applications. The portal also forwards valid applications to the Director (LM), manages storage, and provides real-time updates and reminders to both applicants and reviewing officers.
User	1. Business Users / Importers – Interact with the LM Portal for application submission and tracking. 2. LM Director (Centre) – Reviews applications forwarded by the system. 3. System Admins – Oversee workflow operations, audit logs, and automated notifications.
Pre - Condition	1. User has logged in and filled out the Importer Certificate application form. 2. All mandatory documents are uploaded. 3. The application is ready for submission.
Sequence of Operations	1. Validation: a. System checks that all mandatory fields and document uploads are complete and correctly formatted. 2. Submission Logging: a. Generates a unique Application ID. b. Captures timestamp of submission. c. Updates status to “Under Review”. 3. Forwarding: a. Routes the application to the Director of Legal Metrology for review. 4. Document & Activity Logging: a. Stores uploaded documents securely with encryption and access control. b. Maintains a full audit trail of: c. User actions (submit, update, resubmit) d. System events (status change, forwarding, reminders) 5. User & Officer Interaction Support: a. Displays action history in the portal. b. Enables real-time application tracking by applicant. c. Sends auto-reminders to be pending stakeholders (e.g., applicant or reviewer). d. Flags duplicate submissions based on PAN, GST, or firm name.

Post - Condition	1. Application is validated, submitted, assigned an ID, and forwarded to the LM Director. 2. Application status is updated and visible to stakeholders. 3. Notifications and audit logs are triggered.
Validity Checks	1. All mandatory fields must be completed before submission is accepted. 2. Application ID must be system-generated and unique. 3. Timestamp, user ID, and status must be recorded for each major event. 4. Uploaded files must: a. Be securely stored b. Be associated with the correct application c. Pass format and size checks 5. Duplicate applications must be flagged using: a. PAN number b. GSTIN c. Registered firm name 6. Audit log must capture: a. Each user/system activity b. Timestamps c. Actions and transitions
Input	1. Submitted application form 2. Uploaded supporting documents 3. User ID and session data
Output	1. Unique Application ID 2. Submission timestamp 3. Application status = "Under Review" 4. Secure document storage 5. Visible application tracker and action history 6. Auto-reminder notifications 7. System alerts for duplicates (if any)
Exception Handling	1. Missing/invalid fields: "Some required fields are incomplete or incorrectly filled." 2. File upload failure: "File type/size invalid. Please re-upload in PDF/JPG (max 5MB)." 3. Duplicate application: "An application with similar details already exists." 4. Audit trail or storage issue: "Error logging action or saving document. Please contact admin." 5. Notification failure: "Auto-reminder could not be sent. Will retry in the next cycle."

3.6.3 Director (LM) Role #FR003

Description	This module defines the workflow responsibilities of the Director of Legal Metrology (LM) for processing importer certificate applications. The Director reviews applications submitted by business users, checks completeness and correctness of submitted data and documents, and either accepts, rejects, or forwards them to the relevant State/UT Controller for verification. After receiving feedback from the Controller, the Director takes a final decision and issues a digitally signed certificate through the LM Portal. The system records the decision, updates the application status, and notifies the applicant.
User	1. Director, Legal Metrology (Centre) – Performs application review, coordination with State Controllers, and final approval/rejection. 2. State/UT Controllers – Provide field-level verification and remarks. 3. Business Users (Importers) – Submit applications and receive final outcome. 4. System Admins – Oversee workflow, digital signature integration, and certificate generation.
Pre - Condition	1. Application is submitted by the applicant and forwarded by the system. 2. All mandatory fields and documents are present. 3. LM Director has secure access and dashboard visibility.
Sequence of Operations	1. LM Director logs into the LM Portal. 2. Views pending applications in the dashboard. 3. Opens and reviews application form and uploaded documents. 4. Based on review: a. Forwards to Controller (State/UT) with comments, or b. Rejects application with remarks, or c. Returns to applicant if incomplete 5. Controller receives the application and submits verification report/remarks. 6. LM Director reviews Controller's feedback. 7. Based on findings: a. Approves or rejects the application b. Adds final remarks 8. If approved: a. Digitally signs certificate using integrated signing tool b. Triggers certificate issuance command in the system 9. Final status is updated on LM Portal. 10. Applicant is notified and can download the certificate from their dashboard.

Post - Condition	Application is either: - Forwarded to State/UT Controller, - Returned to applicant for clarification, or - Rejected - After Controller feedback, final approval or rejection is issued. - Certificate is generated and digitally signed, if approved. - Application status is updated in the system.
Validity Checks	- Application must be reviewed only by users with Director (LM) role. Remarks must be provided for: - Rejection - Return to applicant - Forwarding to Controller - Controller feedback must be received before final decision. Digital certificate must: - Be signed with a valid digital signature - Contain Application ID, applicant details, date, and issuing authority System must: - Log each action with user ID and timestamp - Store digital certificates securely - Update application status and notify user in real time
Input	- Application data and documents - Comments from State/UT Controller - Director's decision and remarks - Digital signature input (certificate approval)
Output	- Application status update (Approved, Rejected, Returned) - Notification to applicant - Forwarding trail to/from Controller - Digitally signed importer certificate - Log entries of all actions taken
Exception Handling	Missing documents: "Application incomplete. Cannot proceed without mandatory documents." No remarks on rejection/return: "Remarks required to send back or reject an application." Controller response delay: "Escalation triggered. No response received within defined timeframe." Digital signature failure: "Certificate could not be signed. Please retry or contact system admin." System error during status update: "Decision saved but status not updated. Please refresh and confirm."

3.6.4 Controller (States/UT) Role #FR004

Description	This module defines the responsibilities of the Controller of Legal Metrology at the State/UT level in reviewing importer certificate applications forwarded by the Director (LM) at the Centre. The Controller verifies the applicant's operational compliance, performs on-site checks if necessary, and validates submitted documents. Based on findings, the Controller drafts a compliance report or rejection remarks and uploads it to the LM Portal, returning the application to the Director for final action.
User	1. Controller (State/UT Legal Metrology Department) – Reviews, verifies, and reports on importer applications. 2. Director, Legal Metrology (Centre) – Receives the Controller's report to take final decision. 3. System Admins – Ensure access, secure report uploads, and workflow tracking.
Pre - Condition	1. Application has been reviewed and forwarded by the LM Director to the respective State/UT Controller. 2. The Controller has secure access to the LM Portal and jurisdiction-mapped application queue.
Sequence of Operations	1. Controller logs into the LM Portal and accesses the assigned application queue. 2. Reviews: a. Firm details b. Location/warehouse/shop data c. Uploaded regulatory documents (PAN, GST, etc.) 3. Performs field verification, if required. 4. Drafts a report summarizing: a. Compliance status b. Document findings c. Field verification results d. Recommendation (Accept / Reject) 5. Uploads the signed report to the LM Portal. 6. If rejected: a. Enters detailed remarks b. Flags the application as "Rejected – State Controller" 7. Submits the application back to the LM Director for final review and decision.
Post - Condition	1. Controller submits a compliance report or rejection remarks to the LM Portal. 2. Director receives the report and proceeds with final approval/rejection.

Validity Checks	1. Only authorized Controllers within the mapped State/UT jurisdiction may access and review assigned applications. 2. Reports must: a. Be uploaded in allowed format (PDF or structured input) b. Include date, Controller name/ID, and signature (if applicable) c. Contain explicit remarks for rejections 3. System must: a. Log each Controller action with timestamp and user ID b. Update application status (e.g., "Verified by State Controller", "Rejected by Controller") c. Notify the LM Director once submission is completed
Input	1. Application details (firm name, license, location) 2. Uploaded documents 3. Field verification findings (if applicable) 4. Controller's remarks/report
Output	1. Compliance report (PDF or system form) 2. Application status update 3. Notification to LM Director 4. Application tagged with jurisdictional verification stamp
Exception Handling	1. Missing access to application: "No application found, or permission denied. Check jurisdiction." 2. Document mismatch or unreadable: "Unable to verify documents. Please request resubmission." 3. Verification not possible: "Location could not be verified. Add comment and return to LM Director." 4. Upload failure: "Error uploading report. Please try again or contact support." 5. Remarks missing on rejection: "Please enter reasons for rejection before submission."

3.6.5 System (Post Approval) #FR005

Description	The post-approval system module automates the final steps of the importer certificate process after a decision is made by the Director of Legal Metrology (Centre). If approved, the system generates a digital certificate, assigns a unique certificate number and validity period, and delivers the certificate via email and user dashboard. It ensures secure archival, real-time status updates, and provides access to certificate records for both applicants and regulators.
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User	1. Importers / Business Users – Receive and download their approved certificates 2. LM Director (Centre) – Views list of issued certificates and final status per application. 3. System Admins – Ensure proper functioning of certificate generation, archival, and notification processes.
Pre - Condition	1. LM Director has taken final action on the application (Approved or Rejected). 2. All required compliance checks and controller reviews are complete. 3. Digital signature workflow is completed (for approved applications).
Sequence of Operations	1. LM Director marks the application as “Approved” or “Rejected”. 2. System: a. Updates application status in backend b. For approvals: c. Generates a digital certificate in PDF format d. Assigns a unique certificate number and validity date e. Signs the certificate digitally (if not already signed by Director) 3. Sends certificate to applicant via: a. Email attachment or secure download link b. Display in user dashboard under "My Certificates" 4. Enables permanent access to certificate for future downloads. 5. Archives: a. Application data b. Certificate copy c. Approval trail 6. Updates Director's dashboard to include newly issued certificate with status and download option.
Post - Condition	1. Certificate is digitally generated and accessible. 2. Application and certificate are securely archived. 3. Status is reflected to all stakeholders in real time.

Validity Checks	- Only approved applications trigger certificate generation. Each certificate must include: - Unique Certificate Number - Applicant Name - Validity Period - Digital Signature Certificate must be: - Downloadable anytime from dashboard - Archived with original application and approval data All actions must be: - Logged with timestamps and user IDs - Available in audit trail Director's dashboard must accurately reflect: - Issued certificates - Application status - Certificate download/report options
Input	- Final decision (Approved/Rejected) - Applicant details - Certificate metadata (ID, validity, digital signature) - Document template
Output	- Application status change to "Approved" or "Rejected" - PDF certificate with unique ID and validity - Certificate download link (email + dashboard) - Archived records for audit - Certificate listing in Director dashboard
Exception Handling	Certificate generation failure: "Unable to generate certificate. Please contact admin." Missing metadata: "Certificate number or validity missing. Cannot proceed." Email delivery failed: "Certificate generated but email could not be sent. Notify user manually." Dashboard not updated: "Certificate issued but not visible. Refresh or report to system admin." Duplicate certificate ID detected: "Certificate number conflict. Regenerate with new ID."

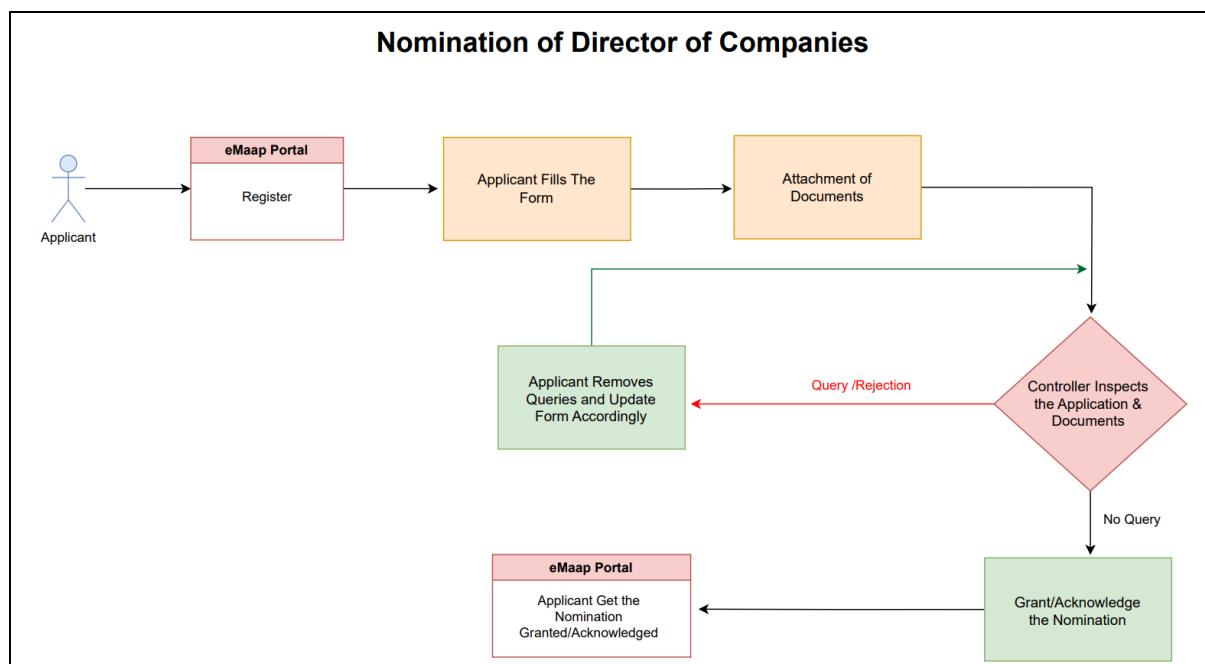
3.6.6 Use Cases (Certificates of Importers of weight and measures)

Field Name	Description
Use Case ID	eMaap_CIWM_008

Use Case Name	Certificate of Importers for Weights and Measures
Reference	FRS Number: FR001 to FR005
Created By (Date)	eMaap functional team, (16-06-2025)
Last Update (Date)	16-06-2025
Primary Actor	Business User / Importer
Secondary Actor	Director (Legal Metrology), State Controller (LM)
Trigger Activating Use Case	Submission of Importer Certificate Application on eMaap Portal
Inter Linkages	Dashboard module, Document Upload module, Notification module, Certificate generation module
Normal Event Flow	1. User logs in to the portal. 2. Navigates to Importer Certificate module. 3. Fills application form and uploads required documents. 4. Submits the application. 5. Director reviews and forwards to Controller. 6. Controller verifies and returns remarks. 7. Director approves and generates certificate. 8. Certificate available in user dashboard.
Alternate Event Flow	1. If incomplete, Director returns for update. 2. User re-submits with corrections. 3. If Controller raises issue, Director informs user. 4. User may reapply after corrections.
Frequency of Use	As per business need, typically once per importer annually or when changes occur.
Assumptions	1. User has valid login credentials. 2. Internet connectivity is stable. 3. Uploaded documents meet size and format criteria.
Additional Notes	1. Certificate is digitally signed. 2. Notifications are sent via SMS/email. 3. All actions are audit logged for transparency.

3.7 Nomination of Directorates of companies for center and states

This module will allow businesses to designate an individual (who is a director of the company) to represent their company's interests. This representative (director) will act as the business's point of contact (POC) for addressing any issues/concerns identified by the Legal Metrology Division. Additionally, this module will provision businesses to update the details of director appointed in cases where the current representative steps down or is replaced.



The Nomination of Director of Companies process through the eMaap Portal begins with the applicant accessing the portal to register themselves on the system. Once registered, the applicant proceeds to fill out the nomination form with relevant details. Following this, the applicant uploads all required supporting documents for the nomination.

The application, along with the documents, is then forwarded to the Controller for inspection. The Controller examines both the application form and the attached documents for completeness, accuracy, and compliance with the applicable regulations. If any discrepancies, errors, or incomplete details are identified, the Controller raises a query or rejection and sends it back to the applicant through the portal.

Upon receiving the query, the applicant is required to remove the deficiencies and update the form accordingly. This updated application is then resubmitted via the portal for re-evaluation by the Controller. If the application passes inspection with no further queries, the Controller proceeds to grant or acknowledge the nomination of the Director of Companies. Finally, the eMaap Portal notifies the applicant, and the acknowledgment or nomination approval becomes available for viewing or download.

3.7.1 Login and Application form Filling #FR001

Description	This module enables registered applicants (such as company representatives) to submit nominations for Directors of companies as per legal metrology regulations at both the central and state levels. Through the LM Portal, users access a dedicated Nomination of Director form that captures essential company and director information. The
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	system allows users to fill, validate, and save the form, ensuring data integrity and user flexibility.
User	1. Applicants / Company Representatives – Submit Director nomination details for regulatory approval. 2. Legal Metrology Officers (Centre/State) – Review submitted nominations and take further action. 3. System Admins – Manage access, data validation rules, and workflow configuration.
Pre - Condition	1. User is logged in with valid credentials. 2. Applicant has an active business profile on the LM Portal. 3. The "Nomination of Directors" module is enabled for the applicant's dashboard.
Sequence of Operations	1. Applicant logs into the LM Portal. 2. System redirects to their personalized dashboard. 3. User selects the "Nomination of Directors" module. 4. A structured form opens with the following fields: a. Company Name (text field) b. Corporate Identification Number (CIN) (text field) c. Director Name (text field) d. Director Identification Number (DIN) (text field) e. Nomination Details (e.g., nature, duration, purpose) 5. Additional input types: a. Dropdowns (e.g., state, type of business) b. Date pickers (for nomination dates) c. Checkboxes (e.g., declaration of accuracy) 6. System performs client-side and server-side validation for: a. Format compliance (e.g., CIN/DIN) b. Mandatory field completion c. Correct data types 7. User options: a. Save as Draft b. Submit for Review 8. If saved, data is stored, and resume option is enabled on dashboard.
Post - Condition	1. The nomination form is saved as a draft or submitted for review. 2. Application data is stored securely and is accessible for resubmission or updates.

Validity Checks	System must validate: - CIN and DIN pattern/length (e.g., CIN = 21 characters) - Nomination date must not be in the future - All mandatory fields are filled before enabling submission - Only users with authenticated business accounts can access the nomination module. Form saves must generate: - Draft timestamp - Application ID (if submitted) - Data must be stored securely and accessible for review, submission, or deletion by the applicant only.
Input	- Company and director identification details - Nomination metadata (dates, role type) - Form interaction events (save, submit)
Output	- Draft or submitted application entry - Validation messages - Application ID (on submission) - Status updates and dashboard view
Exception Handling	Missing required fields: "Please complete all mandatory fields before submitting." Invalid DIN/CIN format: "Please enter a valid Corporate/Director Identification Number." Save error: "Draft could not be saved. Please try again." Submission timeout: "Session expired. Please log in again." Module access blocked: "You are not authorized to use this module. Contact admin."

3.7.2 Document upload and submission #FR002

Description	This module enables applicants to upload and submit mandatory documents (e.g., Board Resolution, Director Consent Form) as part of the Nomination of Director of Companies process under Legal Metrology at both the central and state levels. The portal supports secure file upload with format and size validation, preview, and virus scanning. After uploading, applicants can submit the complete application, which generates a unique Application ID and sends a confirmation notification.
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User	1. Applicants / Company Representatives – Upload supporting documents and submit nomination applications. 2. Legal Metrology Officers (Centre/State) – Receive and review submitted applications and attached documents. 3. System Admins – Ensure document storage, virus scanning, and notification services function properly.
Pre - Condition	1. Applicant is authenticated and has filled out the nomination form. 2. All required documents are prepared and ready for upload. 3. Upload section is active and accessible.
Sequence of Operations	1. Applicant navigates to the Document Upload section of the nomination form. 2. System displays required documents list (e.g. Board Resolution, Director Consent Form). 3. Applicant uploads documents using multi-upload feature. 4. System validates each file for: a. Format: PDF, JPEG, PNG b. Size: ≤ 5 MB c. Security: Virus scan 5. Applicant can: a. Preview files b. Remove or replace any uploaded file before submission 6. After uploading all documents, applicant submits the form. 7. System: a. Saves form and file data b. Generates a unique Application ID c. Updates status to “Submitted” d. Sends confirmation email/SMS to the applicant
Post - Condition	1. Uploaded documents are securely stored and linked to the application. 2. System assigns a unique Application ID. 3. Application status is updated to “Submitted.” 4. Applicant receives a confirmation via Email/SMS.

Validity Checks	Each file must be: - In approved format (PDF, JPEG, PNG) ≤ 5 MB in size - Free of malware (post virus scan) - Mandatory files (e.g., Board Resolution, Consent Form) must be present. - Preview/removal must be possible before submission. Application must not be submitted unless: - All required documents are uploaded - Form data is valid Upon submission: - Application ID must be generated - Submission timestamp must be recorded - Status must change to “Submitted” - Audit logs must record user and action - Notification must be sent to registered contact details
Input	- Document files (PDF, JPEG, PNG) - Completed form data - Applicant session and credentials
Output	- Uploaded and scanned documents - Application ID - Submission status update - Email/SMS confirmation
Exception Handling	Unsupported file type: “Invalid format. Only PDF, JPEG, or PNG files are allowed.” File too large: “File exceeds the 5 MB size limit.” Virus scan failure: “Upload blocked. File flagged as unsafe.” Missing required files: “All mandatory documents must be uploaded before submission.” Submission failure: “Unable to submit. Please try again or contact support.” Notification failure: “Application submitted, but confirmation could not be delivered.”

3.7.3 Controller review and Inspection #FR003

Description	This module defines the workflow for Controller-level review and inspection of applications submitted for Nomination of Directors of Companies under the Legal Metrology framework at the Centre. Upon submission, the application is routed to the designated Controller/Inspector, who verifies the form entries and documents. The Controller can either approve the nomination, raise a query, or reject the application, with all actions logged and communicated to the applicant via automated notifications.
User	1. Controllers / Inspectors (Centre Legal Metrology) – Review applications, inspect documentation, and take action (Approve/Query/Reject). 2. Applicants / Company Representatives – Respond to Controller queries or view rejection status. 3. System Admins – Monitor routing, status changes, and notification integrity.
Pre - Condition	1. Application has been submitted successfully with all required fields and documents. 2. System has routed the application to the appropriate Controller based on jurisdiction. 3. Controller has access to the backend dashboard.
Sequence of Operations	1. System routes the submitted application to the designated Controller/Inspector. 2. Controller logs in and accesses the application through their dashboard. 3. Reviews: a. Form data (e.g., Director Name, DIN, Company Info) b. Uploaded documents (Board Resolution, Consent Form, etc.) 4. Based on review: a. If valid and complete, approves the application. b. If issues are found, either: c. Raises a query, or d. Rejects the application 5. Controller enters mandatory remarks when raising a query or rejecting. 6. System: a. Updates application status to “Query Raised” or “Rejected” b. Sends notification (email/SMS/portal alert) to the applicant with Controller’s comments

Post - Condition	Application status is updated to either: - Approved (ready for next stage) - Query Raised - Rejected - Applicant is notified with Controller's remarks if applicable.
Validity Checks	- Only authorized Controllers can access assigned applications. All actions must be logged with: - User ID (Controller) - Timestamp - Action type (Approve/Reject/Query) - Remarks are mandatory for queries and rejections. Once a query is raised: - Applicant must be notified - System must prevent further edits by Controller until applicant responds If rejected: - Application must be locked from further edits by applicant - Controller's rejection remarks must be permanently recorded
Input	- Submitted application and documents - Controller's decision (Approve/Query/Reject) - Remarks (for queries/rejections)
Output	- Updated application status - Remarks saved in the system - Notification to applicant - Audit trail of Controller action
Exception Handling	Document not loading: "Unable to access uploaded file. Please check file integrity." No remarks provided: "Remarks are mandatory for rejection or query." Review timeout/session expired: "Your session has expired. Please log in again." Incorrect status update: "System error while updating status. Retry or escalate to admin." Notification failure: "Action taken but notification not delivered. Resend or notify manually."

3.7.4 Query Resolution by applicant #FR004

Description	This module facilitates applicant-side resolution of queries raised by the Controller during the review of director nomination applications. Upon receiving a query, the applicant can log in to the LM Portal, view the remarks, and update the form or re-upload required documents. The applicant then re-submits the corrected application, which updates the status and routes the application back to the Controller for re-inspection.
User	1. Applicants / Company Representatives – View queries, update data/documents, and re-submit applications. 2. Controllers (Centre) – Re-inspect resubmitted applications and take final action. 3. System Admins – Monitor the query handling workflow, resubmission logs, and notification services.
Pre - Condition	1. Application has been reviewed and marked as "Query Raised" by the Controller. 2. Remarks have been entered by the Controller and made available in the portal. 3. Applicant account is active with proper access.
Sequence of Operations	1. Applicant logs in to the LM Portal. 2. Navigates to the "My Applications" section and views the application with "Query Raised" status. 3. Reviews Controller's remarks/comments displayed alongside the application. 4. Edits form fields as necessary and uploads/replaces documents. 5. Clicks "Re-submit" after completing all updates. 6. System: a. Validates the updated inputs b. Updates application status to "Resubmitted after Query" c. Routes the application back to the same Controller d. Controller receives notification for re-inspection.
Post - Condition	1. Application is resubmitted with corrections. 2. Status is updated to "Resubmitted after Query". 3. Application is routed back to the same Controller for review.

Validity Checks	1. Only applications in “Query Raised” status are eligible for resubmission. 2. Applicant must: a. View and address Controller’s remarks b. Edit form and/or update at least one document 3. System must: a. Log all resubmission activity (timestamp, user ID, changes) b. Enforce validations on mandatory fields and document formats c. Route resubmitted application back to the original Controller 4. Status must clearly reflect "Resubmitted after Query" to differentiate from fresh applications.
Input	1. Updated form data 2. Re-uploaded/replaced documents 3. Applicant credentials and session
Output	1. Status = "Resubmitted after Query" 2. Controller’s dashboard updated with resubmitted entry 3. Submission timestamp and audit trail 4. Notification sent to Controller
Exception Handling	1. No form edits made: “No changes detected. Please update at least one field or document.” 2. Mandatory fields missing: “All required fields must be completed before resubmission.” 3. Upload failure: “File upload failed. Check format/size and try again.” 4. Resubmission timeout: “Session expired. Log in again to re-submit.” 5. Routing error: “Unable to forward application to Controller. Please contact support.”

3.7.5 Final approval or acknowledgement #FR005

Description	This module facilitates the final approval and acknowledgment issuance process after a successful review of a Director Nomination application. Once the Controller verifies that all queries have been resolved and documentation is complete, the system allows the Controller to approve the nomination, triggering the generation of a digitally signed acknowledgment certificate. This certificate becomes available to the applicant via their dashboard and is accompanied by a notification of approval.
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User	1. Controllers (Centre Legal Metrology) – Approve finalized nominations and trigger certificate generation. 2. Applicants / Company Representatives – Receive acknowledgment and download the nomination certificate. 3. System Admins – Oversee digital signature integration, status updates, and notification services.
Pre - Condition	1. Applicant has responded to all queries (if any), and the application has been re-inspected. 2. The Controller has determined that the application is complete and compliant. 3. Digital certificate template is configured in the system.
Sequence of Operations	1. Controller reviews the final version of the application. 2. If all criteria are met: a. Controller selects "Approve". b. System updates status to "Approved". 3. System generates a digitally signed nomination acknowledgment certificate, including: a. Company Name b. Corporate Identification Number (CIN) c. Director's Name d. Director Identification Number (DIN) e. Issuance Date f. Controller's Digital Signature 4. Certificate is stored securely and linked to the application ID. 5. Applicant receives: a. Email/SMS notification of approval b. Access to download certificate from dashboard under "My Applications" 6. System logs the approval and issuance in the audit trail.
Post - Condition	1. Application status is updated to "Approved". 2. A digitally signed nomination acknowledgment certificate is generated and archived. 3. Applicant receives notification and gains access to the certificate from the portal.

Validity Checks	1. Application must be in a valid state (i.e., all queries resolved, and no pending fields). 2. Only authorized Controllers can approve and generate certificates. 3. Certificate must: a. Contain mandatory fields (Company, Director details, Issuance Date) b. Be signed using a valid digital signature c. Be linked to the application and user profile 4. System must: a. Log timestamp, Controller ID, and certificate ID b. Notify the applicant c. Archive the certificate securely for regulatory reference
Input	1. Final Controller approval 2. Verified applicant and company data 3. Digital signature key/token (if used by Controller)
Output	1. Application status: "Approved" 2. Digitally signed PDF certificate 3. Certificate available in applicant's portal 4. Notification sent to applicant 5. Archived certificate and action log
Exception Handling	1. Digital signature error: "Certificate could not be signed. Please retry or contact admin." 2. Template not loaded: "Certificate template missing or misconfigured." 3. System error during status update: "Approval saved but status not updated. Please retry." 4. Notification failure: "Approval registered, but notification could not be sent. Please notify manually."

3.7.6 post-processing and audit trail #FR006

Description	This module ensures robust post-processing management and traceability for the Nomination of Directors of Companies applications. The system maintains a comprehensive, timestamped audit trail of all application lifecycle events, provides dashboards for monitoring by Controllers, triggers escalation alerts for unattended applications, and allows applicants to track progress in real time. Additionally, the system archives approved or rejected applications and documents after a defined retention period for regulatory compliance.
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User	1. Controllers (Centre and State) – Monitor and manage status-based reports, access audit logs. 2. Applicants / Company Representatives – Track real-time application status and download documents. 3. System Admins – Configure audit trail, archival rules, and escalation parameters.
Pre - Condition	1. Application has been submitted and is actively moving through the review cycle. 2. User roles and privileges are defined in the system. 3. Audit and archival subsystems are properly configured.
Sequence of Operations	1. System logs each event (submission, review, query, resubmission, approval, download) with: a. Timestamp b. User ID c. Action taken 2. Controllers access dashboards showing: a. Pending, Approved, and Rejected applications b. Filters by status, date, and applicant 3. If an application remains in pending status beyond a configured threshold (e.g., 15 days): a. System sends escalation alerts via portal/email to the responsible officer 4. Applicant logs in to track the real-time status of their application (e.g., Under Review, Query Raised, Approved). 5. After a defined period (e.g., 1-year post-closure): a. The system archives applications and attached documents b. Archived files are stored securely and flagged as non-editable/read-only 6. Authorized users can access audit logs and archive records as needed.
Post - Condition	1. Audit trail is generated and accessible for regulatory and administrative purposes. 2. Escalation alerts are triggered for inaction or delays. 3. Archived applications and documents are stored securely and marked as read-only.

Validity Checks	Every system or user action must be: - Timestamped - Linked to user ID - Logged in an immutable audit trail Status changes must: - Be reflected instantly in the applicant dashboard - Be included in Controller's report views Escalation triggers must be: - Configurable (e.g., 15-day limit) - Sent only to authorized roles Archived records must: - Be retained securely for at least 1 year (or per policy) - Be non-editable and accessible only by authorized users - Audit logs and archival data must be exportable in standard formats (PDF, CSV) for audits
Input	- Application events (form actions, status updates) - User activity logs - Configurable parameters (retention period, escalation time)
Output	- Time-stamped audit trail entries - Dashboard reports for Controllers - Escalation alerts (email/system) - Application status view for applicants - Archived applications and documents
Exception Handling	Audit log failure: "System unable to record event. Action aborted. Please retry or contact admin." Escalation not triggered: "Escalation alert not sent due to configuration error. Review system settings." Archive access denied: "You do not have permission to access this archived file." Real-time status not updating: "Status refresh failed. Please reload or check back later." Dashboard loading error: "Unable to fetch reports. Check filters or contact support."

3.7.7 Use cases (Nomination of Directorates of companies)

Field Name	Description
Use Case ID	eMaap_NDC_009
Use Case Name	Nomination of Directorates of companies
Reference	FRS Number: FR001 to FR006
Created By (Date)	eMaap functional team, (16-06-2025)

Last Updated (Date)	16-06-2025
Primary Actor	Applicant / Business User
Secondary Actor	Legal Metrology Controller, System Admin
Trigger (Activating Use Case)	Business logs in to nominate a new director
Inter Linkages	User Registration, Document Upload Module, Notification System
Normal Event Flow	Typical sequence of steps: 1. Applicant logs in 2. Fills nomination form 3. Uploads documents 4. Submits to Controller 5. Controller reviews 6. Application approved or query raised.
Alternate Event Flow	Steps taken when issues arise: 1. Controller raises query 2. Applicant modifies & resubmits form 3. Re-inspection happens 4. Approval or rejection based on corrections.
Assumptions	Preconditions or expectations: 1. Applicant is a registered user 2. Director details and documents are valid 3. Legal Metrology Controller has access to verify documents.
Additional Notes	Any special notes, such as: 1. Acknowledgement is digitally available 2. Nomination is subject to Legal Metrology Division's jurisdiction. 3. Notifications are system-generated via portal dashboard/email/SMS.

3.8 End-to-End Processing by user

This module addresses the backend processing of all the services provided by eMaap such as registration, licenses, verification etc. The module will be configured to meet the requirements of all 36 states and Centre, incorporating a comprehensive approval workflow that aligns with the hierarchical structure of each state and the Centre. Additionally, the module will integrate APIs from each state, which will be gathered by the Legal Metrology Department (LMD) and displayed on the eMaap portal for seamless access and operation.

3.8.1 User Interface on eMaap Portal #FR001

Description	This module enables end users to access and submit service requests through the eMaap Portal. It provides a unified user interface (UI) that supports the complete flow from service selection, form filling, document upload, and submission to acknowledgment generation. It ensures ease of navigation, real-time feedback, and proper system logging for all user-initiated processes related to services such as registration, model approval, importer license, and more.
User	1. Applicants / Business Users – Access the portal to apply for services like registration, approval, or licensing. 2. Department Officials – Receive and process user-submitted applications. 3. System Admins – Maintain UI modules, ensure functionality and form configurations.
Pre - Condition	1. The user has a registered account on the eMaap Portal. 2. The selected service is active and available in the system. 3. The user has access to required documentation for upload.
Sequence of Operations	1. User logs in to the eMaap Portal using valid credentials. 2. From the dashboard, user selects a desired service: a. Registration b. Model Approval c. Importer License d. Certificate Nomination, etc. 3. System loads the corresponding application form. 4. User fills in all mandatory and optional fields as per the selected service. 5. Uploads supporting documents (PDF, JPEG, PNG) as specified. 6. System performs real-time validation of entries and files. 7. User clicks Submit. 8. System: a. Saves the application b. Generates a unique application reference number c. Updates status to "Submitted" d. Sends an acknowledgment notification (via email/SMS) 9. The application appears in the user dashboard under "My Applications".
Post - Condition	1. Application is successfully submitted and recorded in the system. 2. An acknowledgment with a unique application reference number is generated. 3. Application status becomes "Submitted" and is available in the user dashboard.

Validity Checks	1. User must be authenticated to access the portal. 2. Service-specific forms must: a. Validate mandatory fields b. Enforce file format and size limits 3. Application cannot be submitted unless: a. All required fields are completed b. All required documents are uploaded 4. System must: a. Assign a unique reference ID b. Record the submission timestamp c. Update the status to "Submitted" d. Display the application in user dashboard e. Trigger acknowledgment notification with application ID
Input	1. Login credentials 2. Service selection 3. Form entries 4. Uploaded files (e.g., PAN, GST, certificates)
Output	1. Validation prompts and form feedback 2. Application reference number 3. Acknowledgment screen and notification 4. Status update in user dashboard
Exception Handling	1. Invalid login: "Incorrect username or password. Try again." 2. Form validation error: "Please fill all mandatory fields and correct errors." 3. File upload error: "Unsupported format or file too large (max 5 MB)." 4. Service unavailable: "The selected service is currently disabled." 5. Submission failure: "Application could not be submitted due to server error. Please retry."

3.8.2 Backend processing workflow engine #FR002

Description	This functionality defines the behavior of the backend processing workflow engine that automatically handles routing and validation of submitted applications on the eMaap Portal. After a user submits a service request (e.g., license, registration, model approval), the system forwards the application to the workflow engine, which identifies the applicant's State/UT, applies state-specific validation rules, and checks for mandatory completeness, document validity, and duplicate entries before progressing the request to the appropriate departmental reviewer.
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User	1. System (Workflow Engine) – Executes backend validations and routing. 2. State/UT Controllers and Inspectors – Receive routed applications for further action. 3. Applicants – Indirectly affected by validation outcomes. 4. System Admins – Configure state-specific workflows, rules, and routing logic.
Pre - Condition	1. A user has submitted a valid application via the eMaap Portal. 2. Backend rules and hierarchies are pre-configured for each State/UT. 3. Workflow engine is online and integrated with the submission system.
Sequence of Operations	1. Application is submitted by the user and enters the workflow engine. 2. System identifies the applicant's State/UT using location or service mapping. 3. Workflow engine determines the routing hierarchy (e.g., State Controller → Inspector). 4. Validation process runs: a. Mandatory field check b. Document validity (format, presence, size) c. Duplicate detection (based on applicant ID, service type, CIN/GST) 5. Workflow engine applies state-specific validation rules stored in backend. 6. Based on validation results: a. If successful: Application is routed to the assigned State/UT authority. b. If failed: Application is flagged, and system updates status (e.g., "Validation Failed – Incomplete"). 7. If routed, a task is created in the receiving officer's dashboard.
Post - Condition	1. Application is either: a. Successfully routed to the correct State/UT authority, or b. Flagged for missing fields, invalid documents, or duplicates. 2. System updates application status accordingly (e.g., "Routed," "Validation Failed").

Validity Checks	Workflow engine must: - Accurately identify State/UT from applicant metadata - Match routing configuration to the state's hierarchy Enforce: - Field-level completeness - Document presence and format - Uniqueness of the application (no duplicate by applicant/service type) All validations must: - Be logged with timestamp and application ID - Update application status on pass/failure - Be traceable in audit logs - Only validated and routed applications can proceed to the next stage of inspection/review
Input	- Submitted application data - Uploaded documents - Applicant's location/state metadata - Configured state rules and hierarchy
Output	- Routing decision and target authority - Validation status messages - Internal logs for processing - Application status update ("Routed", "Flagged")
Exception Handling	Missing or invalid state mapping: "Unable to determine jurisdiction. Contact support." Backend rule misconfiguration: "State-specific rules could not be applied. Escalate to admin." Validation engine error: "System error during validation. Please retry submission." Duplicate application found: "An existing request with similar details already exists." Routing failure: "Application could not be forwarded to authority. Try again later."

3.8.3 Hierarchical Review Process (State/Centre) #FR003

Description	This feature defines the multi-level hierarchical review process for user requests (e.g., license, registration, model approval) submitted through the eMaap Portal. It enables structured processing through designated roles within the State or Central Legal Metrology Department, starting from initial clerical scrutiny to in-depth inspection and final approval/rejection by senior officers. Each step in the workflow includes
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	review, comments, status updates, and conditional progression to the next level.
User	1. Officer 1 – Clerk/Assistant (State/Centre) – Performs initial review of form and documents. 2. Officer 2 – Inspector/Field Officer – Conducts technical/documental verification or site visits. 3. Officer 3 – Controller/Director LM (State or Centre) – Takes final decision on approval, rejection, or clarification. 4. Applicants – Indirectly affected through query resolution and status updates. 5. System Admins – Manage user roles, workflow routing, and escalation configurations.
Pre - Condition	1. Application has passed initial backend validations and been routed to the appropriate State/Centre office. 2. Officer hierarchy and routing configuration is properly set in the system. 3. Officers have role-based access to their respective dashboard modules.
Sequence of Operations	1. Officer 1 (Clerk/Assistant): a. Receives application in dashboard. b. Reviews submitted form and uploaded documents. c. Adds initial remarks or flags missing data. d. Forwards to Officer 2 (Inspector). 2. Officer 2 (Inspector): a. Reviews forwarded application. b. Conducts deeper verification (e.g., field/site inspection if applicable). c. Uploads inspection report or documents. d. Either forwards to Officer 3 or raises a query to applicant. 3. Officer 3 (Controller/Director LM): a. Receives the completed file with all lower-level remarks. b. Reviews entire case. c. Takes final action: 1. Approve request (triggers certificate generation). 2. Reject with mandatory reason (application closed). 3. Return for clarification (status = Query Raised, sent back to applicant).
Post - Condition	1. Application is either approved, rejected (with reason), or returned for clarification. 2. Status updates and remarks are logged at each stage. 3. Final outcome is recorded and visible to the applicant.

Validity Checks	Each officer can: - Only access applications assigned to their role and jurisdiction. - Perform only their allowed actions (view, comment, forward, approve, reject). System must: - Enforce mandatory remarks for rejection or query Log every action with: - Officer ID - Timestamp - Action Type - Prevent escalation without mandatory inputs (e.g., forwarding without review) Final action (approval/rejection) must: - Lock the application from further edits - Trigger certificate or rejection notice - Notify the applicant with final status
Input	- Application data and uploaded documents - Remarks/decisions from Officers 1 and 2 - Inspection findings (if any) - Final decision and comments by Officer 3
Output	- Application status updates (In Review, Forwarded, Query Raised, Approved, Rejected) - Notification to applicant - Approval trigger or query escalation - Action history log at each level
Exception Handling	Officer not assigned or unavailable: "No designated reviewer found for this level. Contact admin." Incorrect role attempting action: "You do not have permission to perform this action." Missing remarks: "Remarks required before forwarding or rejecting application." Field report not uploaded: "Inspection result missing. Please upload required documents." System routing error: "Workflow routing failed. Please escalate to system administrator."

3.8.4 Query Handling (If Applicable) #FR004

Description	This Functionality enables the system-driven query management process for applications submitted through the eMaap Portal. If a query is raised by a reviewing officer (at any hierarchical level), the system notifies the applicant and enables them to log in, review the query remarks, make necessary updates, and resubmit the application. Once resubmitted, the workflow resumes from the last point of interruption, ensuring seamless end-to-end tracking and revalidation.
User	1. Applicants / Business Users – View and respond to queries, update applications, and resubmit. 2. Reviewing Officers (Clerk, Inspector, Controller) – Raise queries and re-review resubmissions. 3. System Admins – Monitor query handling logic, status flags, and notifications.
Pre - Condition	1. A query has been raised by a reviewing officer and is recorded in the system. 2. Applicant has been notified (email/SMS/portal alert). 3. The application status is marked as "Query Raised" and remains editable for resubmission.
Sequence of Operations	1. Reviewing officer raises a query in the eMaap backend. 2. System updates the application status to "Query Raised". 3. Applicant receives an email/SMS notification with a summary or link to the portal. 4. Applicant logs in and: a. Opens the application from the dashboard. b. Reviews query remarks from the officer. c. Updates the form fields or uploads additional/corrected documents. 5. Applicant clicks "Re-submit". 6. System action: a. Validates the updated entries b. Updates the status to "Resubmitted After Query" c. Resumes the workflow by forwarding the application back to the officer who raised the query d. Logs the query response and submission time
Post - Condition	1. Application is resubmitted with updated information. 2. Status is updated to "Resubmitted After Query". 3. The application is re-routed to the appropriate reviewing officer.

Validity Checks	Queries must: - Include officer remarks - Be associated with a specific application and reviewer Applicant must: - Only be able to edit applications in "Query Raised" status - Be prevented from submitting until all required fields/documents are corrected System must: - Log all resubmission actions - Retain full query history with timestamps - Ensure workflow resumes from the same officer who raised the query - Notify relevant parties upon resubmission
Input	- Query remarks from officer - Updated form data and/or new documents - Applicant login and submission
Output	- Updated application record - Status update ("Resubmitted After Query") - Officer dashboard updated with resubmission - System logs with timestamp and changes - Notification to officer
Exception Handling	Applicant does not respond: "Auto-closure alert after deadline. No response received." Invalid document format during upload: "File format not supported. Please upload PDF/JPEG/PNG." Incomplete resubmission: "All mandatory fields must be updated before resubmission." Routing failure after resubmission: "Unable to forward to reviewer. Contact support." Notification delivery failure: "Query alert not sent. Will retry via alternate channel."

3.8.5 Certification Generation & Delivery #FR005

Description	This Functionality handles the automated generation, digital signing, and delivery of certificates or licenses after final approval of an application. Once the reviewing officer (at the Centre or State level) approves a request, the system generates a digitally signed certificate, makes it accessible via the applicant's dashboard, and sends a notification through email/SMS. Certificates are secured with a digital signature key (Centre or State authority) to ensure authenticity.
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User	1. Applicants / Business Users – Receive and download approved certificates from the portal. 2. Controllers / Directors (Centre and States) – Finalize approvals, authorize certificate generation. 3. System Admins – Manage digital certificate templates, signature integration, and document archival.
Pre - Condition	1. Application has completed the full review workflow. 2. Approval has been granted by the authorized Controller/Director. 3. The applicant's identity and request details are verified. 4. Digital signature infrastructure (Centre/State) is available and configured.
Sequence of Operations	1. Final approval is recorded by the authorized officer. 2. The system: a. Fetches approved application data b. Populates predefined certificate/license template 3. Applies digital signature using Centre or State cryptographic key. 4. Generates a PDF certificate/license with: a. Applicant name b. Reference/Application number c. Approval date d. Certificate/license number e. Validity period 5. Saves the certificate in a secure repository. 6. Makes certificate downloadable from applicant dashboard. 7. Sends notification via email/SMS with: a. Certificate download link b. Confirmation of approval 8. Updates application status to "Certificate Issued". 9. Logs action in audit trail.
Post - Condition	1. A digitally signed certificate or license is generated and stored. 2. The certificate is made downloadable in the applicant's dashboard. 3. Notification is sent to the applicant with a link to access the certificate. 4. The certificate is archived and logged in the audit trail.

Validity Checks	1. Only approved applications may trigger certificate generation. 2. Certificate must: a. Include unique certificate/license number b. Be signed with valid Centre/State digital signature key c. Match the approved applicant and request data 3. System must: a. Log certificate generation with timestamp and approver ID b. Archive certificates securely c. Make the document available only to the authenticated applicant - d.) Ensure notification is delivered to registered contact information
Input	1. Final approval trigger 2. Application data (name, ID, validity) 3. Certificate template 4. Digital signature key (Centre/State)
Output	1. Digitally signed certificate (PDF) 2. Application status = "Certificate Issued" 3. Email/SMS notification to applicant 4. Dashboard update for download access 5. Audit log entry
Exception Handling	1. Digital signature error: "Unable to apply digital signature. Please retry or contact admin." 2. Template not found: "Certificate template missing. Cannot generate certificate." 3. Notification failure: "Certificate generated, but alert could not be sent. Check communication settings." 4. PDF rendering issue: "Error generating certificate PDF. Contact technical support." 5. File access error: "Certificate could not be saved or displayed. Please refresh or retry."

3.8.6 Post-Processing & Monitoring #FR006

Description	This functionality supports ongoing monitoring, auditing, and reporting of user requests after submission and processing within the eMaap portal. It ensures that every action taken on an application is logged in the backend audit trail, provides officers and administrators with dynamic reporting tools, and allows applicants to track their application status in real time. The module strengthens transparency, enables
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	operational oversight, and facilitates performance reviews across departments and jurisdictions.
User	1. Applicants / Business Users – Track real-time status of applications via dashboard. 2. Department Officers / Reviewers – Filter, manage, and review submitted applications by criteria. 3. System Administrators – Access full audit logs and system-wide performance reports
Pre - Condition	1. Application must be submitted and active in the system. 2. Role-based access is configured for officers/admins. 3. Audit logging and reporting engines are active.
Sequence of Operations	1. Audit Trail Logging: a. Each event (submission, query, approval, document upload, status change) is automatically logged with: b. Timestamp c. User ID d. Action type e. Application reference number 2. Monitoring by Officers/Admins: a. Officers access a dashboard or reporting module. b. Can filter applications by: 1. Service type 2. Application status (pending, approved, rejected) 3. State/UT 4. Date range 5. Assigned reviewer c. Can generate daily, weekly, or custom reports. 3. Applicant Tracking: a. Applicant logs into portal. b. Views status history of their application with: 1. Last action taken 2. Pending level (e.g., under review, query raised) 3. Approval timeline (if available) 4. System Notifications (optional): a. Admins or officers receive alerts for specific trends (e.g., high pending load).

Post - Condition	1. Logs are captured for all major events. 2. Officers and admins can access visual or tabular reports. 3. Applicants receive up-to-date application progress on their dashboard.
Validity Checks	1. Every significant user/system action must be: a. Logged with a timestamp b. Tagged to a user ID and application ID 2. Report filters must only access applications within the officer's role or jurisdiction. 3. Applicants should only see their own application history and status. 4. Reports must include: a. Count summaries (e.g., daily approvals) b. Exportable data (CSV/PDF) c. Date/time of report generation
Input	1. User actions (submit, approve, reject, comment) 2. Application metadata (state, type, officer assignment) 3. Report parameters (filters, date ranges)
Output	1. Audit logs with detailed entries 2. Officer reports (tabular/exportable) 3. Applicant status tracker view 4. Visual dashboards for monitoring workloads
Exception Handling	1. Audit log failure: "Error recording event. Check system logging module." 2. Report generation timeout: "Report is too large or parameters invalid. Refine filters." 3. Applicant status not loading: "Real-time tracking unavailable. Try again later." 4. Filter misconfiguration: "Invalid report filters. Reset and try again."

3.8.7 Use Cases (End-to-End Processing)

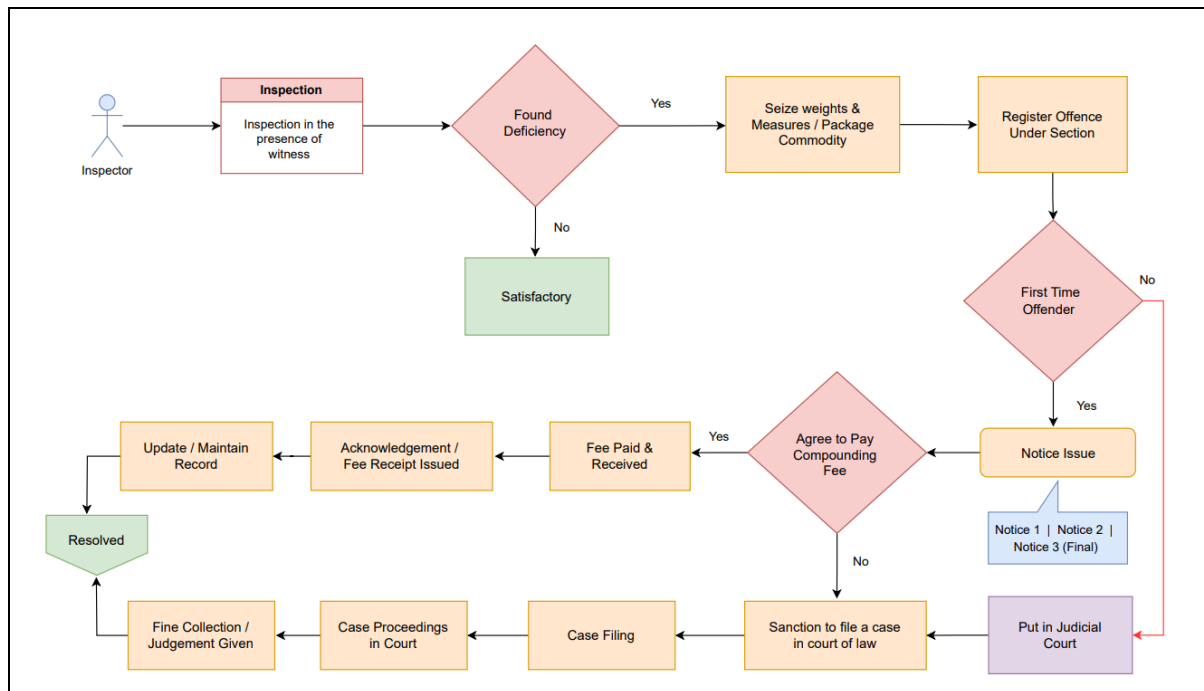
Field Name	Description
Use Case ID	eMaap_EEP_010
Use Case Name	End to End Processing
Reference	FRS Number: FR001 to FR006
Created By (Date)	eMaap functional team, (16-06-2025)
Last Updated (Date)	16-06-2025
Primary Actor	eMaap Backend System / LMD Backend Operator.
Secondary Actor	State API Systems, Controllers, Admins.
Trigger (Activating Use Case)	Registration, license, verification via eMaap portal.

Inter Linkages	User Dashboard, State Approval Workflows, API Integration Layer, Notification System.
Normal Event Flow	Typical sequence of backend actions: 1. Request submission by user 2. Routed to appropriate state/center backend 3. Approval workflow initiated 4. API call made 5. Status updated and displayed.
Alternate Event Flow	If a state's API is unresponsive or fails: 1. Manual intervention logged 2. Notification sent to backend admin 3. Retry or escalate depending on configuration.
Frequency of Use	Frequently used as all user-initiated services pass through backend processing.
Assumptions	All state APIs are functional and accessible. Approval hierarchy is properly mapped for each state and Centre.
Additional Notes	Module supports centralized monitoring. All logs are stored for audit purposes. Integrated notification for real-time status updates.

3.9 Real Time Tracking of Enforcement activities

This module will be designed to ensure effective regulatory oversight and streamlined operations through real time tracking and monitoring. This module will allow the Legal Metrology Officers will undertake enforcement activities such as case logging, conducting inspections, raids and seizures, imposing penalties under weights & measures laws, generating tickets against compoundable fee etc. Geo-tagging will facilitate precise location identification during inspections and raids, while case logging will allow for systematic recording of enforcement details. Ticket generation will support the creation and management of penalty or violation notices, ensuring accountability and compliance with legal standards. Together, these capabilities will strengthen regulatory oversight and enhance operational transparency.

Through this module, businesses will get real time updates via SMS/email to monitor penalties imposed under these laws, track inspections, raids, and seizures. The businesses will also be able to log appeals against orders issued by inspectors, assistant controllers, deputy controllers, or controllers.



The enforcement process illustrated in the provided image outlines a structured approach to handling deficiencies found during official inspections. The process begins with an inspector conducting an inspection in the presence of a witness. This ensures transparency and accountability in the initial stages of assessment. Following the inspection, the next critical step is determining whether a deficiency has been found. If the inspection result is satisfactory, the case is considered resolved with no further action required.

However, if a deficiency is detected, the process escalates. The inspector is required to seize the weights, measures, or packaged commodities that are non-compliant. This is followed by registering the offence under the relevant legal section, marking the beginning of formal enforcement procedures. The system then assesses whether the violator is a first-time offender. If the offender has previous infractions, the matter is immediately taken to judicial court to avoid leniency for repeat violations.

For first-time offenders, a notice is issued, which may include up to three stages: Notice 1, Notice 2, and Notice 3 (Final). These progressive notices aim to provide the offender an opportunity to comply or respond before harsher measures are taken. If the offender responds positively and agrees to resolve the issue, the system then checks if they are willing to pay a compounding fee, which is a form of settlement to avoid further legal action.

If the offender agrees to pay the compounding fee, and the fee is received, an acknowledgment or fee receipt is issued. The case details are then updated or maintained in the record system, and the case is marked as resolved. This approach allows for an efficient and less resource-intensive resolution, particularly beneficial for minor or first-time violations.

On the other hand, if the offender refuses to pay the compounding fee, the inspector must seek a sanction to file a case in a court of law. Once sanctioned, the case proceeds with formal filing, leading to court proceedings. Depending on the outcome, a fine is collected, or a judgment is issued, ultimately resolving the case. This outcome, too, is entered into the records to maintain accurate data for future reference.

This enforcement workflow balances administrative resolution and legal action. It prioritizes compliance and education for first-time violators, while ensuring repeated or non-compliant offenders face legal consequences. It emphasizes documentation, accountability, and progressive enforcement, thereby reducing unnecessary court burden and promoting efficient governance. The flow also underscores the importance of transparency and structured decision-making at each stage, creating a robust mechanism to uphold standards and ensure fairness in regulatory enforcement.

3.9.1 User Management & Authentication #FR001

Description	This functionality governs the secure access and identity management features of the enforcement system within the eMaap Portal. It provides role-based access control (RBAC), user profile maintenance, secure login/logout logging, and authentication mechanisms including password resets via OTP (email/SMS). Administrative users can manage user roles and permissions via the privilege editor to ensure only authorized personnel can perform specific enforcement functions.
User	1. Inspectors – Field users conducting enforcement activities. 2. Supervisors – Monitor and manage field-level enforcement data. 3. Legal Officers – Handle notices, hearings, and legal reviews. 4. Admins (Department/IT) – Manage user accounts, role privileges, and access control.
Pre - Condition	1. Users must be registered in the eMaap system with an assigned role. 2. Authentication setup (password, contact details) must be configured. 3. Admin must define and approve roles and privileges before user activation.

Sequence of Operations	1. User Registration and Role Assignment (Admin): a. Admin creates new user or edits existing profile. b. Assigns role: Inspector / Supervisor / Legal Officer / Admin. c. Sets up initial login credentials. 2. User Login: a. User enters credentials (username + password). b. System authenticates and loads role-specific dashboard. 3. Password Reset: a. User requests password reset. b. System sends OTP via registered email/SMS. c. Upon verification, user sets a new password. 4. User Profile Management: a. User can view/edit profile (name, contact, department). b. Admins can edit roles and department affiliations. 5. Access Logging and Auditing: a. Every login/logout is logged with: 1. Timestamp 2. User ID 3. IP address (if applicable) 6. Role Privilege Editing: a. Admin accesses role management console. b. Configures or updates role-based access privileges.
Post - Condition	1. Users can log in and access system features based on assigned role. 2. User activities (login/logout) are logged. 3. Profile updates and password resets are recorded with timestamps.
Validity Checks	1. Only authenticated users may access the system. 2. Role-based access control must: a. Enforce restrictions per user role b. Prevent privilege escalation 3. OTP for password reset must: a. Be valid only for a short duration (e.g., 5–10 minutes) b. Be linked to the registered email/phone 4. All access events must be: a. Logged with timestamps b. Available for audit by admin 5. Role privilege changes must be: a. Allowed only for admin users b. Logged in the system with user ID and timestamp

Input	1. User credentials 2. Role assignment 3. OTP (for password reset) 4. Profile information
Output	1. Authenticated session 2. Role-based dashboard and feature access 3. Password reset confirmation 4. Login/logout logs 5. Role update audit entries
Exception Handling	1. Invalid credentials: "Incorrect username or password." 2. OTP expired or invalid: "OTP is invalid or has expired. Please request a new one." 3. Unauthorized access attempt: "You do not have permission to access this module." 4. Role edits denied (non-admin): "Only admins can modify user privileges." 5. System error on login: "Login failed due to server issue. Try again later."

3.9.2 Inspection Initiation #FR002

Description	This functionality enables field inspectors to initiate and record real-time inspection activities through the eMaap portal. It provides a dedicated interface to begin a new inspection, capturing vital information such as location, date/time, inspector details, category of inspection, evidence, and witness data. The system supports GPS-based tracking, photo/document attachments, and remarks, and allows users to save drafts or submit finalized reports. A unique Inspection ID is generated for every initiated inspection for tracking and audit purposes.
User	1. Field Inspectors – Initiate inspections, upload data, and submit reports. 2. Supervisors/Controllers – Review inspection logs and monitor field activity. 3. System Admins – Configure inspection categories, monitor logs, and manage inspection IDs.
Pre - Condition	1. Inspector is logged in with valid credentials. 2. The inspection initiation module is accessible based on the user's role and jurisdiction. 3. GPS is enabled or manual location entry is permitted.

Sequence of Operations	1. Inspector accesses the "Initiate Inspection" module. 2. System loads a structured inspection form with the following: a. Auto-generated Inspection ID b. Inspector's name (auto-filled from login) c. Date and time (auto-captured) d. GPS location (auto-captured) with manual override option 3. Inspector selects inspection category from a dropdown: a. Weights b. Measures c. Packaged Commodities, etc. 4. Inputs additional details: a. Witness details (name, mobile) b. Remarks or notes 5. Attaches photos or documents as evidence (PDF, JPEG, PNG). 6. Options to: a. Save as Draft b. Submit for final storage and processing 7. Upon submission: a. Data is validated b. Stored in the backend database c. Marked with status "Inspection Initiated"
Post - Condition	1. A new inspection record is created and stored in the database. 2. The inspection is assigned a unique Inspection ID. 3. Data can be reviewed or edited before final submission (if saved as draft).
Validity Checks	1. Inspection must: a. Be associated with a logged-in inspector b. Have a system-generated unique Inspection ID c. Capture either GPS or manually entered location d. Include at least one inspection category 2. Mandatory fields: a. Inspector name b. Location c. Date/time d. Category 3. Attachments (if any) must: a. Be in allowed formats b. Pass size and virus scan validations c. Drafts must be editable until submitted d. All actions must be logged with timestamps for audit tracking

Input	1. Inspector's name (from session) 2. GPS/manual location data 3. Inspection category 4. Date/time 5. Witness information 6. Notes and attachments (photos/documents)
Output	1. Unique Inspection ID 2. Inspection status ("Draft" or "Initiated") 3. Entry stored in database 4. Confirmation message to inspector 5. Record visible in inspector dashboard
Exception Handling	1. GPS unavailable: "GPS location not accessible. Please enter location manually." 2. Missing mandatory fields: "Please fill in all required fields before submitting." 3. Attachment format error: "Only PDF, JPEG, or PNG files up to 5MB allowed." 4. Save failure: "Unable to save draft. Check internet connection or retry." 5. Submission error: "Inspection data could not be submitted. Contact admin."

3.9.3 Evaluation of Inspection Findings #FR003

Description	This functionality supports the structured evaluation and documentation of inspection findings during enforcement activities. Inspectors are required to assess whether any regulatory deficiencies were identified. The system provides an intuitive interface to mark inspections as Satisfactory or Deficient, collect detailed information about violations, and link them to relevant Legal Metrology clauses, with the ability to attach supporting evidence and digitally confirm the report.
User	1. Field Inspectors – Evaluate findings, log deficiencies, and confirm inspections. 2. Supervisors / Controllers – Review evaluation summaries and take further legal or administrative action. 3. System Admins – Manage clause mappings, evidence policies, and evaluation form logic.
Pre - Condition	1. An inspection has been initiated and recorded in the system. 2. Inspector is logged in with valid credentials. 3. Inspection data is saved or ready for evaluation.

Sequence of Operations	- Inspector accesses the evaluation form linked to an initiated inspection. Selects checkbox: “Deficiency Found – Yes/No” Based on selection: If No: System marks inspection as “Satisfactory” Auto-stamps decision time If Yes: Additional fields appear for: Deficiency type Quantity Severity level Linked regulatory clause (dropdown or auto-suggest) Inspector attaches supporting documents/photos - Inspector completes remarks and reviews entered data. - Inspector digitally signs or confirms the evaluation. System logs: - Status: “Deficiency Noted” or “Satisfactory” - Timestamp of evaluation - Inspector ID and digital signature - Evaluation record is stored in the backend and visible in dashboards.
Post - Condition	- Inspection is marked as either Satisfactory or Deficient. - Deficiency data, evidence, and references are recorded and logged. - Digital signature or confirmation is captured for audit integrity.
Validity Checks	Evaluation must: - Include one mandatory checkbox selection (Yes/No) - Auto-generate timestamp for the decision If “Yes” is selected: - Deficiency type, quantity, severity, and clause must be completed - At least one attachment must be uploaded (if required by configuration) - Only the assigned inspector may submit the evaluation All evaluations must be: - Digitally signed or confirmed - Logged in the audit trail with timestamp and inspector ID

Input	1. Deficiency checkbox selection 2. Details of deficiency (type, quantity, severity) 3. Linked regulatory clause 4. Attachments (evidence) 5. Inspector's digital signature
Output	1. Inspection status ("Satisfactory" or "Deficiency Noted") 2. Timestamped evaluation record 3. Documented deficiency log 4. Linked clause references 5. Updated inspection log
Exception Handling	1. No selection made: "Please specify whether a deficiency was found." 2. Deficiency marked but details missing: "Enter all required fields when deficiency is marked as Yes." 3. Clause linkage missing: "Select at least one regulatory clause related to the violation." 4. Attachment failure: "Evidence file could not be uploaded. Please retry." 5. Digital signature not applied: "Evaluation cannot be completed without inspector confirmation."

3.9.4 Seizure Process #FR004

Description	This functionality facilitates the secure and traceable initiation and documentation of item seizures during legal metrology enforcement inspections. Through a structured interface, inspectors can capture details of seized items, generate official seizure documents (PDF), and link them to the related inspection record. The process includes dynamic item entry, witness acknowledgment, and barcode/QR code generation for inventory management and legal traceability.
User	1. Field Inspectors – Initiate and complete the seizure process during inspections. 2. Supervisors / Controllers – Review seizure records and documents. 3. System Admins – Configure item categories, manage barcode logic, and monitor data storage.
Pre - Condition	1. An inspection must already be initiated and recorded. 2. Inspector is logged in and has access to the inspection record. 3. Seizure action is justified based on inspection findings (e.g., non-compliance).

Sequence of Operations	- Inspector clicks “Initiate Seizure” on the inspection interface. System opens a Seizure Form: Fields for: - Item name - Quantity - Serial numbers (if applicable) - Remarks - Dynamic “Add More Items” function - Inspector inputs witness details and captures their acknowledgment (signature or checkbox). System auto-generates: - Unique Seizure ID - Barcode/QR code per item (if configured) - Inspector reviews data and clicks Submit. System: - Links seizure to the inspection ID - Stores data in the seizure record table Generates and stores a Seizure Document (PDF) including: - Item list - Witness info - Seizure ID and codes - Document becomes accessible to both inspector and supervisor via dashboard.
Post - Condition	- Seizure is documented, assigned a unique ID, and linked to the inspection. - PDF seizure report is generated and stored. - All seized items are logged with barcode/QR codes for traceability.
Validity Checks	Seizure can only be initiated if: - Related inspection is in progress or completed - Inspector is authenticated Each seizure: - Must have a unique Seizure ID - Should include at least one seized item entry - Must capture witness acknowledgment PDF report must: - Include item summary, Seizure ID, inspection reference, timestamp - Be securely stored and retrievable - All actions (initiation, edits, submission) must be logged in audit trail with user ID and timestamp

Input	1. Item details (name, quantity, serial number) 2. Witness details and acknowledgment 3. Inspector's remarks 4. Related inspection ID
Output	1. Unique Seizure ID 2. Barcode/QR codes (optional) 3. PDF seizure report 4. Seizure record linked to inspection 5. Dashboard update for seizure history
Exception Handling	1. Item data missing: "Please complete item name and quantity before adding." 2. Witness not acknowledged: "Seizure cannot proceed without witness confirmation." 3. Barcode generation error: "Barcode service failed. Retry or contact admin." 4. Document generation error: "Unable to generate seizure report. Try again later." 5. Linking failure: "Seizure could not be linked to inspection. Verify Inspection ID."

3.9.5 Offense Registration #FR005

Description	This Functionality enables inspectors or enforcement officers to formally register offenses identified during inspections. The system provides a user interface to select offense types, associate legal sections, auto-fill relevant descriptions, and generate a unique Offense Registration Number (ORN). Offenses are logged against a specific Inspection ID and tracked through a lifecycle status: Filed, In Process, or Closed.
User	1. Field Inspectors / Enforcement Officers – Register offenses during or after inspections. 2. Supervisors / Legal Officers – Review offense logs and initiate legal action if necessary. 3. System Admins – Maintain legal section master list, offense codes, and status tracking rules.
Pre - Condition	1. A valid Inspection ID exists. 2. The inspection has resulted in a non-compliance or violation. 3. The user is authenticated and authorized to register offenses.

Sequence of Operations	1. User accesses the Inspection Record and clicks on “Register Offense”. 2. A registration form appears with fields: a. Offense Type (dropdown) b. Legal Section (dropdown from master list) c. Auto-populated offense description (based on section) d. Remarks/Additional Details (optional) 3. Upon form completion: a. System generates a unique Offense Registration Number (ORN) b. Offense is saved and linked to the Inspection ID c. Default status = "Filed" 4. Offense record is visible in: a. Inspector’s dashboard b. Supervisor’s offense tracking module 5. Status can be updated to: a. In Process (e.g., during legal review) b. Closed (e.g., fine paid, court action completed)
Post - Condition	1. An offense is registered and saved in the system under a unique Offense Registration Number (ORN). 2. The offense is linked to the corresponding Inspection ID. 3. Status of the offense is tracked through predefined stages.
Validity Checks	1. Each offense must be: a. Linked to a valid Inspection ID b. Assigned a unique Offense Registration Number (ORN) c. Associated with a predefined legal section from the master list 2. Auto-filled offense descriptions must match the legal section selected 3. System must: a. Enforce mandatory fields before submission b. Prevent duplicate entries for the same offense/inspection c. Track and allow updates to offense status - d.) Record all actions in the audit trail with timestamp and user ID
Input	1. Offense Type 2. Legal Section (from dropdown) 3. Inspector’s remarks 4. Inspection ID

Output	1. Unique Offense Registration Number (ORN) 2. Auto-filled offense description 3. Offense status ("Filed", "In Process", "Closed") 4. Linked offense-inspection log 5. Dashboard display and audit log entry
Exception Handling	1. No offense type selected: "Please select an offense type before proceeding." 2. Invalid or missing legal section: "Please choose a valid legal section from the list." 3. Inspection ID not linked: "No active inspection found. Cannot register offense." 4. System failure during save: "Unable to register offense. Please try again later." 5. Duplicate offense entry: "This offense is already registered under the selected inspection."

3.9.6 Offender Profile & Validation #FR006

Description	This functionality facilitates the identification, validation, and classification of offenders involved in enforcement violations. Through an integrated search and verification interface, enforcement officers can locate existing offender records by business name, ID, or contact number, review their violation history, and determine whether they are first-time or repeat offenders. The module also enables the creation of a new profile if the offender is not found and supports optional integration with external government databases for identity confirmation.
User	1. Inspectors / Enforcement Officers – Search, validate, and register offender profiles. 2. Supervisors / Legal Officers – Review offender history and determine repeat offenses. 3. System Admins – Configure database integrations and maintain offender search indexes.
Pre - Condition	1. User is logged in with the necessary role and permissions. 2. The offender search interface is accessible as part of an ongoing inspection or offense registration. 3. Offender data (e.g., name, ID, contact) is available from the inspection record.

Sequence of Operations	- Inspector accesses “Offender Lookup” during inspection or offense registration. Inputs one or more of the following: - Business name - Business/Offender ID (e.g., PAN, GSTIN, CIN) - Phone number System searches the local offender database: If a match is found: - Display offender profile - Show violation history (date, offense type, resolution) - Automatically flag as “Repeat Offender” If no match: - Flag as “First-Time Offender” - Option to create new offender profile If integrated: - System queries external government database (e.g., GST, MCA, Digi Locker) for identity validation - Displays verification results or links Upon creating or selecting the offender: - Link the profile to the current inspection and offense record - Update dashboard and logs accordingly
Post - Condition	Offender is either: - Matched and flagged as “Repeat Offender” with violation history displayed. - Not found and flagged as “First-Time Offender”. - Newly registered with a unique offender profile ID.
Validity Checks	- Offender search must require at least one valid parameter (name, ID, or phone). Profiles must be flagged correctly: “Repeat Offender” if past violations exist “First-Time Offender” if no prior record is found New offender profiles must include: - Name, contact info, and unique ID System must: - Log all search queries and outcomes - Prevent duplicate record creation - Securely link offender profiles to inspection and offense logs - Maintain audit trail with timestamps and user IDs
Input	- Offender identifiers: Name, ID, contact number - Inspector query and selection

Output	1. Matched offender profile (if found) 2. Past violations log 3. Offender status: "First-Time" or "Repeat" 4. New offender ID (if created) 5. System linkage to inspection/offense record
Exception Handling	1. Invalid or incomplete search input: "Please enter at least one valid search parameter." 2. Database connection error: "Unable to retrieve offender data. Try again or contact admin." 3. External system timeout: "Government database not responding. Try later or proceed manually." 4. Duplicate profile creation: "An offender with similar credentials already exists." 5. Permission error: "You are not authorized to create or edit offender profiles."

3.9.7 Notice Management #FR007

Description	This functionality enables authorized officers to generate, issue, and track enforcement notices in a structured and auditable manner. The system supports sequential issuance of Notice 1, Notice 2, and Notice 3 (Final) using configurable templates, automatically populating offender and case details. It records delivery methods, maintains acknowledgment status, supports re-issuance with version control, and logs all notices in a dedicated database for regulatory traceability.
User	1. Enforcement Officers / Inspectors – Generate and issue notices during the enforcement lifecycle. 2. Legal Officers / Controllers – Review notice history and take further legal action if required. 3. System Admins – Manage templates, delivery configurations, and audit logs.
Pre - Condition	1. An inspection and offense record must be linked to an offender. 2. The system user has appropriate permissions to issue notices. 3. Notice templates are predefined and configured in the system.

Sequence of Operations	- Officer navigates to the Notice Management interface. Clicks “Generate Notice”, selects: - Notice 1, Notice 2, or Notice 3 (Final) - System restricts sequence (e.g., Notice 2 can only follow Notice 1) System loads the relevant template, auto-filling: - Offender details - Inspection and Offense Reference Numbers - Violation summary - Applicable legal sections - Officer previews the notice and edits optional fields (e.g., hearing date, amount). - Clicks “Confirm & Issue” System: - Stores the notice with a unique Notice ID - Saves issue date and auto-assigns status = Issued - Sends via configured channels (Email, SMS, or manual delivery record) - Links notice to inspection/offense/offender records Officer can later: - Record acknowledgment status - Re-issue the notice (system creates a new version under same Notice Type) - View/download previous versions Each notice is stored in the Notice Table with: - ID, Type, Version, Issue Date, Delivery Method, Status
Post - Condition	- A notice is generated, issued, and stored with a unique reference. - Status of each notice is recorded and monitored (Issued / Delivered / Acknowledged). - Re-issued notices are tracked with version numbers.
Validity Checks	- Only one active version per notice type allowed per offense. Notices must: - Be generated from valid templates - Include linked inspection and offense references - Be assigned a unique Notice ID - Track status transitions and version numbers System must: - Prevent skipping notice levels - Log all delivery actions and timestamps - Allow updates to delivery/acknowledgment statuses only by authorized users - Store notices securely in a searchable Notice Table

Input	1. Notice type selection 2. Linked inspection and offense data 3. Auto-filled template fields 4. Manual inputs (e.g., remarks, hearing schedule)
Output	1. PDF/HTML formatted notice 2. Notice Reference ID and version 3. Status flags (Issued, Delivered, Acknowledged) 4. Audit entry per issuance or re-issuance 5. Notification (Email/SMS) sent to offender (if contact available)
Exception Handling	1. Incorrect notice sequence: "You must issue Notice 1 before proceeding to Notice 2." 2. Missing template: "Notice template not configured. Contact system administrator." 3. Delivery channel failure: "Unable to send notice via SMS/email. Please try again or mark as manually delivered." 4. Re-issue without prior notice: "Cannot re-issue without an existing issued version." 5. Acknowledgment status update failure: "Error updating status. Please retry."

3.9.8 Compounding fee agreement #FR008

Description	This functionality enables enforcement officers to initiate and manage compounding fee agreements in cases where offenders choose to resolve violations by paying a fine rather than proceeding to court. The system allows the offender to agree or decline, automatically calculates the fee based on offense type, supports manual or online payments, and generates a digital compounding fee receipt. Upon payment, the system updates the case status and notifies the offender. If the offender refuses to pay, the workflow shifts to court filing.
User	1. Enforcement Officers / Legal Officers – Record offender decisions, verify payments, and trigger next steps. 2. Offenders / Business Representatives – View fees, make payments, and receive receipts. 3. System Admins – Manage payment gateway configurations and compounding fee logic.
Pre - Condition	1. A valid offense must be registered and linked to an inspection. 2. The offender is contacted and asked to decide on fee compounding. 3. Payment gateway configuration is in place (for online payments).

Sequence of Operations	1. Officer accesses the Compounding Fee Agreement screen linked to the offense. 2. System displays “Agree to Pay Fee – Yes/No” radio buttons. 3. <u>Based on user input:</u> a. <u>If Yes:</u> 1. System calculates fee using offense type and predefined rules. 2. Displays fee amount and payment options (manual/online). 3. For online: - a.) Redirects to payment gateway (e.g., Razorpay, PayU). b. Upon success, captures transaction ID and marks status as Paid. 4. For manual: a. Officer enters manual receipt details and confirms. 5. System: a. <u>Generates digital receipt with:</u> 1. Offender info 2. Amount 3. Transaction ID 4. Date/time b. Links receipt to Inspection ID and Offense ID. - c.) Sends confirmation via email/SMS. b. <u>If No:</u> 1. System triggers court filing workflow. 2. Updates offense status to “Court Action Initiated”.
Post - Condition	1. Offense is either: a. Resolved via compounding fee payment, or b. Escalated to court filing if declined. 2. Receipt is generated and stored. 3. Notifications are sent and status updated.

Validity Checks	1. Only offenses in eligible status can be compounded. 2. Fee must: a. Be calculated based on offense type and predefined rules b. Be displayed before payment initiation 3. If online payment: a. Must be processed via an authenticated gateway b. Response and transaction ID must be captured 4. If manual payment: a. Receipt must include mandatory fields (receipt number, date, amount) 5. All payments must: a. Update offense status to Paid b. Trigger digital receipt generation c. Be stored and linked to the respective inspection/offense d. Send acknowledgment notification 6. If declined, system must auto-trigger court filing initiation
Input	1. Offender's decision (Yes/No) 2. Offense type and ID 3. Payment method (manual/online) 4. Transaction or manual receipt details (if applicable)
Output	1. Compounding fee amount 2. Payment status (Pending/Paid) 3. Digital receipt with transaction ID 4. Case status update (Paid or Court Action) 5. Notifications sent to offender 6. Record stored in database and linked to related IDs
Exception Handling	1. Payment gateway failure: "Transaction failed. Please retry or choose another payment option." 2. Invalid decision input: "Please select an option to proceed." 3. Fee calculation error: "Unable to calculate compounding fee. Check offense type or contact admin." 4. Manual receipt validation failure: "Missing or incorrect manual receipt details." 5. Notification delivery error: "Fee receipt issued, but notification could not be sent."

3.9.9 Case filing and Sanction #FR009

Description	This functionality enables enforcement officers to initiate the case filing workflow when compounding is rejected, or a violation warrants judicial action. The process includes obtaining sanction from a supervisory officer, digitally capturing their approval, generating a court filing document, assigning a case number, and transferring the file to the Legal Department. It ensures transparency, auditability, and timely notification to all stakeholders including the Inspector, Legal Officer, and Offender
User	1. Inspectors / Enforcement Officers – Request case filing and prepare supporting reports. 2. Supervisors / Sanctioning Officers – Review and approve/reject filing requests. 3. Legal Officers – Manage legal documents, update court details, and proceed with litigation. 4. System Admins – Manage workflow, logs, and notifications.
Pre - Condition	1. Offense is marked as "Compounding Declined" or "Court Action Required". 2. All inspection and offense records are completed and linked. 3. The user initiating the case is authorized.

Sequence of Operations	1. Inspector initiates a request for case filing via the “Request Sanction” button. 2. System routes the request to the Sanctioning Officer (Supervisor) for review. 3. Officer: a. Reviews inspection and offense summary b. Enters remarks and applies digital signature for sanction 4. Upon approval: a. System logs Sanctioning Officer’s name, role, timestamp b. Stores entry in the Sanction History Log c. Generates Case Filing Document (PDF) containing: 1. Inspection summary 2. Offense details 3. Sanction approval 4. Evidence attachments 5. System auto-generates a Case Number. 6. Case file is transferred to the Legal Department Dashboard. 7. Legal Officer: a. Uploads additional legal documentation (e.g., affidavits, notices) b. Updates: 1. Court filing date 2. Venue 3. Presiding Judge details 8. System sends notification to: a. Inspector b. Legal Officer c. Offender (via email/SMS)
Post - Condition	1. Case is filed with a unique Case Number. 2. Legal department receives the case package and may upload additional documents. 3. Court details are stored, and notifications are sent. 4. Sanction and filing actions are logged.

Validity Checks	Case can only be filed if: - Compounding is declined - Sanction is digitally approved All actions must be: - Time-stamped - Associated with valid user roles Sanction log must: - Include officer name, designation, and digital signature Case filing document must: - Contain all linked references (inspection, offense, sanction) Case Number must be: - System-generated and unique Legal upload interface must: - Accept documents in approved formats (PDF, JPEG, etc.) - Capture mandatory court details Notifications must reach: - All key users (Inspector, Legal Officer, Offender) Audit logs must capture: - All case transitions, approvals, and uploads
Input	- Inspector request to file case - Sanctioning officer's approval (with digital signature) - Offense and inspection data - Legal officer's updates (documents, court info)
Output	- Sanction approval log entry - Case filing document (PDF) - Unique Case Number - Legal dashboard update - Notifications to all parties - Case status updated to "Filed with Court"
Exception Handling	Missing sanction: "Cannot proceed. Supervisor approval required." Digital signature failure: "Unable to sign. Please retry or contact admin." PDF generation error: "Case document could not be generated." Case number conflict: "Duplicate case number detected. Please retry." File transfer error: "Legal team could not be notified. Manual transfer required."

3.9.10 Judicial Proceedings #FR010

Description	This functionality handles the post-filing judicial workflow in enforcement cases. It allows legal officers to update the system with court schedules, record verdict outcomes, upload judgment files, and set final case resolution statuses. Based on the judgment (e.g., fine, dismissal, warning), the system can trigger next steps, such as payment processing or appeals. A real-time dashboard provides visibility into the status of ongoing and resolved cases.
User	1. Legal Officers – Manage court hearing data, upload judgments, and close cases. 2. Inspectors / Enforcement Officers – View court progress and update outcomes. 3. System Admins – Maintain workflow rules and data integrity for judicial records.
Pre - Condition	1. Case has been officially filed and transferred to the legal module. 2. Court details and documents are available from the legal officer or court authority.

Sequence of Operations	1. Legal Officer opens the case from the Judicial Processing Dashboard. 2. Enters court hearing schedule, including: a. Date b. Court venue c. Judge name (optional) 3. After the hearing: a. Selects judgment outcome from dropdown: b. Fine imposed c. Warning issued d. Case dismissed e. Enters court remarks (text field) f. Uploads judgment file (PDF only) 4. If outcome = Fine: a. Enters fine amount b. Uploads payment receipt if paid 5. System sets final case status based on outcome: a. Resolved b. Pending Appeal c. Closed 6. Based on status: a. System triggers next workflow: 1. Payment confirmation 2. Appeal initiation 3. Dashboard update 7. Status appears on Ongoing vs. Resolved Cases Dashboard for monitoring
Post - Condition	1. Judgment and related documents are stored and visible in the system. 2. Case status is updated accordingly. 3. Next actions (e.g., appeal, payment) are triggered based on judgment type.

Validity Checks	Each case must: - Be linked to a previously filed and sanctioned inspection - Include a valid judgment type selection - Have an associated judgment document (PDF) Fine entry: - Mandatory if outcome = "Fine Imposed" - Receipt upload required to mark payment as completed Status transitions must: - Match the court outcome logic - Be logged with timestamp and user ID All judicial updates must: - Be recorded in the audit trail - Trigger appropriate downstream processes (e.g., notify parties, update dashboards)
Input	- Court hearing schedule (date, location) - Judgment type and remarks - Uploaded PDF judgment file - Fine amount (if applicable) - Payment receipt (optional)
Output	- Updated case record with court outcome - Final status (Resolved, Pending Appeal, Closed) - Judgment document storage - Dashboard update for case monitoring - Triggered next-step workflow (e.g., appeal, penalty closure)
Exception Handling	Invalid file type on upload: "Only PDF format allowed for judgment documents." Missing mandatory fields: "Please enter all required court judgment details." Incorrect fine data: "Fine amount must be numeric and greater than zero." File upload error: "Judgment document upload failed. Please retry." Status mismatch error: "Cannot close case without judgment outcome and document."

3.9.11 Resolution and Record Maintenance #FR011

Description	This functionality finalizes the enforcement case lifecycle by managing the resolution, archival, and exportability of inspection-related legal actions. Upon payment of fees or completion of judicial processing, the system marks the case as “Resolved”, updates the offender’s profile, locks further modifications to the inspection record, archives all documents, and enables authorized export of the complete case file. Additionally, resolved cases are linked to the analytics and reporting engine for compliance insights and regulatory metrics.
User	1. Legal Officers / Enforcement Officers – Finalize case status and manage archival. 2. Supervisors / Controllers – Review, audit, or export finalized case data. 3. System Admins – Manage data archival settings and analytics integration.
Pre - Condition	1. Case has completed all necessary actions: a. Compounding fee paid or b. Court judgment delivered and documented 2. All related documents are uploaded, and the outcome is confirmed. 3. System roles and permissions are defined for closure and archival actions.
Sequence of Operations	1. Officer opens the enforcement case marked “Ready for Closure.” 2. System validates: a. Fee payment status or b. Judgment file and outcome status 3. Officer clicks “Flag as Resolved”. 4. System: a. Updates case status to Resolved b. Locks the inspection record to prevent edits c. Updates offender’s profile with: 1. Case type 2. Resolution type (Fee Paid / Judicial Verdict) 3. Resolution date 4. Archives final documents (inspection, notices, judgment, receipts) into case history 5. Authorized users (e.g., Supervisor, Legal Officer) may: a. Export full case file as PDF bundle or ZIP archive b. View case record in Analytics & Reporting dashboard

Post - Condition	1. Case is marked as Resolved and locked from further edits. 2. Offender's history is updated with resolution details. 3. Case data is archived and linked to the reporting engine. 4. Export and reporting features are enabled.
Validity Checks	1. Case can only be resolved if: a. Fee is confirmed as paid (via receipt or payment record), or b. Judgment is uploaded and marked complete 2. Inspection records must: a. Be locked after resolution b. Remain read-only except to authorized auditors 3. Offender's profile must: a. Reflect the resolution with date and type b. Maintain cumulative offense history 4. Archival process must: a. Include all system-generated and user-uploaded documents b. Be logged with timestamp and user ID 5. Export and reporting linkage must: a. Be role-restricted b. Enable audit-friendly retrieval and dashboard integration
Input	1. Officer confirmation for resolution 2. Linked fee receipt or court judgment document
Output	1. Case status = Resolved 2. Locked inspection record 3. Archived document set 4. Updated offender profile 5. Exportable case file (PDF/ZIP) 6. Case linked to analytics engine
Exception Handling	1. Missing final documents: "Cannot resolve case. Ensure payment or judgment is uploaded." 2. Unauthorized user: "Only authorized roles can mark case as resolved." 3. Archival error: "Error archiving case documents. Please retry." 4. Export failure: "Unable to generate case export. Try again or contact admin." 5. Analytics sync failure: "Case resolved but not linked to reporting module. Escalate to system admin."

3.9.12 Search and Reporting #FR012

Description	This functionality provides a comprehensive search and reporting interface for enforcement-related records within the eMaap platform. It allows users to perform advanced searches on inspections and offenses by multiple parameters, access filterable dashboards, and generate predefined and custom reports. The results can be exported in Excel or PDF format, and reports can be printed or shared, enabling effective monitoring, compliance tracking, and administrative decision-making.
User	1. Inspectors / Enforcement Officers – Search inspections and generate field-level summaries. 2. Supervisors / Controllers – Access trend and compliance reports for their jurisdiction. 3. Legal Officers – Track pending court cases and notice activity. 4. DoCA / State Admin Users – Analyze large-scale patterns, collections, and repeat offenders. 5. System Admins – Maintain report templates, export configurations, and access controls.
Pre - Condition	1. Enforcement data (inspections, offenses, payments, court cases) has been entered and processed in the system. 2. The user is authenticated and has role-based access to search or report features.
Sequence of Operations	1. User navigates to “Search and Reports” section. 2. Selects “Advanced Search” and enters filter criteria: a. Date range b. Location (District/Zone/State) c. Inspection or Case Status (e.g., Resolved, Pending, Court) d. Optional: Offender name, ID, category 3. Clicks “Search” → system displays results in a table. 4. User can: a. View results on-screen b. Export to Excel or PDF 5. For reports: a. Selects from predefined templates: 1. Daily Inspection Summary 2. Offense Trends 3. Fee Collection Report 4. Repeat Offenders List 5. Pending Notices / Court Cases b. Enters filters (optional) c. Clicks “Generate Report”

	- System processes data and generates downloadable report document. - User can print or export the report.
Post - Condition	- Search results are retrieved and viewable/exportable. - Reports are generated and stored or shared. - Dashboards reflect the most recent case activity and enforcement trends.
Validity Checks	Search must: - Accept at least one valid filter - Return only results within the user's access scope Report generation must: - Use accurate and current data - Include report metadata (generated date, officer name, filters used) Export files must: - Be downloadable in Excel or PDF - Be structured and printable System must: - Log every search and report action for audit trail - Update dashboards in near real time based on status changes
Input	- Search parameters (dates, location, status, offender info) - Report type selection - Export format selection (Excel/PDF)
Output	- Filtered search result table - Downloadable reports (PDF/Excel) - Visual dashboard views by status (Resolved, Pending, Court) - Charts/tables for trend and compliance insights
Exception Handling	No results found: "No records match your criteria." Large data volume: "Too many results. Please narrow your filters." Export failure: "Error generating file. Please try again later." Unauthorized access: "You do not have permission to view this data." Report generation timeout: "Report took too long to generate. Refine filters."

3.9.13 Notification and Alert #FR013

Description	This functionality enables the system to send real-time SMS and email alerts to stakeholders at key points in the enforcement process. Notifications are triggered upon events such as notice issuance, payment due, court date updates, and fee receipt issuance. The system ensures that recipients are promptly informed of obligations, updates, and outcomes, improving compliance and communication transparency.
User	1. Offenders / Business Users – Receive SMS/email notifications related to their enforcement case. 2. Inspectors / Legal Officers – Monitor delivery of notices and alerts. 3. Supervisors / Admins – Configure notification templates and track delivery status. 4. System Admins – Maintain gateway integrations and log delivery outcomes.
Pre - Condition	1. Valid mobile number and/or email address is available in the offender's profile. 2. A system event (notice issued, payment due, court date set, receipt generated) has occurred. 3. Notification templates are configured and enabled.
Sequence of Operations	1. A triggering event occurs: a. Notice Issued b. Payment Due c. Court Date Scheduled d. Fee Receipt Generated 2. System fetches: a. Recipient's contact information (email, phone) b. Relevant template for the event 3. Personalizes message using event data: a. Reference numbers, dates, fee amount, location, etc. 4. Sends notification via: a. SMS gateway b. Email server 5. Logs the alert in the system: a. Timestamp b. Recipient c. Delivery status (Sent/Failed) 6. User (officer) can view history in notification log or case record.

Post - Condition	1. Notification is sent to the designated recipient(s). 2. Delivery status is recorded. 3. Notification entry is logged and visible in the notification history.
Validity Checks	1. Notifications must be: a. Tied to valid triggering events b. Sent only to verified contact details 2. Templates must: a. Be available and active for the event type b. Use valid placeholders and formatting 3. System must: a. Log every alert with timestamp, recipient, delivery status b. Allow review and export of notification history c. Retry or flag failed messages for escalation 4. All sensitive data (e.g., notice numbers) must be securely included using masked or authorized templates
Input	1. Triggering event (e.g., Notice generated) 2. Recipient contact details 3. Message template 4. Dynamic data (notice number, court date, etc.)
Output	1. SMS or Email sent to recipient 2. Status log (Delivered / Failed) 3. Entry in case's notification history 4. Optional user interface banner/message
Exception Handling	1. Missing contact details: "Notification failed. Email or phone number not available." 2. Template not found: "Notification template missing for this event." 3. Gateway/API error: "SMS/Email service unavailable. Retry scheduled." 4. Invalid message content: "Failed to personalize message. Check data placeholders." 5. Delivery failure: "Alert not delivered. Marked for retry or manual notification."

3.9.14 Admin Panel and Configuration #FR014

Description	The Admin Panel provides centralized controls for configuring, managing, and auditing all critical elements of the enforcement tracking system. It allows system administrators to manage notice templates, inspection types, user roles, and fee thresholds, while also supporting module enablement per state, data backup/restore, and full access to the audit trail. This module ensures flexibility, policy alignment, and administrative oversight for seamless operations across jurisdictions.
User	1. System Administrators – Manage system-wide settings, configurations, and audit trails. 2. Department Admins (State/Centre) – Customize features based on state-specific policies. 3. IT Support Officers – Perform backups, restores, and emergency module reconfiguration.
Pre - Condition	1. User is authenticated with admin privileges. 2. Administrative features are enabled in the system's configuration schema. 3. All core enforcement modules are installed.

Sequence of Operations	- Admin logs into the Admin Panel dashboard. Available configuration options: Notice Template Management - Add/edit/delete templates - Use placeholders for autofill Inspection Category Management - Define types (Weights, Measures, Packaged Goods) - Add jurisdiction-specific rules Role and Privilege Settings - Create/assign roles - Set module access levels Fee Calculation Configuration - Define thresholds for offenses - Link to offense type or severity Enable/Disable Modules - Based on state/UT policy - E.g., disable model approval in a non-participating state Audit Log Viewer - Filter logs by action, user, date - Export logs to Excel/PDF Backup & Restore - Schedule daily backups - Perform manual or automatic restore - Admin confirms any changes. - System applies settings and logs configuration events.
Post - Condition	- Changes to system configuration are saved and reflected across user roles. - System logs changes to audit trail. - Templates, categories, and thresholds are updated system-wide or by state, as applicable.

Validity Checks	Templates must: - Use valid placeholders - Be mapped to specific notice types Role settings must: - Follow the principle of least privilege - Be auditable and change-tracked Fee thresholds must: - Align with predefined offense codes - Trigger correct auto-calculation Backup & restore functions must: - Include integrity verification - Be logged with timestamps and user IDs All admin actions must be: - Recorded in an immutable audit log - Available for export or audit review
Input	- Notice template data - Inspection category definitions - Role-permission matrix - Fee thresholds - State policy flags - Backup/restore preferences
Output	- Updated templates, categories, and user settings - System-level flags for module activation - Audit log entries - Backup files or restored datasets
Exception Handling	Invalid template syntax: "Template contains invalid placeholders. Please correct." Duplicate role or category: "This entry already exists in the system." Permission error: "You do not have access to modify this setting." Backup failure: "Backup process failed. Retry or contact IT admin." Restore conflict: "Data conflict during restore. Manual review required."

3.9.15 Security and Compliance #FR015

Description	This functionality ensures that all enforcement-related operations within the system adhere to the highest standards of data security, integrity, and regulatory compliance. It includes role-based access control, secure storage and transmission of data, logging of all critical user actions, and enforcement of privacy and authentication protocols such as CAPTCHA, 2FA, session management, and adherence to national data protection laws. It also includes daily backups and access auditing for document integrity and disaster recovery.
User	1. All Users (Inspectors, Officers, Admins, Legal Officials) – Interact with the system under controlled access policies. 2. System Administrators – Configure and monitor security settings, logs, and compliance status. 3. Auditors / Compliance Officers – Review logs and confirm adherence to policy.
Pre - Condition	1. User authentication is successful via login credentials. 2. System security policies are enabled and enforced. 3. Access rights are defined per role.

Sequence of Operations	1. User Login: a. Enters credentials b. CAPTCHA or 2FA triggered for sensitive roles or actions 2. Session Management: a. User logs in b. System auto-logs out after inactivity (e.g., 15 min idle timeout) 3. Data Access: a. Based on role permissions, user views only permitted records b. Sensitive documents are accessed via secure interface c. All document views/downloads are logged 4. Data Transmission: a. All form submissions, uploads/downloads occur via SSL/TLS encryption 5. Database Integrity: a. Daily scheduled backups are triggered automatically b. Backup logs are available for audit 6. Audit Logging: a. Any edit/delete actions are logged with: 1.) User ID 2.) Timestamp 3.) Action type 4.) Affected record 7. Compliance Controls: a. Data handling and access comply with: 1.) Indian IT Act 2.) DPDP (Digital Personal Data Protection Act) 3.) Other relevant local/state policies
Post - Condition	1. User activity is logged with user ID and timestamp. 2. Documents are securely stored and accessed. 3. Data privacy and protection standards are upheld. 4. Audit trails are generated for review.

Validity Checks	Access rights must: - Be mapped to user roles and enforced system-wide All documents: - Must be stored in secure, access-controlled repositories - Must have view/download logs per user All sensitive actions: - Must require CAPTCHA/2FA - Must be logged with timestamp, user ID, and action type System must: - Encrypt all data in transit using SSL/TLS - Backup data daily with automated verification - Log access attempts, including failed login and timeout events Audit logs must be: - Tamper-proof - Available for export to auditors
Input	- User credentials (username/password) - User actions (view, edit, delete, upload, download) - CAPTCHA and 2FA tokens
Output	- Role-validated access to screens/documents - Encrypted data transmission logs - Audit logs of sensitive operations - Backup snapshots with time and status - Compliance reports (optional)
Exception Handling	Unauthorized access: "Access denied. You do not have permission for this action." Session timeout: "Your session has expired. Please log in again." 2FA failure: "Invalid verification code. Try again." CAPTCHA error: "Incorrect CAPTCHA. Please try again." Backup failure: "Database backup failed. Admin has been notified." Edit log missing: "Change not logged. Check logging configuration."

3.9.16 Audit & History Tracking #FR016

Description	This functionality provides full visibility into all actions performed within the enforcement system by implementing detailed audit trails, version control, and status progression tracking. It captures every user interaction (add/edit/delete/view), maintains historical versions of notices and court documents, and allows authorized users to view and export a complete case history, including document attachments and
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	workflow milestones. It also includes a timeline viewer for administrators to trace the lifecycle of an enforcement event from inspection to resolution.
User	1. Inspectors / Legal Officers / Enforcement Officers – Their actions are logged and traceable. 2. Supervisors / Controllers – View audit logs and case history. 3. System Administrators – Manage logs, version control policies, and enable access to timeline viewer. 4. Auditors / Compliance Officers – Use case history exports for review or legal inquiry.
Pre - Condition	1. Enforcement actions (inspection, seizure, notices, offenses, case resolutions) are initiated and ongoing. 2. Users perform activities under authenticated sessions. 3. Logging and versioning modules are active.
Sequence of Operations	1. User performs any action (e.g., add/edit/delete/view record). 2. System logs: a. Action type b. Timestamp c. User ID and role d. Affected entity (e.g., Inspection ID, Notice ID) 3. If action involves a document (e.g., notice or court file): a. System saves a new version with version number and timestamp. b. Earlier versions remain accessible (read-only). 4. Supervisor or admin accesses the Audit History Panel: a. Filters actions by user, date, entity, or module b. Views a timeline progression of inspection status: c. Inspection > Seizure > Offense > Notice > Case Filing > Court Outcome > Resolution 5. From the case record, user clicks “Download Full Case History”: a. System generates a PDF bundle: b. Case summary c. Status logs d. Attached documents e. Version logs (if enabled) 6. All logs and timelines are stored for audit review and compliance checks.
Post - Condition	1. All actions are recorded in immutable logs. 2. Previous versions of legal documents are retained. 3. Full case timeline is viewable and exportable in PDF format.

Validity Checks	Every user action (add/edit/delete/view) must be: - Logged with timestamp and user ID - Linked to the relevant entity (e.g., Inspection ID) Notices and legal documents must: - Be version-controlled - Retain all previous versions with timestamps Status changes must: - Follow sequential flow (Inspection → Seizure → Offense → etc.) - Be viewable in a timeline format Case history exports must: - Include all documents, logs, and actions - Be accessible only by authorized roles - Logs must be immutable and retained for the required legal duration
Input	- User actions (create/edit/delete/view) - Legal documents (notices, court files) - Case identifiers
Output	- Action logs (with timestamp, user, role, action type) - Versioned document records - Status change timeline - Downloadable case history file (PDF with attachments)
Exception Handling	Log entry failure: “Audit log not saved. Please retry or contact admin.” Version mismatch: “Unable to retrieve prior version. Integrity check failed.” Export generation error: “Case history download failed. Try again later.” Unauthorized timeline access: “Only administrators can access the full progress timeline.” Overwritten data attempt: “Versioned record. Editing requires new version creation.”

3.9.17 Use Cases (Time tracking Enforcement)

Field Name	Description
Use Case ID	eMaap_TTE_011
Use Case Name	Time Tracking Enforcement
Reference	FRS Number: FR001 to FR016
Created By (Date)	eMaap functional team, (16-06-2025)
Last Updated (Date)	16-06-2025
Primary Actor	Inspector, Assistant Controller

Secondary Actor	Business Representative, Witness, Controller, Judicial Authority, eMaap Notification System.
Trigger (Activating Use Case)	Starts when an inspector begins an inspection or logs an enforcement activity via the portal.
Inter Linkages	Connected with: 1. Geo-tagging Service 2. Notification Module 3. Case History Database 4. Compounding Fee Payment System 5. Court Case Filing Module.
Normal Event Flow	1. Inspector conducts inspection with witness 2. Deficiency check 3. If compliant, case closed 4. If not, seizure and offence logging 5. Check repeat offence 6. For first-time, notices issued (up to 3) 7. Compounding fee requested 8. On payment, issue receipt and close case 9. If refused, escalate to court and update records.
Alternate Event Flow	1. If Geo-tag fails – fallback to manual entry 2. If compounding not accepted – judicial sanction required 3. If court judgment delays – system marks case as pending and notifies admin.
Frequency of Use	Frequent – used by all enforcement officers during routine and complaint-based inspections.
Assumptions	1. Officers have real-time access to eMaap 2. Business representatives are reachable via email/SMS 3. Geo-tagging service is operational Legal framework supports digital compounding.
Additional Notes	Supports transparency, accountability, and progressive enforcement Appeal logging by businesses is integrated Data available for audit and reporting purposes.

3.10 Others Module (Dashboard Module)

The system should allow the user to analyze and graphically represent data, and to generate various Management Information System (MIS) reports categorized on criteria such as periodicity, district-wise, zone-wise, status wise etc. Few reports to be generated, but not limited to are as follows:

- i) Customized MIS reports
- ii) Monthly progress reports
- iii) Revenue reports
- iv) Inspection/verification reports with target versus actual
- v) Dashboard for various level of users and officers for regular monitoring of data & information
- vi) Ticketing reports

The dashboard module in the eMaap portal for analytics and reporting is designed to provide comprehensive, role-based insights for effective monitoring and decision-making. It allows users at various administrative levels—including applicants, inspectors, zonal officers, and department heads—to access real-time data visualizations and generate a wide range of Management Information System (MIS) reports. These reports can be customized and categorized based on parameters such as time period (daily, monthly, quarterly), geographical area (district-wise, zone-wise), and status (application progress, inspection outcomes, etc.). Key reports include customized MIS reports, monthly progress reports, revenue reports, inspection and verification performance reports (target versus actual), and ticketing reports. The dashboard presents data in the form of interactive charts, graphs, and tables, with drill-down capabilities for granular analysis. Users can export reports in various formats such as PDF, Excel, and CSV, and can also schedule automated report generation and delivery. Each user role is presented with a tailored dashboard view that highlights the most relevant metrics and information for their function. This model ensures enhanced transparency, data-driven governance, and efficient tracking of operational performance across the portal.

3.10.1 Dashboard Module #FR001

Description	The Dashboard Module provides a centralized, real-time interface that dynamically adjusts based on the user's role within the eMaap system (e.g., Applicant, Inspector, Admin, Legal Officer). It offers personalized widgets, quick access links, analytics charts, alerts, and reports. This ensures that all users can efficiently monitor and manage enforcement activities, system tasks, and performance metrics through an intuitive and responsive interface.
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User	1. Applicants / Business Users – View license-related statuses, initiate applications, and track submissions. 2. Inspectors / Enforcement Officers – Monitor assigned inspections, daily schedules, and route maps. 3. Administrative Officers / Controllers – View zone/district performance, application processing metrics, SLA compliance. 4. System Admins – Configure access, manage widgets, view audit logs, and track platform usage.
Pre - Condition	1. User is successfully authenticated. 2. Role-based permissions are pre-configured in the system. 3. Relevant enforcement data (inspections, applications, notices) exists in the system. 4. Reporting and charting engines are operational.
Sequence of Operations	1. Login & Routing: a. System authenticates the user. b. User is redirected to a role-based dashboard interface. 2. Dashboard Rendering by Role: a. Applicant: Quick links to apply for new/renewal/amendment/duplicate licenses; real-time application status. b. Inspector: Assigned inspections, daily task list, deadlines, and route maps. c. Admin/Officer: Charts for SLA metrics, number of inspections, approval rate, pending workload, district/zonal insights. 3. Interactive Widgets: a. Real-time alerts/notifications for: 1. Notice issuance 2. Payment dues 3. Court dates 4. New assignments - b.) Recent activity log (e.g., last actions by user) 4. Report Generation: - a.) Users generate reports: 1. MIS (customized) 2. Monthly performance 3. Revenue collected 4. Ticketing status 5. Inspection outcomes b. Reports exportable in PDF, Excel, CSV formats. c. Option to schedule recurring reports via email. 5. Analytics & Visualization: a. Graphs, charts, and tables showing real-time metrics.

	b. Drill-down capabilities to view regional/category data. c. Dashboards auto-refresh every 5 minutes. 6. User Actions: a. Switch roles (if assigned multiple roles). b. Update personal profile or upload documents. c. Submit support tickets via integrated feedback/ticketing tool. 7. Security & Logging: a. All interactions (views, report generation, exports) are logged. b. Role-based access control (RBAC) ensures only authorized data is shown.
Post - Condition	1. Users access a personalized dashboard. 2. Reports are generated or exported. 3. User activities are logged, and dashboards reflect real-time data
Validity Checks	1. Users must only see dashboard components based on assigned roles and permissions. 2. Reports: a. Must reflect accurate, real-time data from the enforcement system. b. Should contain metadata (generated date, user ID). 3. Scheduled reports: a. Must trigger at configured times b. Should send emails to valid, authorized recipients 4. All actions (report viewing, downloads, interactions) must: a. Be logged in audit trail with user ID and timestamp 5. Dashboard must be: a. Responsive across devices (desktop, tablet, mobile) b. Compatible with major browsers c. Auto-refreshing at configured intervals (default: every 5 mins)
Input	1. User role and credentials 2. Data filters for reports (date, district, status) 3. User profile data updates
Output	1. Dynamic dashboard widgets and reports 2. Graphical analytics (charts, KPIs) 3. Notifications and recent actions 4. Exported reports (PDF, Excel, CSV)
Exception Handling	1. Unauthorized access: "Access denied. You do not have permission to view this section." 2. Report generation failure: "Report generation failed. Please retry later." 3. Auto-refresh error: "Dashboard refresh error. Check your network or try again."

4. **Widget load failure:** “Unable to load widget data. Contact support.”
5. **Email scheduling issue:** “Report schedule setup failed. Verify recipient address or try again.”

3.10.2 Use Cases (Dashboard Module)

Field Name	Description
Use Case ID	eMaap_DM_012
Use Case Name	Role-Based Dashboard and Reporting
Created By (Date)	eMaap functional team (16-06-2025)
Last Updated (Date)	16-06-2025
Primary Actor	Applicant, Inspector, Zonal Officer, Department Head
Secondary Actor	System Scheduler, Admin, Report Viewer, Notification System.
Trigger (Activating Use Case)	Initiated when a user logs in to the portal and accesses the dashboard module.
Inter Linkages	Connected to: 1. MIS Reporting Service 2. Data Aggregation Layer 3. Scheduling & Export Module 4. User Role Management.
Normal Event Flow	1. User logs into eMaap portal 2. Navigates to dashboard 3. Selects report parameters (role, date, region, status, etc.) 4. System retrieves and visualizes data 5. User views/downloads or schedules reports.
Alternate Event Flow	1. If data source unavailable – show alert with retry option 2. If report exceeds size limit – user advised to refine filters 3. If export fails – allow retry or report issue to admin.
Frequency of Use	Frequent – used daily by officers, weekly/monthly by department heads for performance and revenue monitoring.
Assumptions	1. Data is synchronized in real-time 2. Users have assigned roles and access rights

	3. System can handle large data volume and simultaneous requests.
Additional Notes	Supports drill-down and graphical reports Exportable in PDF/Excel/CSV Custom MIS reports and auto-scheduling supported.

End